Chapter 8
Plane Electromagnetic Waves

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Contents

8.1 Plane Waves in Lossless Media

Basic Issues

- **Time-harmonic field**

$$A(\vec{r},t) = A_0(\vec{r})\cos[\omega t + \phi(\vec{r}) + \phi_0]$$

- **Infinite single media space**

- **EM wave propagation**

In passive region

$$\nabla^2 \vec{E}(\vec{r}) + k_c^2 \vec{E}(\vec{r}) = 0$$

Related concepts

1） Amplitude

2） phase

- **time phase**
- **space phase**
- **Argument**
- **Initial Phase**

3） Equal-phase Surface

4） Equal-Amplitude Surface

Related concepts

$$\vec{E}(\vec{r},t) = \mathrm{Re}[\vec{E}(\vec{r})e^{j\omega t}]$$

$$\vec{E}(\vec{r}) = \vec{e}_x E_x(\vec{r})e^{j\phi_x(\vec{r})} + \vec{e}_y E_y(\vec{r})e^{j\phi_y(\vec{r})} + \vec{e}_z E_z(\vec{r})e^{j\phi_z(\vec{r})}$$

Complex vector contains the changes of field quantities at any time in space

Problems

$$\frac{\partial}{\partial t} = j\omega$$

Analysis of Time-harmonic EM Wave

The law of the field quantities change with spatial location

√ **Plane Wave**

Cylindrical wave

Spherical wave

(The complex vector functions of fixed time)

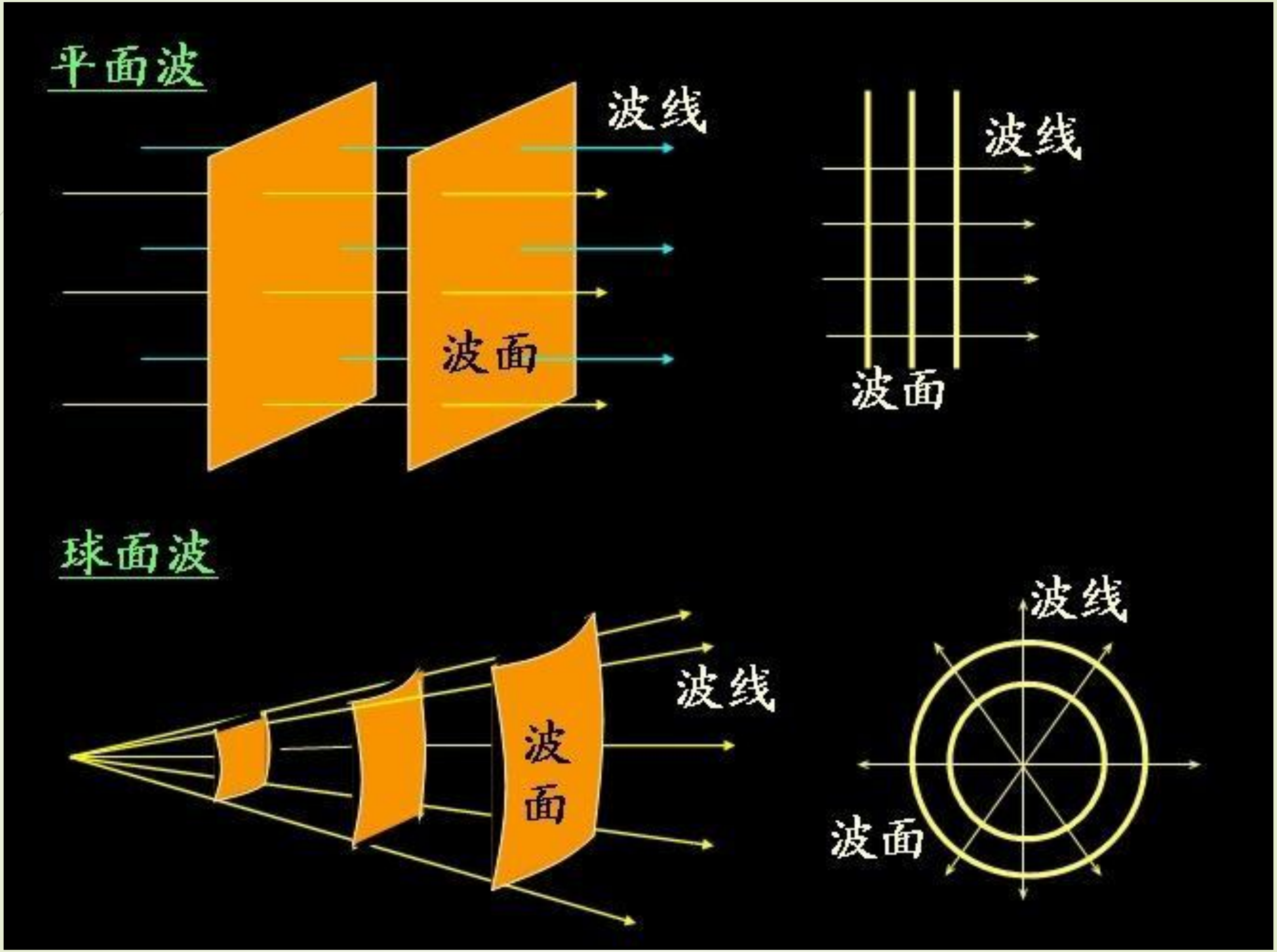
The law of the field quantities change with time

√ **Linear polarized wave**

√ **Circular polarized wave**

√ **Elliptical polarized wave**

(A fixed position of the instantaneous changes)


平面波
波线
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球面波
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The definition of uniform plane wave

- **Plane wave:** At any time Equiphase Surface (wavefronts) is a flat sheet (plane)
- **Uniform:** The amplitude of the EM field in the equiphase surface is fixed
- **Uniform plane wave**

The equiphase surface of EM wave is plane, and also the amplitude of EM field on the equiphase surface is a constant.

Features

- Equi-phase surface and Equal-Amplitude Surface of uniform plane wave are coincidence or parallel
- The electric field complex vector in equiphase surface(Eqs) is a constant vector
- The amplitude and direction of the EM field in equi-phase surface are fixed at any time

$\vec{E}(\vec{r})\big|_{equi}$

Q : At different moments , the EM field in equi-phase surface of uniform plane wave is the sam

$$\nabla^2\vec{E}(\vec{r})+k_c^2\vec{E}(\vec{r})=0$$

The ideal dielectric

$$\sigma=0$$

$$k=\omega\sqrt{\mu\varepsilon}=\beta$$

$$\nabla^2\vec{E}(\vec{r})+k^2\vec{E}(\vec{r})=0$$

Conductive medium

$$\sigma\neq 0$$

$$k_c=\omega\sqrt{\mu\varepsilon_c}=\beta-j\alpha$$

1. The Electromagnetic Field of Uniform Plane Wave

Skills: establishing a proper coordinate system！

take equiphase surface as coordinate plane ,such as x-y plane , thus: $\vec{E}(\vec{r}) = \vec{E}(z)$

$$\nabla^2 \vec{E} + k^2 \vec{E} = 0 \Rightarrow \frac{\mathrm{d}^2 \vec{E}}{\mathrm{d}z^2} + k^2 \vec{E} = 0 \Rightarrow \frac{\partial^2 E_i}{\partial z^2} + k^2 E_i = 0\,, \quad i = x, y, z$$

Solution: $E_i(z) = A_1 \mathrm{e}^{-\mathrm{j}kz} + A_2 \mathrm{e}^{\mathrm{j}kz}$ **then:** $H_i(z) = B_1 \mathrm{e}^{-\mathrm{j}kz} + B_2 \mathrm{e}^{\mathrm{j}kz}$

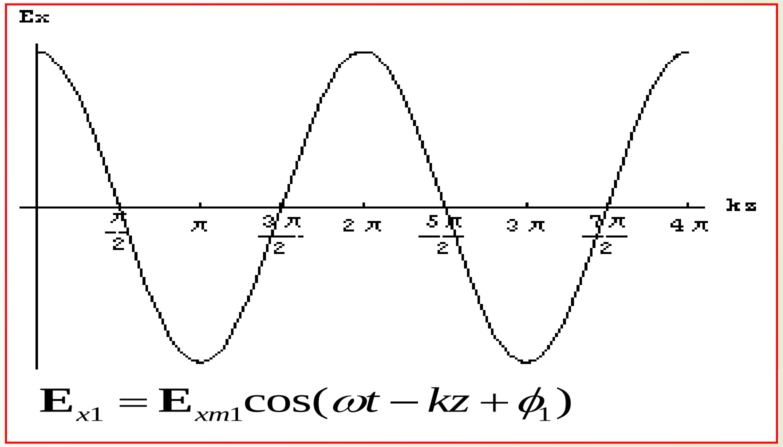
The instantaneous expression of the electric field

$$E_i(z,t) = \mathrm{Re}[E_i(z) e^{j\omega t}]$$

$$= E_{i1}(z,t) + E_{i2}(z,t)$$

$$E_{i1}(z,t) = E_{im1}\cos(\omega t - kz + \phi_1)$$

$$E_{i2}(z,t) = E_{im2}\cos(\omega t + kz + \phi_2)$$



The EM wave propagate along the direction of the spatial phase lag

Important features

$\nabla \cdot \vec{E} =$

$\nabla \cdot \vec{H} =$

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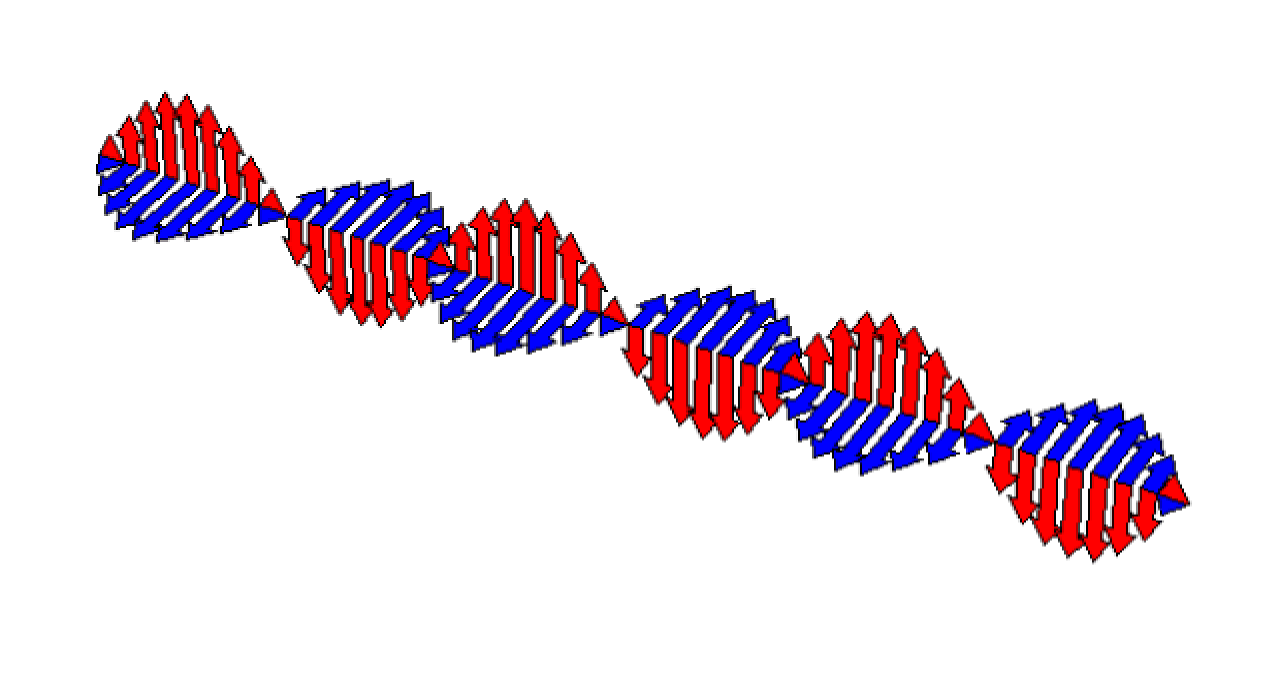
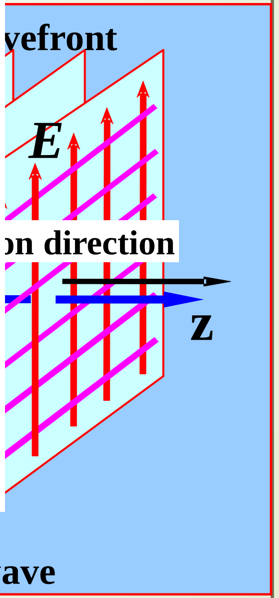




Summary

- **The complex electromagnetic field vector of uniform plane wave that propagating along the z direction can express as :**

$$\vec{E}(z) = \vec{E}_{\mathrm{m}} \mathrm{e}^{-\mathrm{j}kz} \qquad \overrightarrow{H}(z) = \overrightarrow{H}_{\mathrm{m}} \mathrm{e}^{-\mathrm{j}kz}$$

- **Uniform plane wave is transverse electromagnetic waves（TEM wave）**

$$\vec{e}_z \bullet \overrightarrow{E} = 0, \quad \vec{e}_z \bullet \overrightarrow{H} = 0$$

- **The EM wave propagate along the direction of the spatial phase lagging.**

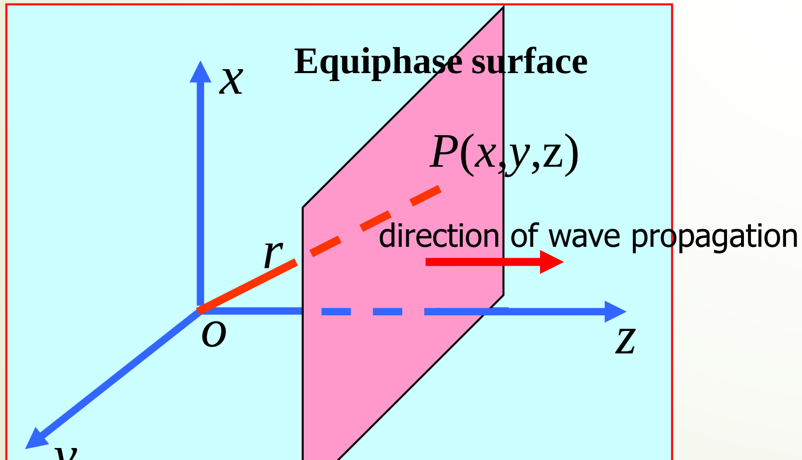
Solution for uniform plane wave propagating along arbitrary direction

To assume wave propagation direction: $\vec{e}_n$

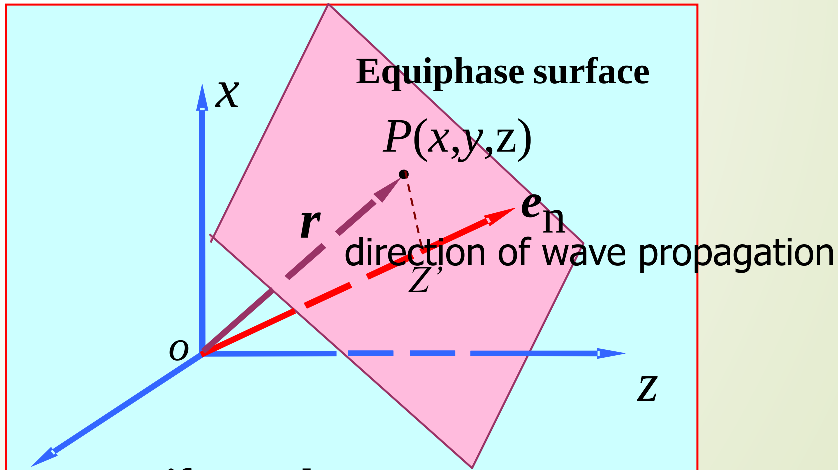
So that: $$\vec{E} = \vec{E}_{\mathrm{m}} e^{-jkz'} = \vec{E}_{\mathrm{m}} e^{-jk\vec{e}_n \bullet \vec{r}}$$

For the convenience of said to define new physical quantities, Wave vector: $\vec{k} = k\vec{e}_n$

so $\vec{E} = \vec{E}_m e^{-j\vec{k}\bullet\vec{r}}$ **as the same** $\overrightarrow{H} = \overrightarrow{H}_m e^{-j\vec{k}\bullet\vec{r}}$



uniform plane wave propagatiing along +Z direction



uniform plane wave propagatiing along arbitrary direction

The composition of solution that corresponding UP-wave EM field

For the uniform plane wave propagate along $\vec{e}_n$, the expression for electromagnetic field solutions:

The complex EM field vector: $\vec{E}(\vec{r}) = \vec{E}_m e^{-j\vec{k}\bullet\vec{r}}$ $\quad \vec{H} = \vec{H}_m e^{-j\vec{k}\bullet\vec{r}}$

where wave vector: $\vec{k} = k\vec{e}_n$

The complex amplitude vector

$$\vec{E}_m = E_{mx}e^{j\varphi_{ex}}\vec{e}_x + E_{my}e^{j\varphi_{ey}}\vec{e}_y + E_{mz}e^{j\varphi_{ez}}\vec{e}_z$$

$$\vec{H}_m = H_{mx}e^{j\varphi_{hx}}\vec{e}_x + H_{my}e^{j\varphi_{hy}}\vec{e}_y + H_{mz}e^{j\varphi_{hz}}\vec{e}_z$$

Transient solution of E:

$$\begin{aligned}\vec{E}(\vec{r},t) &= \mathrm{Re}[\vec{E}(\vec{r})e^{j\omega t}] \\ &= \vec{e}_x E_{mx}\cos(\omega t - \vec{k}\bullet\vec{r} + \varphi_{ex}) \\ &\quad + \vec{e}_y E_{my}\cos(\omega t - \vec{k}\bullet\vec{r} + \varphi_{ey}) \\ &\quad + \vec{e}_z E_{mz}\cos(\omega t - \vec{k}\bullet\vec{r} + \varphi_{ez})\end{aligned}$$

The relationship between $\vec{E}_m$ and $\vec{H}_m$?

The technique of analyzing uniform plane wave and the EM field expression using complex amplitude

Solution for EM field expressed using complex amplitude , when uniform plane wave propagating along $\vec{k}$

$$\vec{E}(\vec{r}) = \vec{E}_m e^{-j\vec{k}\bullet\vec{r}} \qquad \vec{H}(\vec{r}) = \vec{H}_m e^{-j\vec{k}\bullet\vec{r}}$$

Since:

$$\nabla e^{-j\vec{k}\bullet\vec{r}} = -j\vec{k}e^{-j\vec{k}\bullet\vec{r}} \Rightarrow \nabla = -j\vec{k}$$

So:

$$\begin{cases} \nabla\times\vec{H}(\vec{r}) = j\omega\vec{D}(\vec{r}) \\ \nabla\times\vec{E}(\vec{r}) = -j\omega\vec{B}(\vec{r}) \\ \nabla\cdot\vec{D}(\vec{r}) = 0 \\ \nabla\cdot\vec{B}(\vec{r}) = 0 \end{cases} \Rightarrow \begin{cases} -j\vec{k}\times\vec{H}(\vec{r}) = j\omega\vec{D}(\vec{r}) \\ -j\vec{k}\times\vec{E}(\vec{r}) = -j\omega\vec{B}(\vec{r}) \\ -j\vec{k}\cdot\vec{D}(\vec{r}) = 0 \\ -j\vec{k}\cdot\vec{B}(\vec{r}) = 0 \end{cases} \Rightarrow \begin{cases} \vec{k}\times\vec{H}_m = -\omega\vec{D}_m \\ \vec{k}\times\vec{E}_m = \omega\vec{B}_m \\ \vec{k}\cdot\vec{D}_m = 0 \\ \vec{k}\cdot\vec{B}_m = 0 \end{cases}$$

The relationship between E and H of uniform plane wave

$$\begin{cases} \vec{E}_m = -\eta \vec{e}_n \times \vec{H}_m \\ \vec{H}_m = \dfrac{1}{\eta} \vec{e}_n \times \vec{E}_m \end{cases} \xrightarrow{e^{-j\vec{k}\cdot\vec{r}}} \begin{cases} \vec{E}(\vec{r}) = -\eta \vec{e}_n \times \vec{H}(\vec{r}) \\ \vec{H}(\vec{r}) = \dfrac{1}{\eta} \vec{e}_n \times \vec{E}(\vec{r}) \end{cases}$$

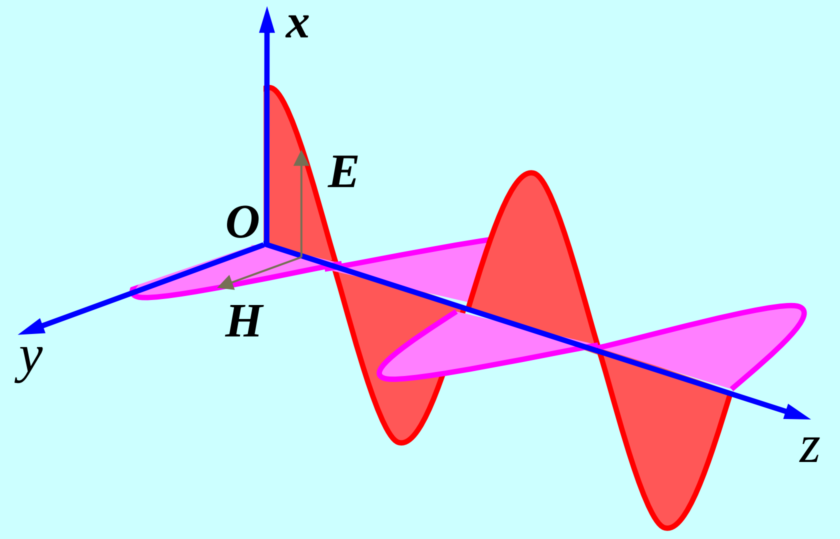
$$\xrightarrow{e^{j\omega t}} \begin{cases} \vec{E}(\vec{r},t) = -\eta \vec{e}_n \times \vec{H}(\vec{r},t) \\ \vec{H}(\vec{r},t) = \dfrac{1}{\eta} \vec{e}_n \times \vec{E}(\vec{r},t) \end{cases}$$

- $\vec{E}$、$\vec{H}$、$\vec{k}$ perpendicular to each other
- Electric and magnetic fields in phase
- Amplitude difference of η times

where $\eta = \sqrt{\dfrac{\mu}{\varepsilon}} \quad (\Omega)$

is the intrinsic impedance of the medium, also called wave impedance

in vacuum $\eta_0 = \sqrt{\dfrac{\mu_0}{\varepsilon_0}} \approx 377 \quad (\Omega)$



Uniform plane wave in ideal medium

Summary for Uniform Plane Wave

- **Solution for electromagnetic field complex vector :**

$$\vec{E}(\vec{r}) = \vec{E}_{\mathrm{m}} \mathrm{e}^{-\mathrm{j}\vec{k}\cdot\vec{r}} \qquad \vec{H}(\vec{r}) = \vec{H}_{\mathrm{m}} \mathrm{e}^{-\mathrm{j}\vec{k}\cdot\vec{r}}$$

- **The direction of $\vec{E}$、$\vec{H}$、$\vec{k}$ meet the Right-handed Corkscrew Rule**

- **Transverse electromagnetic wave（TEM wave）**

$$\vec{k}\cdot\vec{E} = 0, \quad \vec{k}\cdot\vec{H} = 0, \quad \vec{E}\cdot\vec{H} = 0$$

- **The direction of propagation along the space phase lag**

- **Electric and magnetic fields in phase, amplitude difference of η times**

1、Propagation parameters of uniform plane wave

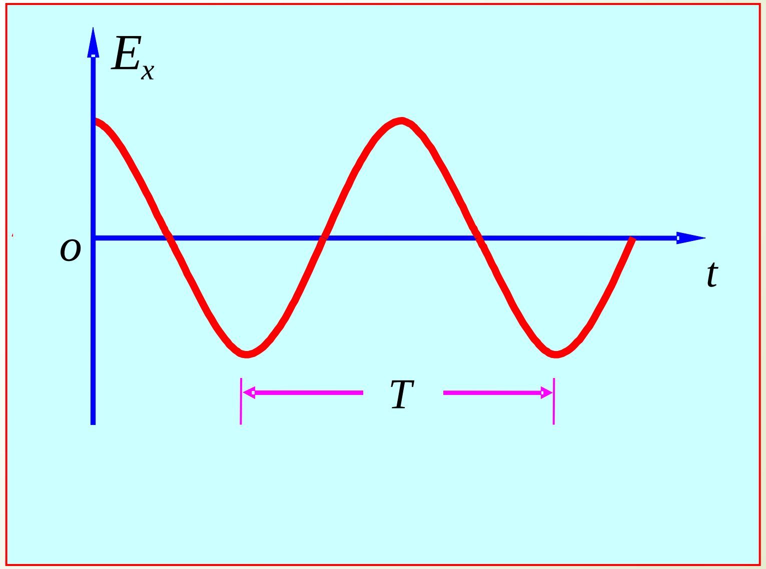
（1）Angular frequency, Frequency and Period

Angular frequency ω： refers to the change of the phase per unit time, unit: rad /s

Period T： At the same location, the time interval corresponding phase change 2π

$$\omega T = 2\pi \Rightarrow T = \frac{2\pi}{\omega} \ (\mathrm{s})$$

Frequency f： $$f = \frac{1}{T} = \frac{\omega}{2\pi} \ (\mathrm{Hz})$$



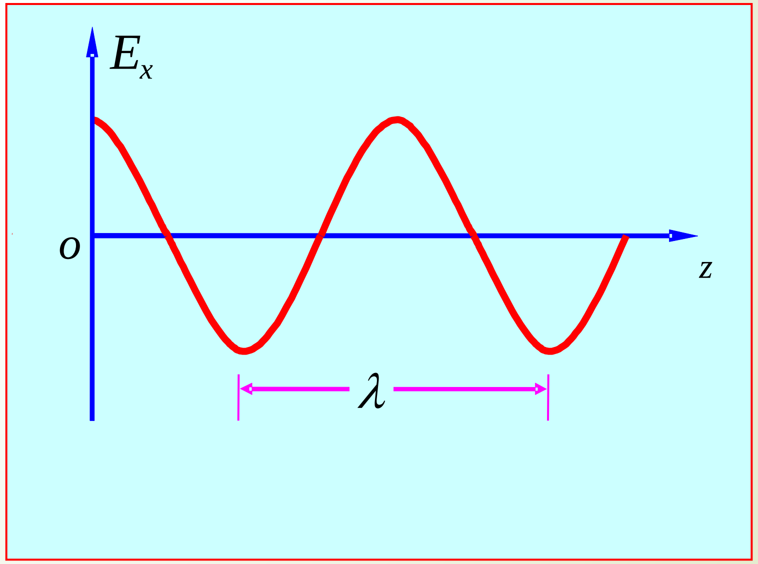
（2）Wave length and Phase constant

Wave length λ： At the same location, the space interval corresponding phase change 2πof equiphase surface

$$kl = 2\pi \quad \Rightarrow \quad \lambda = l = \frac{2\pi}{k} = \frac{1}{f\sqrt{\mu\varepsilon}} \quad (\text{m})$$

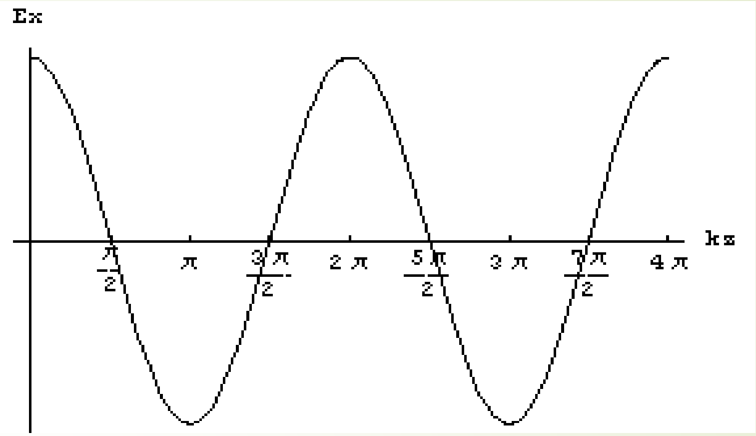
Phase constant k： refers to the change of the phase per unit distance.（wave number）

$$k = \frac{2\pi}{\lambda} \quad (\text{rad/m})$$



（3）Phase velocity

Phase velocity v：The velocity of equiphase surface moved in space.



As $\omega t - kz = C \Rightarrow \omega \mathrm{d}t - k\mathrm{d}z = 0$

Get the phase velocity of uniform plane wave is

$$v = \frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\omega}{k} = \frac{\omega}{\omega\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu\varepsilon}} \quad (\mathrm{m/s})$$

independent with the frequency of EM waves

In vacuum： $v = c = \dfrac{1}{\sqrt{\mu_0\varepsilon_0}} = \dfrac{1}{\sqrt{4\pi\times10^{-7}\times\dfrac{1}{36\pi}\times10^{-9}}} = 3\times10^{8}(\mathrm{m/s})$

Summary for Uniform Plane Wave in Ideal Medium

- **Solution for electromagnetic field complex vector:**

$$\vec{E}(\vec{r}) = \vec{E}_{\mathrm{m}} \mathrm{e}^{-\mathrm{j}\vec{k}\bullet\vec{r}} \qquad \vec{H}(\vec{r}) = \vec{H}_{\mathrm{m}} \mathrm{e}^{-\mathrm{j}\vec{k}\bullet\vec{r}}$$

- **The direction of** $\vec{E}$、$\vec{H}$、$\vec{k}$ **meet the Right-handed Rule;**
- **Transverse electromagnetic wave （TEM wave）**

$$\vec{k}\bullet\vec{E} = 0, \quad \vec{k}\bullet\vec{H} = 0, \quad \vec{E}\bullet\vec{H} = 0$$

- **The direction of propagation along the space phase lagging**
- **E and H in phase, amplitude difference of** η **times;**
- **Related physical quantities : frequency、period、wave length、phase constant 、 wave number、phase velocity、power velocity**

Ex. The complex expression magnetic field intensity of plane wave in air is

$$\vec{H} = (-\vec{e}_x A + \vec{e}_y 2 + \vec{e}_z 4)\mathrm{e}^{-\mathrm{j}\pi(4x+3z)}$$

Where A is constant . Get the :1)$\vec{k}$; 2）wavelength and frequency; 3）the value of A; 4）The complex expression electric field;

Solution： (1) Due to $\vec{H} = \vec{H}_{\mathrm{m}}\mathrm{e}^{-\mathrm{j}\vec{k}\cdot\vec{r}}$ **, So**

$$\vec{H}_{\mathrm{m}} = -\vec{e}_x + \vec{e}_y 2 + \vec{e}_z 4 \ , \quad \vec{k}\cdot\vec{r} = k_x x + k_y y + k_z z = 4\pi x + 3\pi z$$

So $k_x = 4\pi$、$k_y = 0$、$k_z = 3\pi$， $\vec{k} = \vec{e}_x 4\pi + \vec{e}_z 3\pi$

$$k = \sqrt{(3\pi)^2 + (4\pi)^2} = 5\pi$$

(2) $\lambda = \dfrac{2\pi}{k} = \dfrac{2\pi}{5\pi} = \dfrac{2}{5}\ \text{m}$ ， $f = \dfrac{c}{\lambda} = \dfrac{3\times10^{8}}{2/5} = 7.5\times10^{8}\ \text{Hz}$

(3) $\vec{k}\cdot\vec{H}_{\mathrm{m}} = 4\pi(-A) + 0\times2 + 3\pi\times4 = 0 \Rightarrow A = 3$

(4) $\vec{E}(\vec{r}) = \eta_0\vec{H}(\vec{r})\times\vec{e}_{\mathrm{n}}$

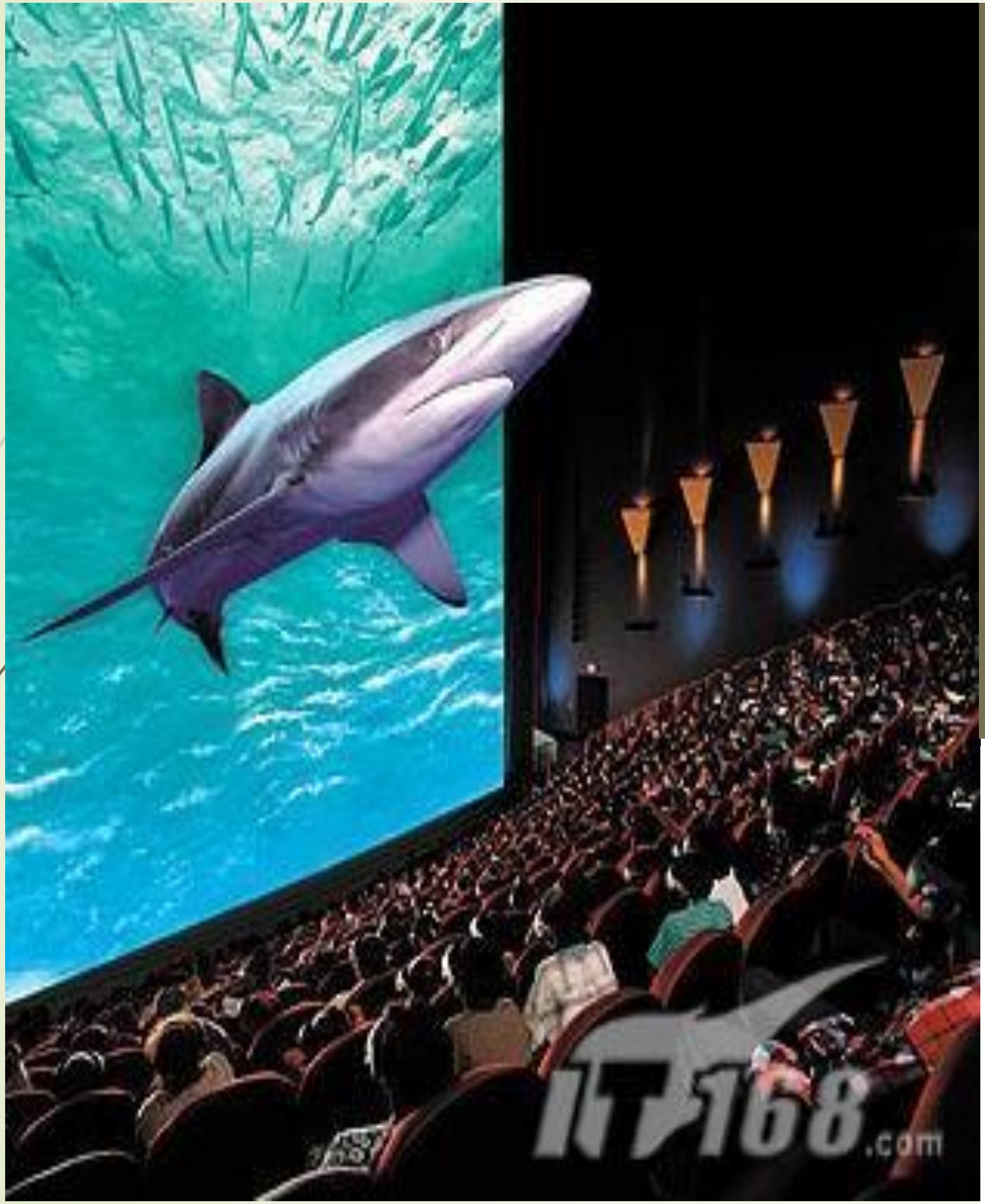
$$= 120\pi(-\vec{e}_x 3 + \vec{e}_y 2 + \vec{e}_z 4)\mathrm{e}^{-\mathrm{j}\pi(4x+3z)} \times (\vec{e}_x\frac{4}{5} + \vec{e}_z\frac{3}{5})$$

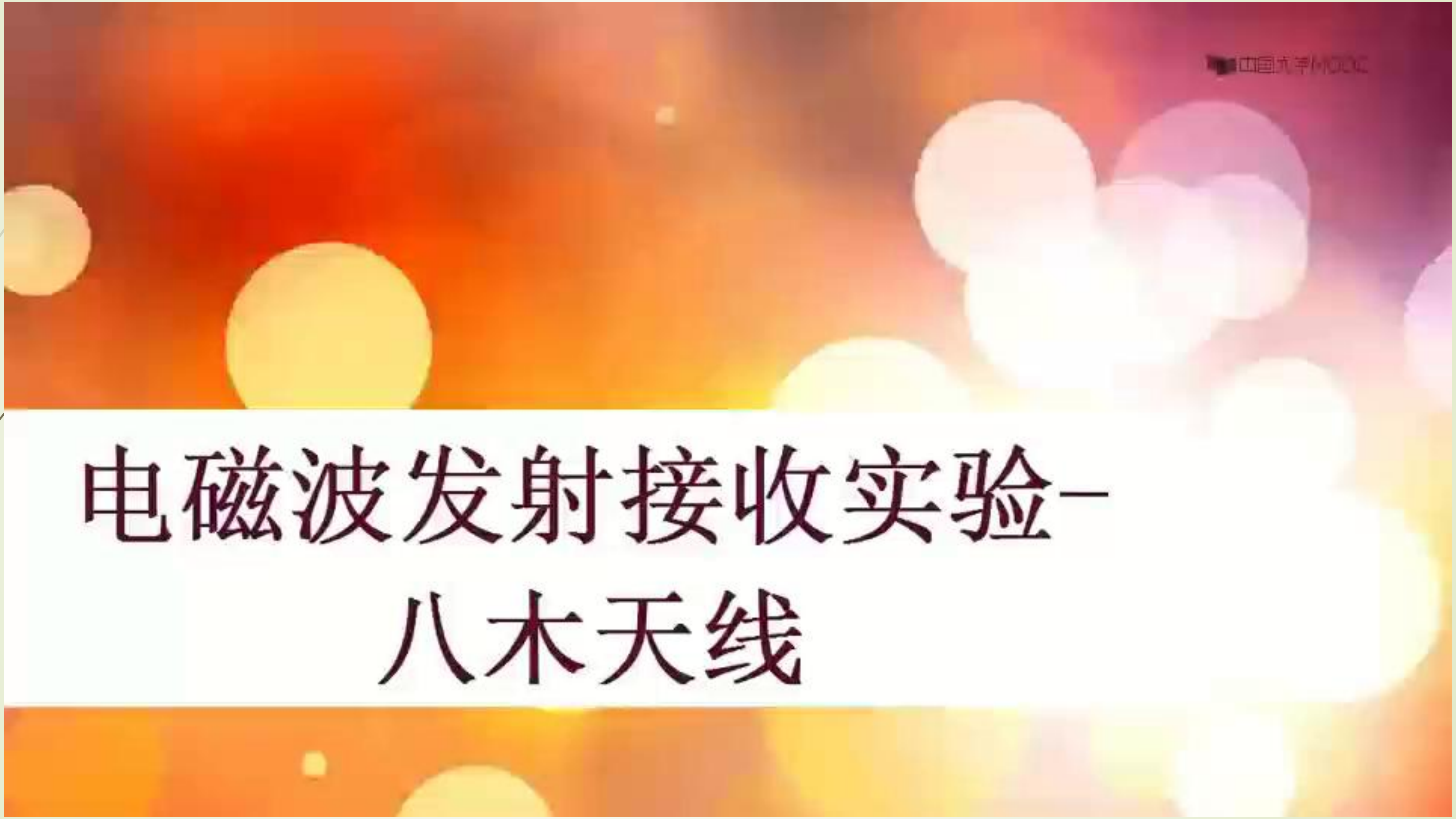
$$= 120\pi(\vec{e}_x 1.2 + \vec{e}_y 5 - \vec{e}_z 1.6)\mathrm{e}^{-\mathrm{j}\pi(4x+3z)}$$

8.1.2 polarization of electromagnetic wave (transient expression)

1. The concept of the polarization
2. Linear polarized wave
3. Circular polarized wave
4. Elliptical polarized wave
5. Polarized wave decomposition
6. Application of Polarization

3D film

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The basic problem :

For time varying electric field $\vec{E}(\vec{r},t)$,at the fixed point ,to study the law of Electric field change with time, such as $\vec{E}(\vec{r}=\vec{r}_0,t)$

Key points

1: for harmonic field, due to the independent variable of the temporal variation can be separated , i.e. The law of the change of time and space are independent of each other. So ,when studying the law of time variation, the value of $\vec{r}_0$, can take any value, such as : $\vec{r}_0 = 0$

2: $\vec{E}(t)$ changing over time behave as its magnitude and direction is changing over time

conclusion :

Studying the law of time variation of harmonic field can be in any spatial location.

The study is focus on the change of the trajectory of the vector of the field over time.

1. The concept of the polarization

- **The polarization of wave**

At fixed point in the space, the trajectory of the E-field vector varies with time

- **Analysis method**

1. **Vector variation, behave as the magnitude variation of the vector components;**
2. **Study of vector component changes over time, starting from the transient expression of field vector. for plane wave:**

$$\vec{E}(\vec{r},t) = \mathrm{Re}[\vec{E}_m \mathrm{e}^{-\mathrm{j}\vec{k}\bullet\vec{r}} \mathrm{e}^{\mathrm{j}\omega t}] = \vec{\mathrm{e}}_x E_{xm}\cos(\omega t - \vec{k}\bullet\vec{r} + \varphi_{ex})$$
$$+ \vec{\mathrm{e}}_y E_{ym}\cos(\omega t - \vec{k}\bullet\vec{r} + \varphi_{ey})$$
$$+ \vec{\mathrm{e}}_z E_{zm}\cos(\omega t - \vec{k}\bullet\vec{r} + \varphi_{ez})$$

conclusion:

1) **The rule of vector end changes over time is depend on the amplitude and initial phase of each component.**

2) **Any polarization can be synthesized by linear polarization!**

The vector end equation

Without loss of generality, consider plane waves that propagate in the +z direction, the expression is : $\vec{\mathbf{E}}(t) = \vec{\mathbf{e}}_x E_x(t) + \vec{\mathbf{e}}_y E_y(t)$

The vector end parametric equations

- **In rectangular coordinate system:**

$$\begin{cases} E_x(t) = E_{xm}\cos(\omega t - kz + \phi_x) \\ E_y(t) = E_{ym}\cos(\omega t - kz + \phi_y) \end{cases}$$

generally, nonlinear polarization

- **In polar coordinates:**

$$\begin{cases} \left|\vec{E}(0,t)\right| = \sqrt{E_{\mathrm{xm}}^2\cos^2(\omega t + \phi_x) + E_{\mathrm{ym}}^2\cos^2(\omega t + \phi_y)} \\ \alpha(t) = \arctan\left[\dfrac{E_{ym}}{E_{xm}}\dfrac{\cos(\omega t - kz + \phi_y)}{\cos(\omega t - kz + \phi_x)}\right] \end{cases}$$

Linear polarized wave

- When: $\varphi_x = \varphi_y$

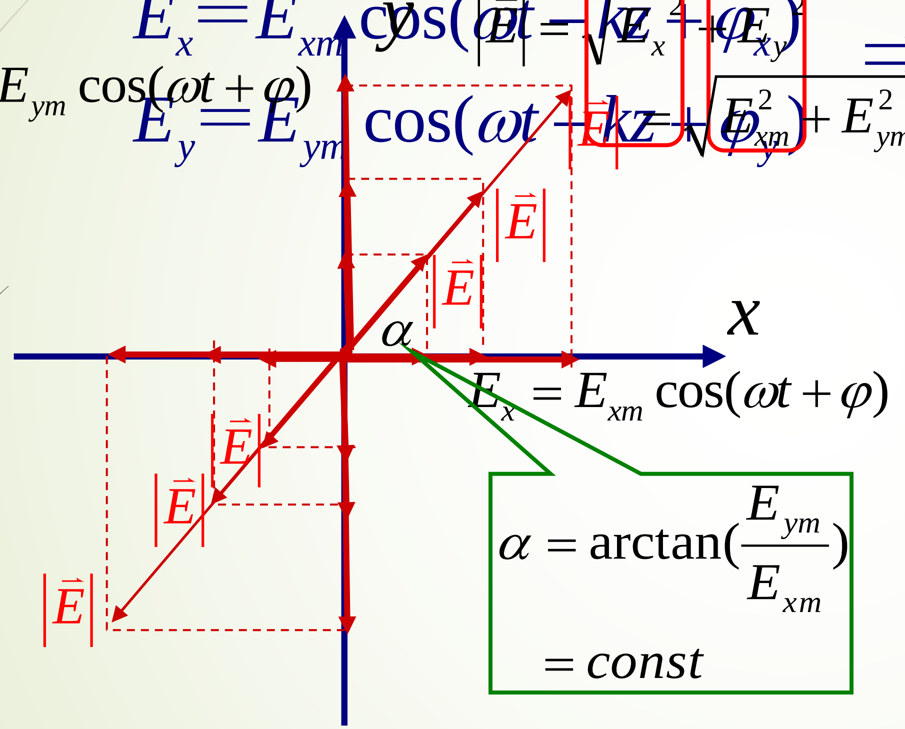
$z = 0 \quad \varphi_x = \varphi_y = \varphi$

$$E_x = E_{xm}\cos(\omega t - kz + \varphi_x)$$

$$E_y = E_{ym}\cos(\omega t - kz + \varphi_y)$$

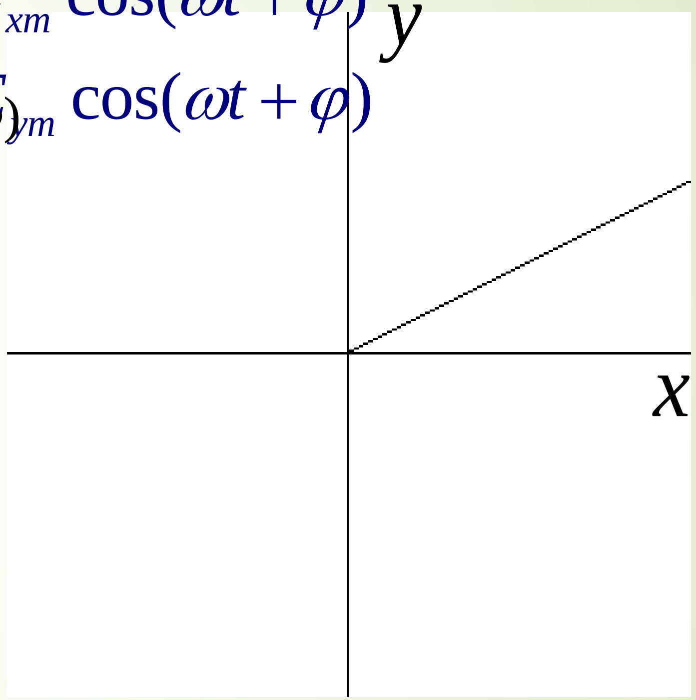
$$|\vec{E}| = \sqrt{E_x^2 + E_y^2} = \sqrt{E_{xm}^2 + E_{ym}^2}\cos(\omega t + \varphi)$$

$$\Rightarrow \begin{cases} E_x = E_{xm}\cos(\omega t + \varphi) \\ E_y = E_{ym}\cos(\omega t + \varphi) \end{cases}$$



$\varphi_x = \varphi_y$

$$\alpha = \arctan\left(\frac{E_{ym}}{E_{xm}}\right) = const$$



$\varphi_x = \varphi_y$

Trajactary of the whole electric field is a line $\longrightarrow$ **Linear polarized wave**

When two linear polarized wave are in phase, it is a linear polarized wave as a whole.

Linear polarized wave

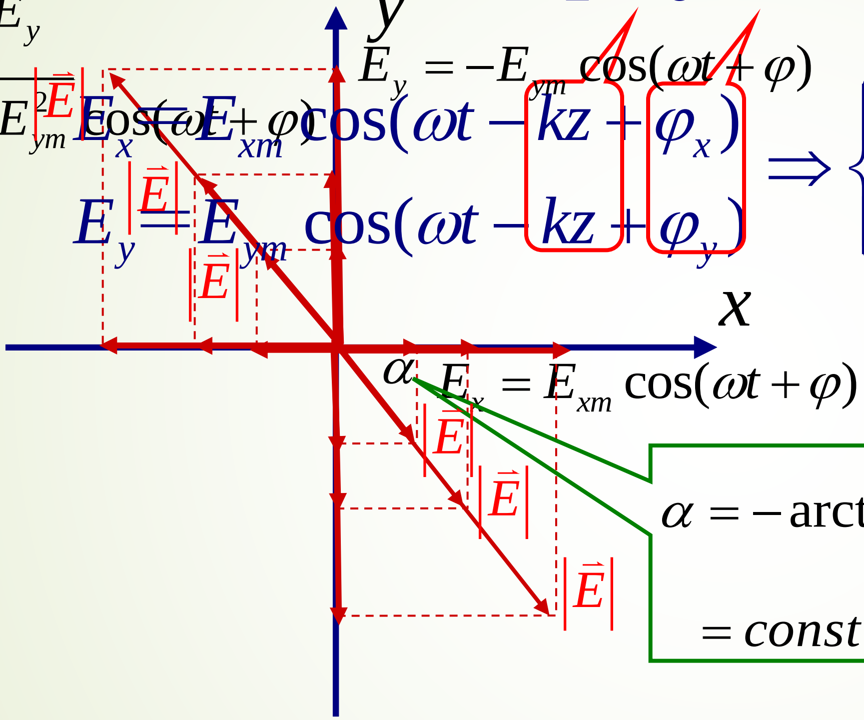
- When: $\varphi_x - \varphi_y = \pm\pi$

$$|\vec{E}| = \sqrt{E_x^{\ 2} + E_y^{\ 2}} = \sqrt{E_{xm}^2 + E_{ym}^2}\,\cos(\omega t + \varphi)$$

$z = 0$

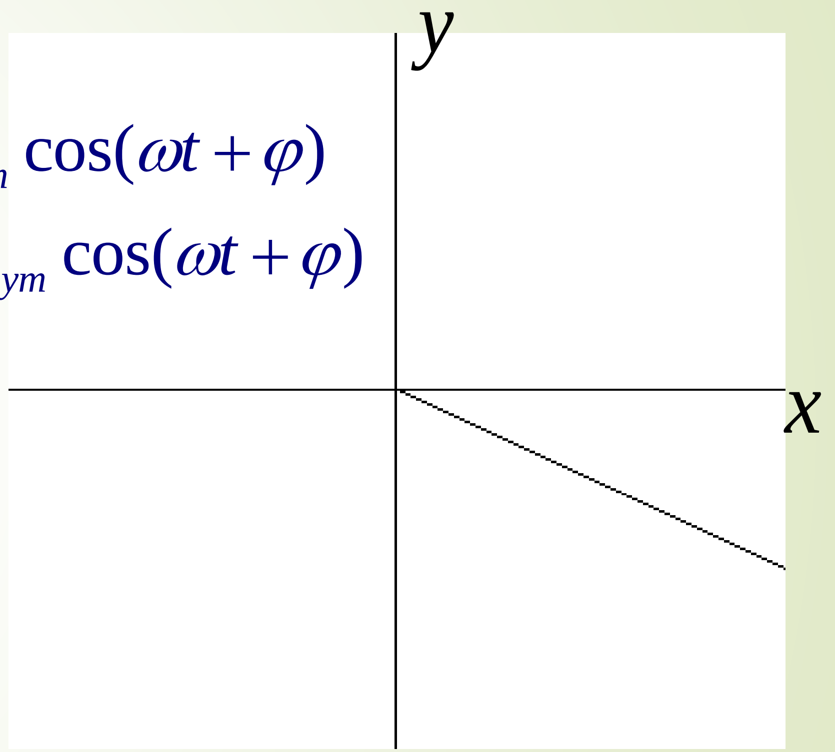
$$E_x = E_{xm}\cos(\omega t - \boxed{kz} + \boxed{\varphi_x})$$
$$E_y = E_{ym}\cos(\omega t - \boxed{kz} + \boxed{\varphi_y})$$

$$\Rightarrow \begin{cases} E_x = E_{xm}\cos(\omega t + \varphi) \\ E_y = -E_{ym}\cos(\omega t + \varphi) \end{cases}$$



$$\alpha = -\arctan\left(\frac{E_{ym}}{E_{xm}}\right) = const$$

$\varphi_x - \varphi_y = \pi$



Trajactary of the whole electric field is a line $\longrightarrow$ **Linear polarized wave**

When two linear polarized wave are in phase, it is a linear polarized wave as a whole。

Conculsion: linear polarized wave

- **condition:** $\Delta\phi = 0$ **or** $\pm\pi$ **time-varying**

then the vector parameter equation was simplified:

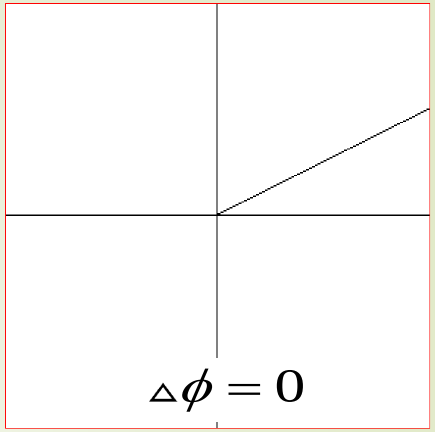
$$\left|\vec{E}\right| = \sqrt{E_x^2(0,t) + E_y^2(0,t)} = \sqrt{E_{xm}^2 + E_{ym}^2}\cos(\omega t + \phi_y)$$

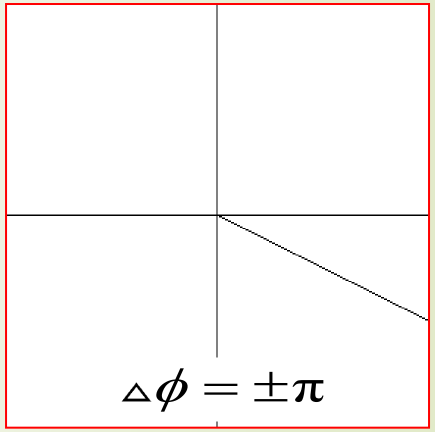
$\Delta\phi = 0$

$$\alpha = \pm\arctan\left(\frac{E_{ym}}{E_{xm}}\right)$$

constant

$\Delta\phi = \pm\pi$





- **Any two Linear polarized waves with the same frequency and direction of propagation, if their polarization direction perpendicular to each other, can be synthesized to a linear polarized wave, when their phase difference is zero or ± π.**

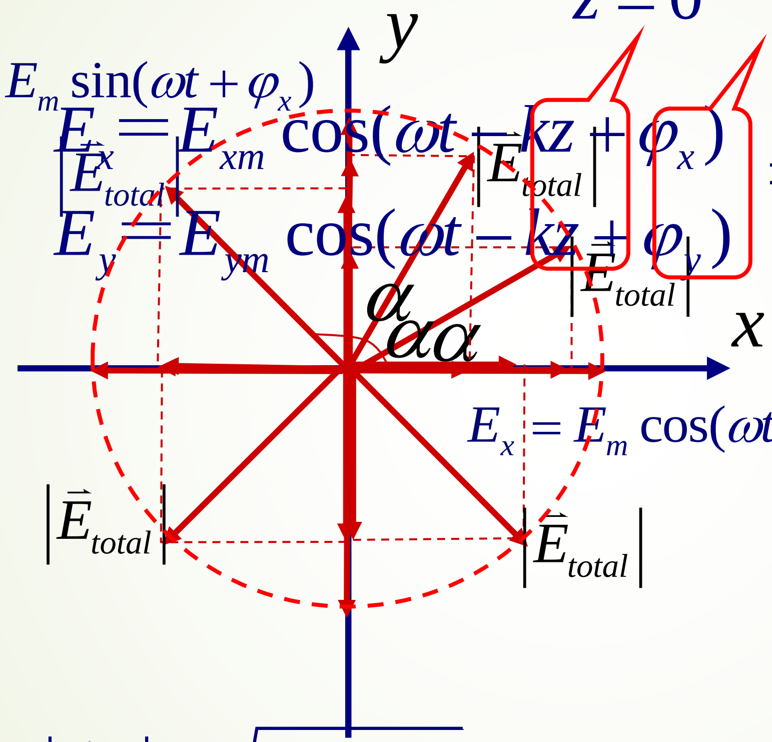
Circular polarized wave

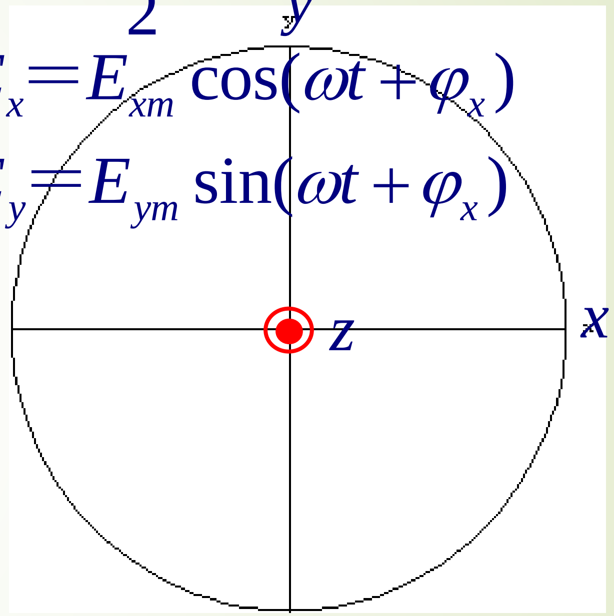
- When $\varphi_y - \varphi_x = -\frac{\pi}{2}$ and $E_{xm} = E_{ym} = E_m$

$z = 0$ $\quad \varphi_y = \varphi_x - \frac{\pi}{2}$

$$E_x = E_{xm}\cos(\omega t - kz + \varphi_x)$$
$$E_y = E_{ym}\cos(\omega t - kz + \varphi_y)$$

$$\Rightarrow \begin{cases} E_x = E_{xm}\cos(\omega t + \varphi_x) \\ E_y = E_{ym}\sin(\omega t + \varphi_x) \end{cases}$$





$$|\vec{E}_{合}| = \sqrt{E_x^2 + E_y^2} = E_m = const$$

$$\alpha = \arctan\left(\frac{E_y}{E_x}\right) = \omega t + \varphi_x$$

Trajactary of the whole electric field is a circle , its rotating direction and transmitting direction obey Right Hand rule ——RHCP

Circular polarized wave

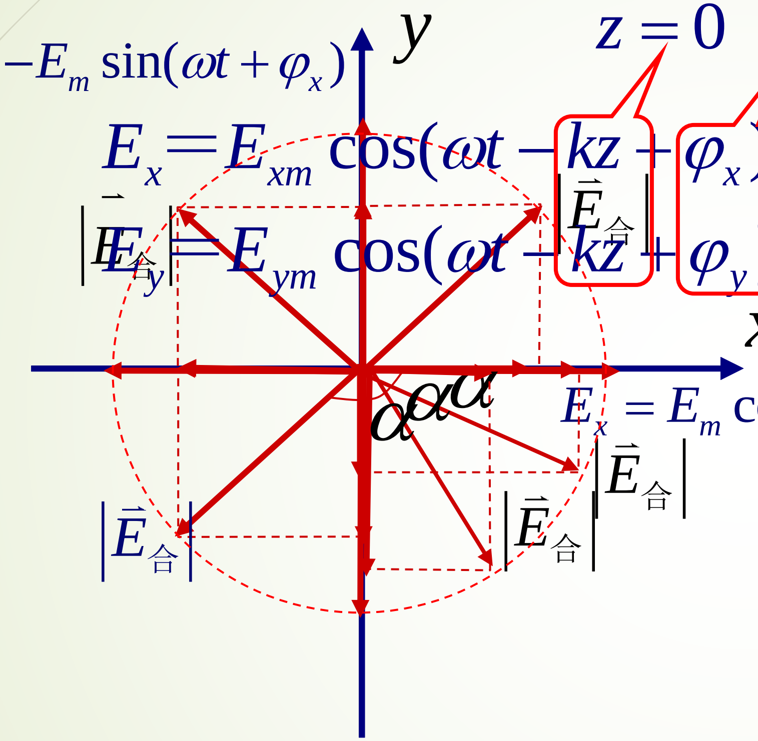
When $\varphi_y - \varphi_x = \dfrac{\pi}{2}$ and $E_{xm} = E_{ym} = E_m$

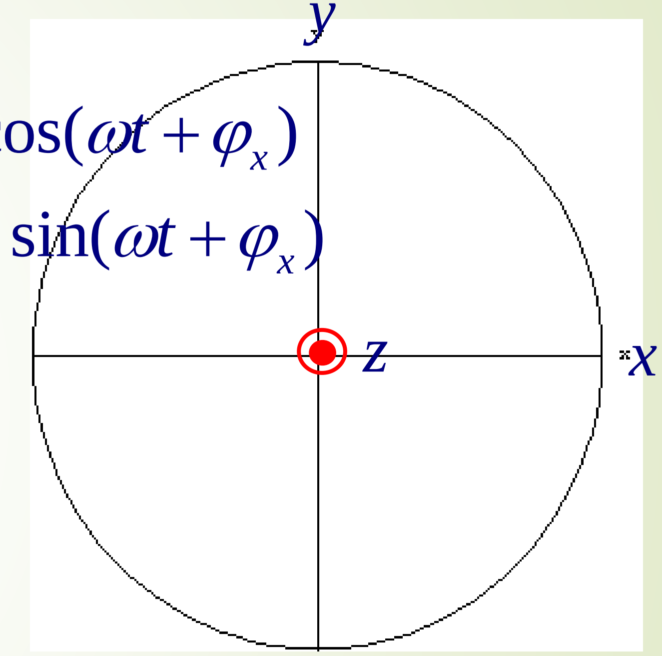
$z = 0 \quad \varphi_y = \varphi_x + \dfrac{\pi}{2}$

$$\begin{aligned} E_x &= E_{xm}\cos(\omega t - \boxed{kz} + \boxed{\varphi_x}) \\ E_y &= E_{ym}\cos(\omega t - \boxed{kz} + \boxed{\varphi_y}) \end{aligned} \Rightarrow \begin{cases} E_x = E_{xm}\cos(\omega t + \varphi_x) \\ E_y = -E_{ym}\sin(\omega t + \varphi_x) \end{cases}$$

$E_y = -E_m\sin(\omega t + \varphi_x)$

$E_x = E_m\cos(\omega t + \varphi_x)$





$$\left|\vec{E}_{合}\right| = \sqrt{E_x^2 + E_y^2} = E_m^2 = const \qquad \alpha = \arctan\left(\frac{E_y}{E_x}\right) = -(\omega t + \varphi_x)$$

Trajactary of the whole electric field is a circle, its rotating direction and transmitting direction obey left Hand rule ——LHCP

Conclusion: Circular polarized wave

- **condition** : $E_{xm} = E_{ym} = E_{\mathrm{m}}$、$\triangle\phi = \pm\pi/2$

constant

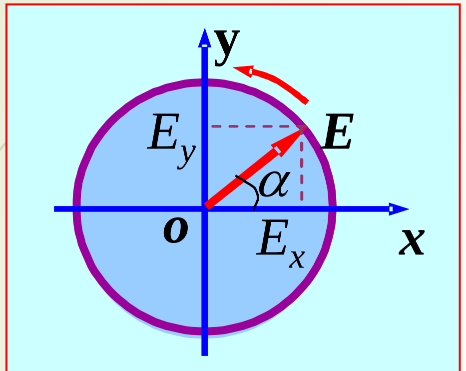
- **The vector end equation**

$$E_x^2 + E_y^2 = E_m^2$$

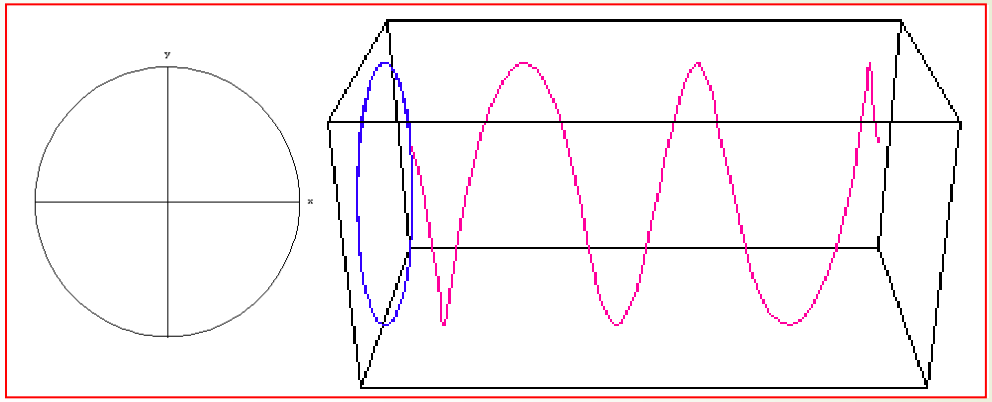
$\triangle\phi = \dfrac{\pi}{2}$	$\Delta\phi = -\dfrac{\pi}{2}$
Left-hand circular polarization wave(LHCP)	**Right-hand circular polarization wave(RHCP)**

- **conclusion: Any two Linear polarized wave that the same frequency and direction of propagation, what's more their polarization direction perpendicular to each other, can be synthesise to circular polarized wave, when their amplitude is the same and their differ phase is ±π/ 2 .**

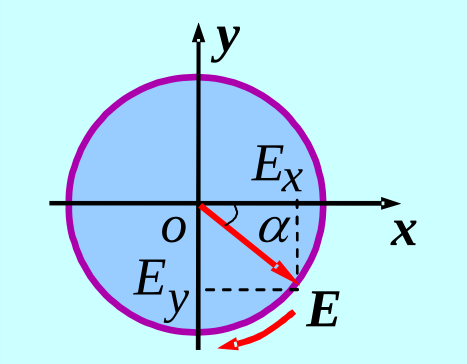
right-hand circular polarization wave： $\Delta\phi = -\pi/2$



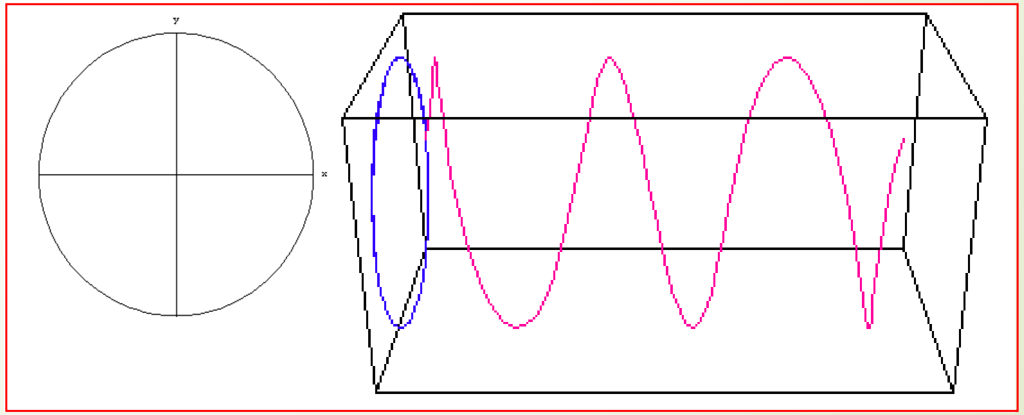
right-hand circular polarization wave



left-hand circular polarization wave： $\Delta\phi = \pi/2$



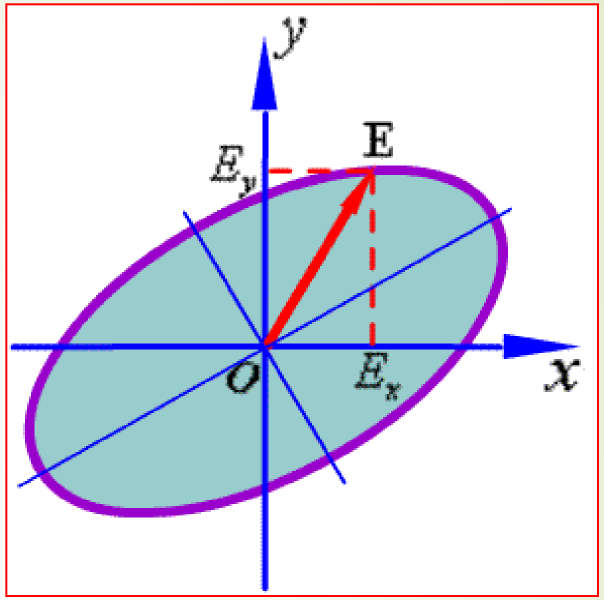
left-hand circular polarization wave

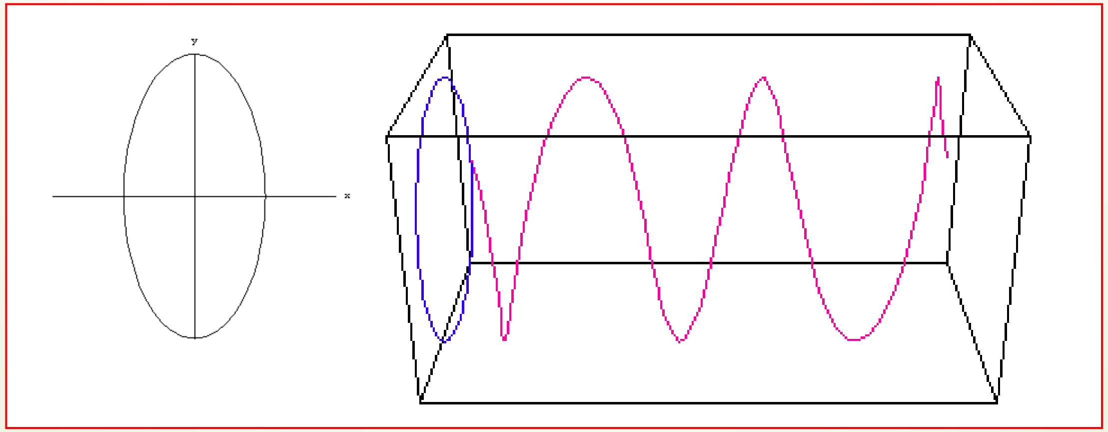


4. elliptical polarized wave

Generally

$$\frac{E_x^2}{E_{xm}^2}+\frac{E_y^2}{E_{ym}^2}-\frac{2E_xE_y}{E_{xm}E_{ym}}\cos\Delta\phi=\sin^2\Delta\phi$$





The mode of the polarization

$$\frac{\mathrm{d}}{\mathrm{dt}}\alpha = \frac{E_{ym}}{E_{xm}\cos^2(\omega t - kz + \phi_x)}(-\omega \sin\triangle\phi)$$

$$\sim -\omega \sin\triangle\phi \,, \qquad \triangle\phi = \phi_y - \phi_x$$

$\triangle\phi = 0 \ or \ \pm\pi:$	$0 < \triangle\phi < \pi:$	$-\pi < \triangle\phi < 0:$
Linear polarization	**left-hand polarization (+Z)** **right-hand polarization (-Z)**	**right-hand polarization (+Z)** **left-hand polarization (-Z)**

conclusion: the mode of polarization for EM waves depends on the difference between initial phases of each component

Summary of composite polarized wave

- The polarized mode of electromagnetic wave is depend on the phase difference($\Delta\varphi = \varphi_y - \varphi_x$) and the amplitudes （$E_{xm}$、$E_{ym}$） of E_x and E_y .
- modes of polarization
  - linear polarization: $\Delta\varphi = 0$、 $\pm\pi$

    $\Delta\varphi = 0$, In quadrant 1, 3; $\Delta\varphi = \pm\pi$, In quadrant 2,4
  - circular polarization: $\Delta\varphi = \pm\pi/2$, $E_{xm} = E_{ym}$

    for the wave in the +z direction :“ + ” correspond to LHCP; “ − ” correspond to RHCP.
  - elliptical polarization: generally

    for the wave in the +z direction : $0 < \Delta\varphi < \pi$, left-hand; $-\pi < \Delta\varphi < 0$, right-hand.

Ex. Illustrate the following uniform plane wave polarization mode.

(1) $\vec{E}=\vec{e}_x E_{\mathrm{m}} \mathrm{e}^{-\mathrm{j}kz}-\vec{e}_y \mathrm{j} E_{\mathrm{m}} \mathrm{e}^{-\mathrm{j}kz}$

(2) $\vec{E}=\vec{e}_x E_{\mathrm{m}} \sin(\omega t-kz+\frac{\pi}{4})+\vec{e}_y E_{\mathrm{m}} \cos(\omega t-kz-\frac{\pi}{4})$

(3) $\vec{E}=\vec{e}_x E_{\mathrm{m}} \sin(\omega t+kz)+\vec{e}_y 2E_{\mathrm{m}} \cos(\omega t+kz)$

Answer:

(1) $E_{x\mathrm{m}}=E_{y\mathrm{m}}$, $\phi_x=0$、$\phi_y=-\frac{\pi}{2}$, $\Delta\phi=-\frac{\pi}{2}$ ⟹ **RHCP**

(2) $\phi_x=-\frac{\pi}{4}$、$\phi_y=-\frac{\pi}{4}$, $\Delta\phi=0$ ⟹ **linear polarized wave**

(3) $E_{x\mathrm{m}}\neq E_{y\mathrm{m}}$, $\phi_x=-\frac{\pi}{2}$、$\phi_y=0$, $\Delta\phi=\frac{\pi}{2}$

⟹ **RH elliptical polarized wave**

Question:

Think about the steps to differentiate the method of polarization of waves.

5. Polarized wave decomposition

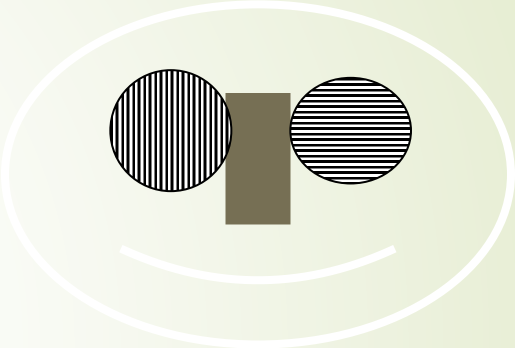
- Any kind of polarization of uniform plane wave can be decomposed into the sum of two linear polarized wave .
- Any kind of polarization of uniform plane wave can be decomposed into the sum of right- and left-hand circularly polarized wave, and their amplitude is different.

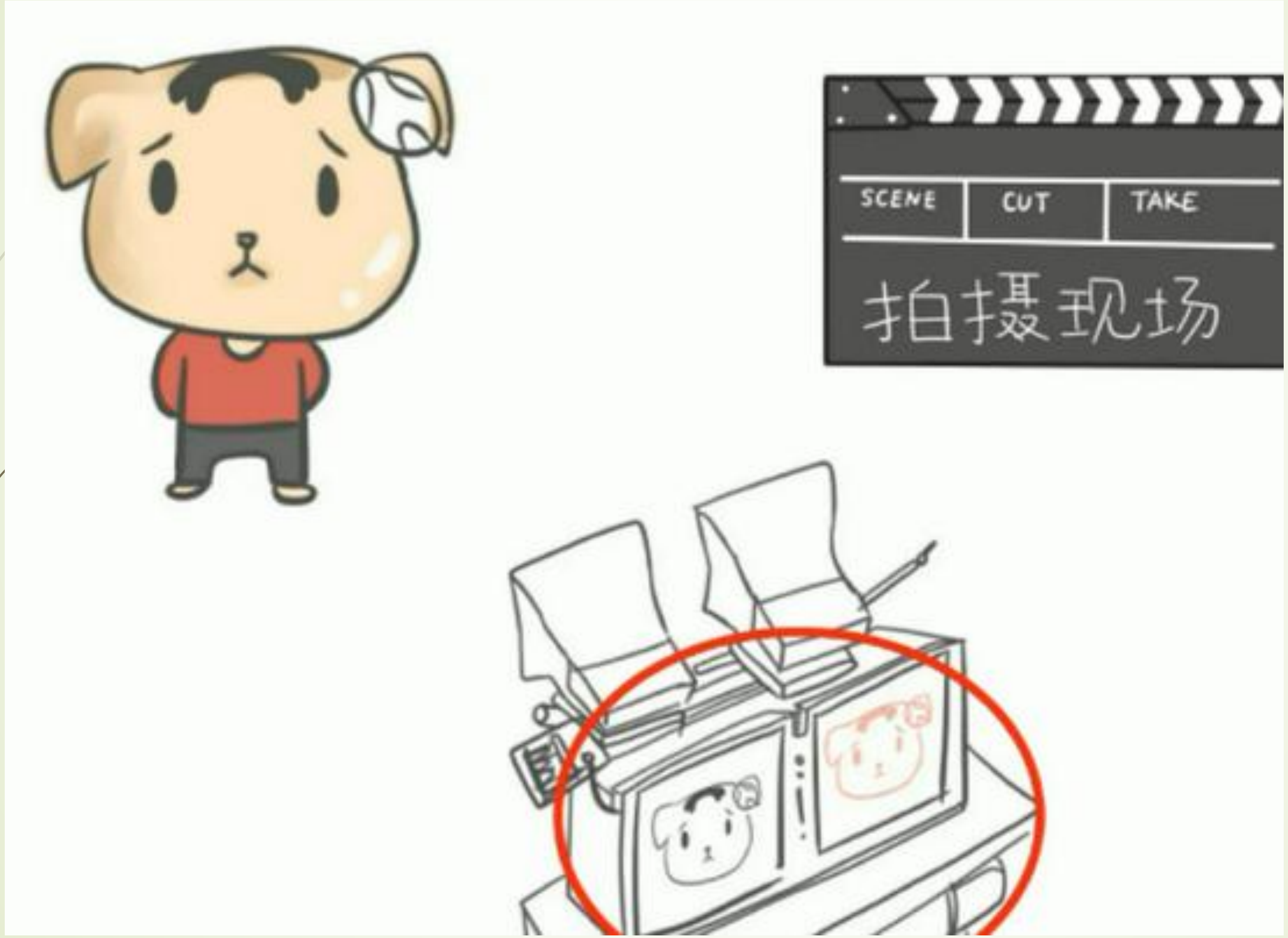
Prove:

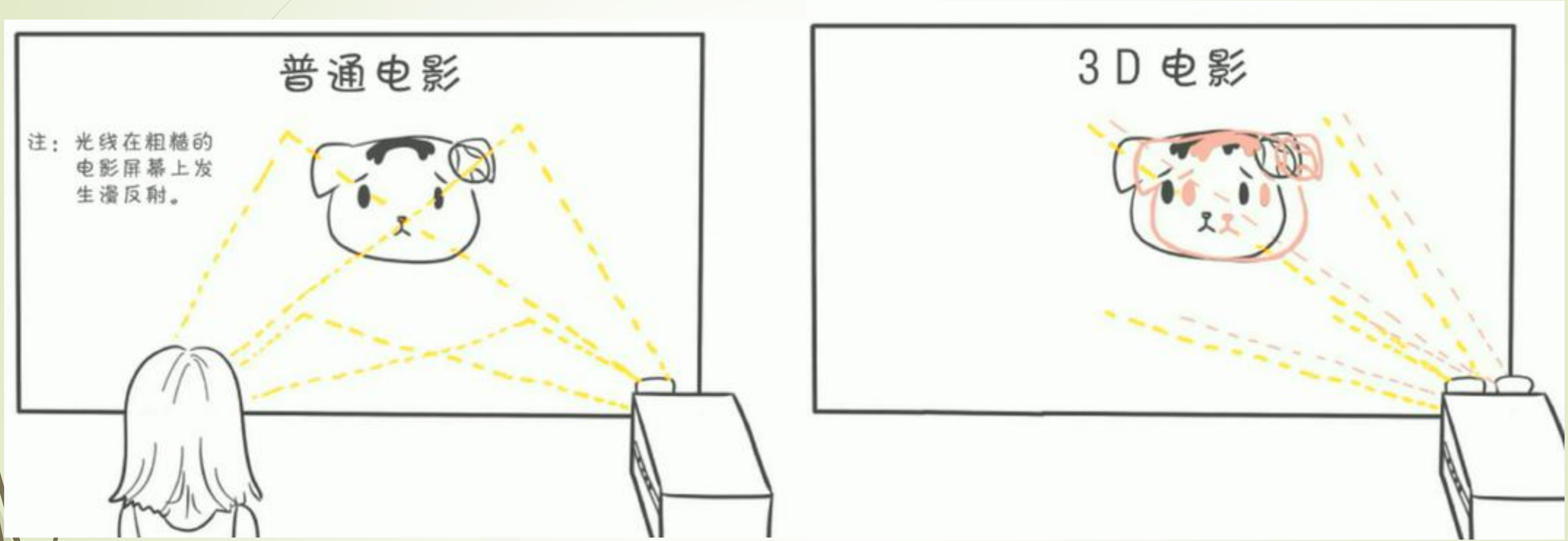
Any linear polarized wave can be decomposed into the sum of right- and left-hand circularly polarized wave, with equal amplitude.

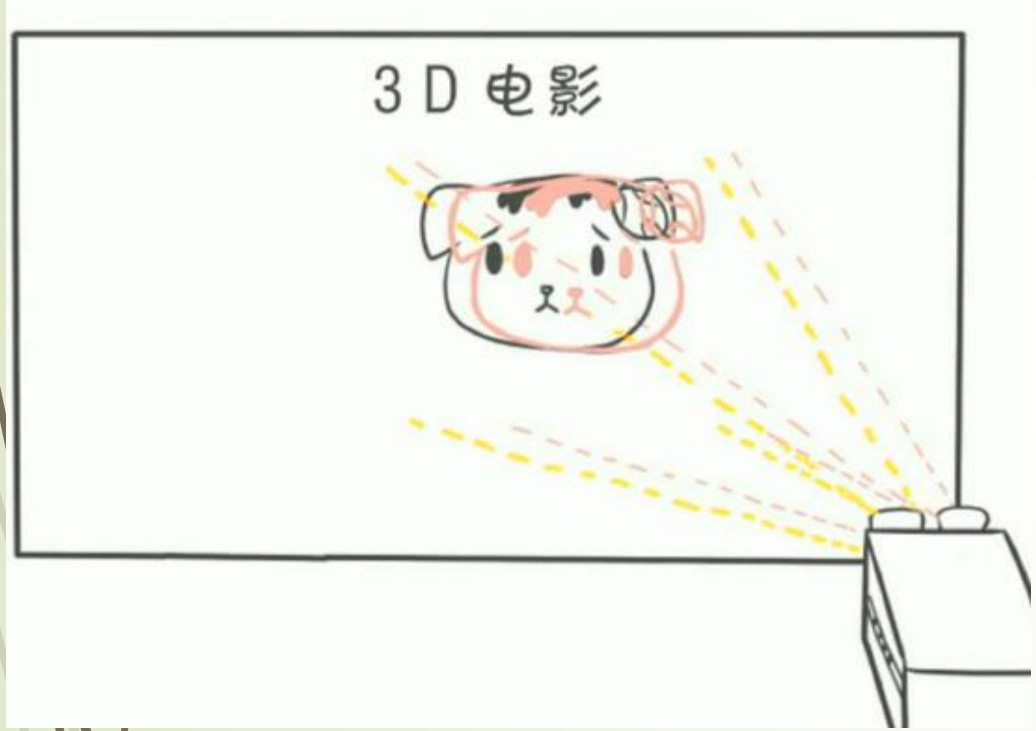
$$\vec{E} = \vec{e}_x E_{\mathrm{m}} \mathrm{e}^{-\mathrm{j}kz} = (\vec{e}_x + \mathrm{j}\vec{e}_y)\frac{E_{\mathrm{m}}}{2}\mathrm{e}^{-\mathrm{j}kz} + (\vec{e}_x - \mathrm{j}\vec{e}_y)\frac{E_{\mathrm{m}}}{2}\mathrm{e}^{-\mathrm{j}kz}$$

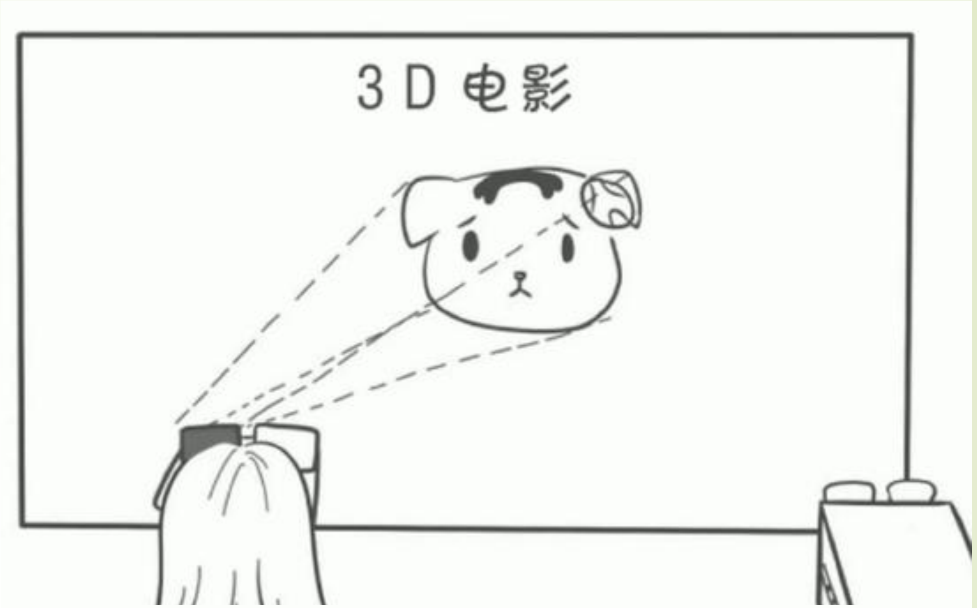
Polarizer

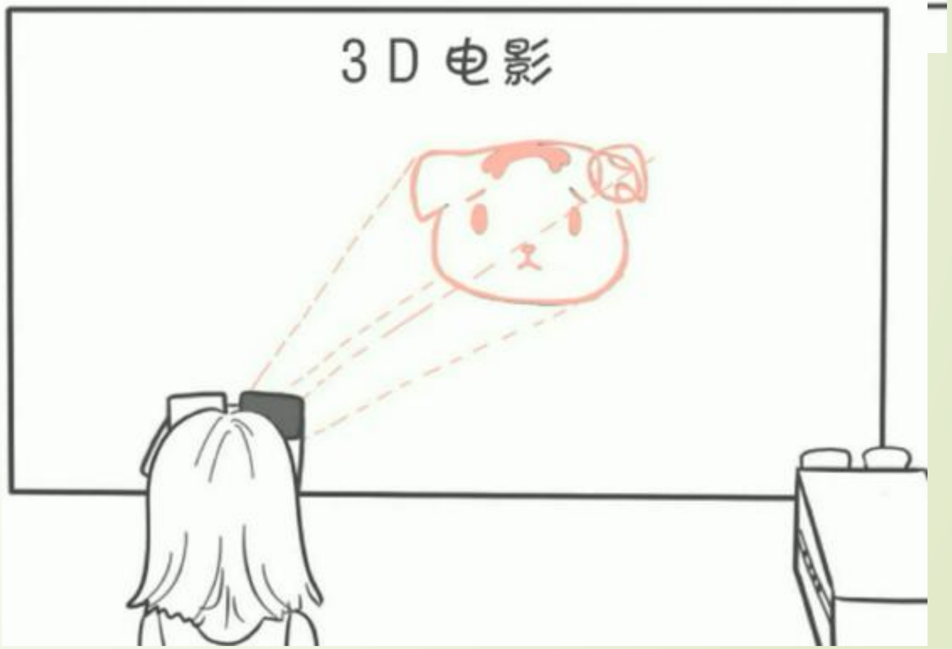



SCENE
CUT
TAKE
拍摄现场

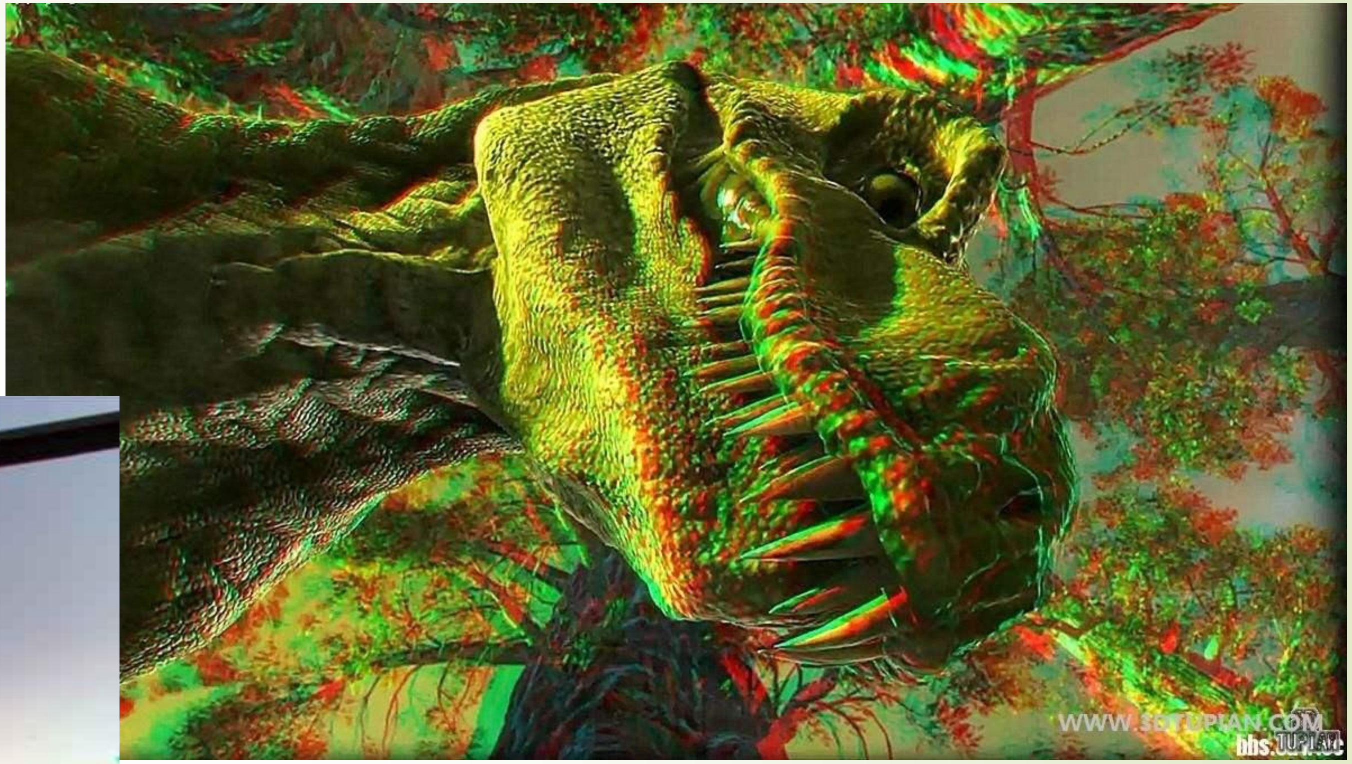

普通电影
注：光线在粗糙的
电影屏幕上发
生漫反射。
3D电影


3D电影


3D电影


3D电影

Anaglyphic Glasses



售后服务：全国联保 店铺三包 其他/...　屏幕比例：16:9

颜色分类：70寸高清电影版【3D】7...　同城服务：同城卖家送货上门

电视类型：OLED电视　刷屏率：60Hz

操作系统：阿里YunOS　亮度：700cd/m^2

视频显示格式：2160p　3D类型：色差式3D

上市时间：2021-01　屏幕尺寸：75英寸

采购地：中国大陆　保修期：12个月

语音遥控类型：遥控器语音遥控　存储容量：2GB+8GB

电视类型：OLED电视 LED电视 全面屏　刷屏率：120Hz

操作系统：阿里YunOS　亮度：450cd/m^2

视频显示格式：2160p　3D类型：偏光式3D

堆码层数极限：3层　毛重：15kg

智能类型：其他智能　上市时间：2020-05

净重(不含底座)：45kg　含边框整屏尺寸：1470x850x15mm

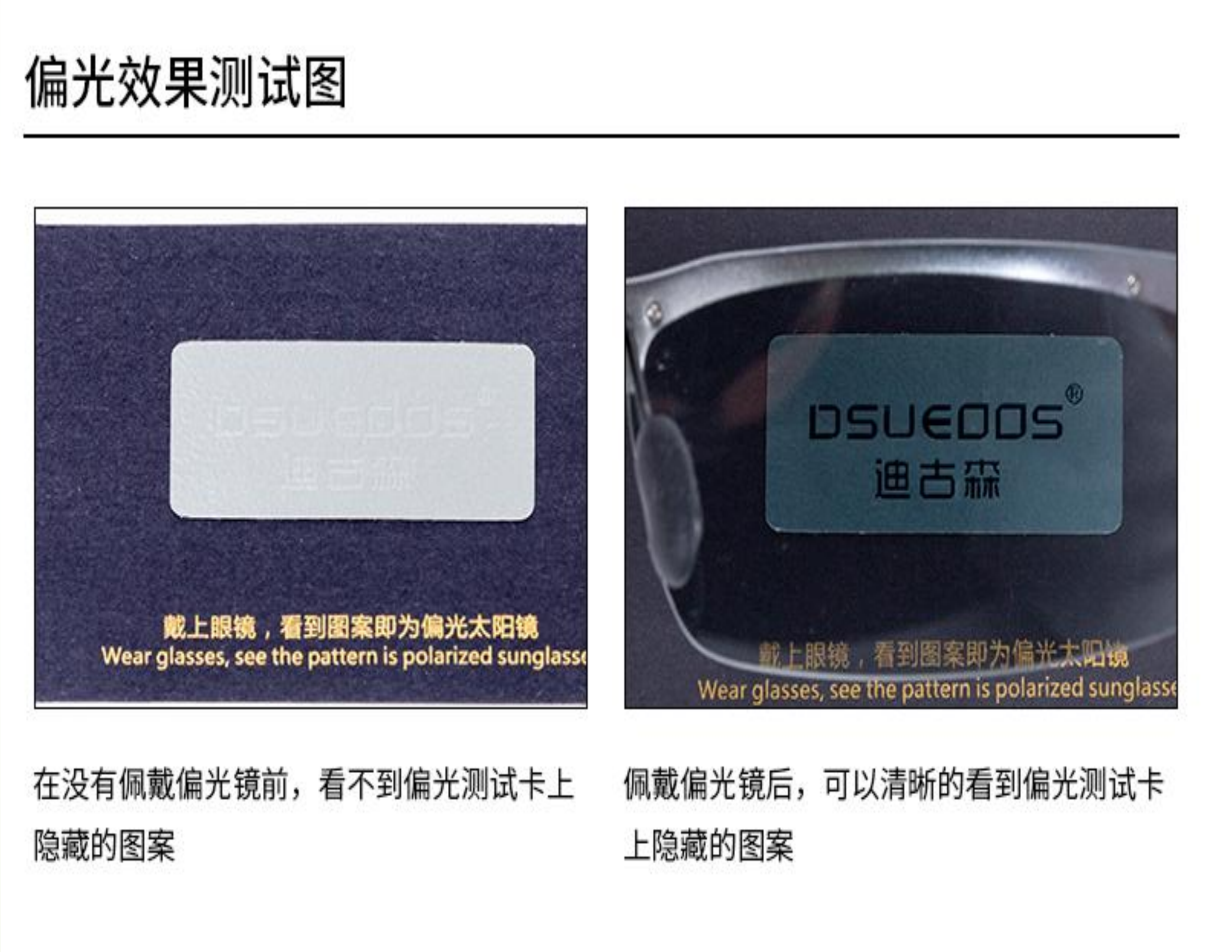
能效备案号：HX50BK7310　电视形态：平板

片源内容：爱奇艺　屏幕厚度：5mm以下

是否内置摄像头：否　语音遥控类型：遥控器语音遥控

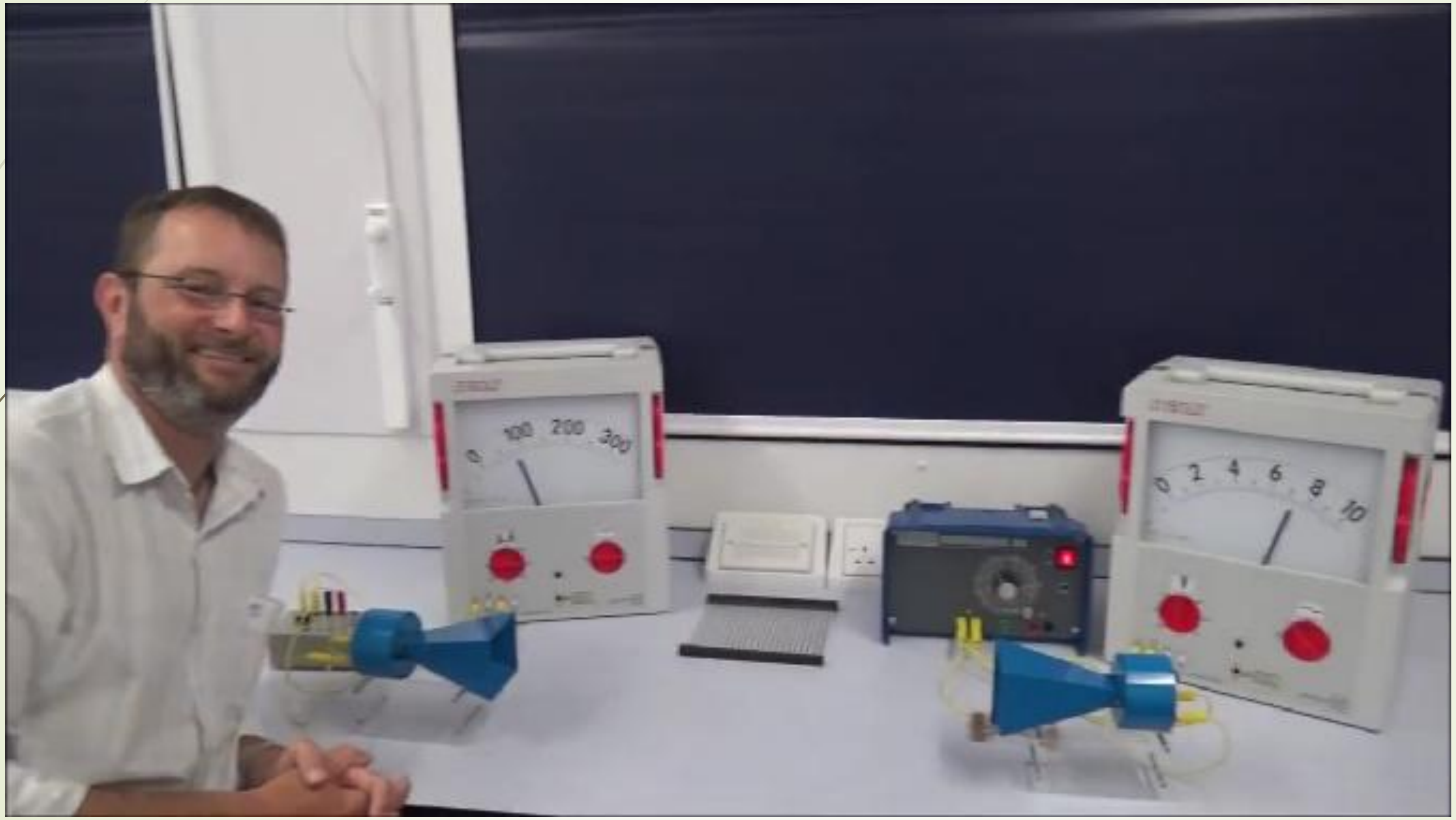
Application: sunglasses

Polarizer：UV-blocking，degrade glare due to reflecting



Application: polarized light and windshield

- The light and windshield both are 45 degree polarized, so the light from the oncoming car will be orthoganal to the polarization method of the windshield.


0 100 200 300
0 2 4 6 8 10

6. The engineering application of polarized wave

The polarization is wide used in many fields. such:

- **In radar target detection technology， Using the characteristics of the polarization changed in the process of electromagnetic scattering to achieve target recognition.**
- **In the radio technology， Using the polarized characteristic of the antenna transmitting and receiving electromagnetic waves, to implement the best performance of radio signals transmitting and receiving.**
- **In optical engineering , the different propagation characteristics of different polarized wave in the same material is used to design optical polarizing film, and so on .**

The summary of single medium uniform plane wave

- **Solution for the electromagnetic field complex vector :**

$$\vec{E}(\vec{r}) = \vec{E}_m e^{-j\vec{k}_c \cdot \vec{r}} \qquad \vec{H}(\vec{r}) = \vec{H}_m e^{-j\vec{k}_c \cdot \vec{r}}$$

- **the direction of $\vec{E}$、$\vec{H}$、$\mathrm{Re}(\vec{k}_c)$ perpendicular to each other, and meet the Right-handed Rule**
- **The amplitude of E is η_c times of the amplitude of H.**
- **using the electric field vector trajectory to judge the polarization characteristics(using the relationship between the two vertical component of the electric field)**
- **Related concepts and physical quantities:**

  frequency、period、wavelength、phase constant、wave number、phase velocity、power velocity、dispersion、the phenomenon of skin、skin depth、surface impedance、attenuation constant 、 phase constant、propagation constant、polarization (linear, circular, elliptic polarization; left-handed and right-handed polarization）

Key points of the problem solving method

- ✓ basis on the direction of propagation , write the expression of electromagnetic field
- ✓ **Solving physical quantities, Accord to the definition and the law.**

8.2 Plane Waves in Lossy Media

$$\nabla^2 \vec{E}(\vec{r}) + k_c^2 \vec{E}(\vec{r}) = 0$$

The ideal dielectric

$$\sigma = 0$$

$$k = \omega\sqrt{\mu\varepsilon} = \beta$$

$$\vec{E}(\vec{r}) = \vec{E}_m e^{-jk\vec{e}_n \bullet \vec{r}}$$

Conductive medium

$$\sigma \neq 0$$

$$k_c = \omega\sqrt{\mu\varepsilon_c} = \beta - j\alpha$$

$$\vec{E}(\vec{r}) = \vec{E}_m e^{-jk_c\vec{e}_n \bullet \vec{r}}$$

$$= \vec{E}_m e^{-\alpha\vec{e}_n \bullet \vec{r}} e^{-j\beta\vec{e}_n \bullet \vec{r}}$$

1. Uniform Plane Wave in Conductive Medium

Given by, $\gamma = \mathrm{j}k_c = \alpha + \mathrm{j}\beta$

then uniform plane wave along the z direction is:

$$\vec{E}(z) = \vec{E}_m e^{-\mathrm{j}k_c z}$$

$$= \vec{E}_m e^{-\gamma z} = \boxed{\vec{E}_m e^{-\alpha z}} e^{-\mathrm{j}\beta z}$$

Amplitude attenuate in the form of index , is evanescent wave

The instantaneous electric field as $\vec{E}(z,t) = \vec{E}_m e^{-\alpha z}\cos(\omega t - \beta z)$

γ **is propagation constant of EMW，unit: 1/m**

$e^{-\alpha z}$ **is attenuation factor,** α **is attenuation constant，unit: Np/m (nepers per meter)**

$e^{-\mathrm{j}\beta z}$ **is phase factor ,** β **is phase constant，unit: rad/m （radian per meter）**

- **The corresponding magnetic field**

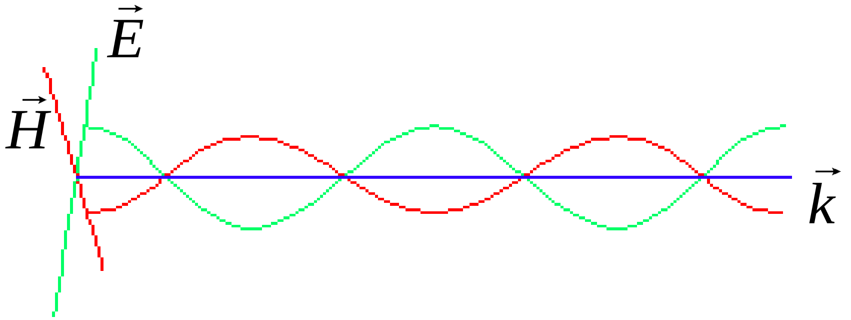
$$\vec{H}(z)=\frac{1}{\eta_c}\vec{e}_z\times\vec{E}(z)=\frac{1}{|\eta_c|}\vec{E}_m e^{-\alpha z}e^{-j(\beta z+\phi)}$$

$$\vec{H}(z,t)=\frac{\vec{e}_z\times\vec{E}_m}{|\eta_c|}e^{-\alpha z}\cos(\omega t-\beta z-\phi)$$

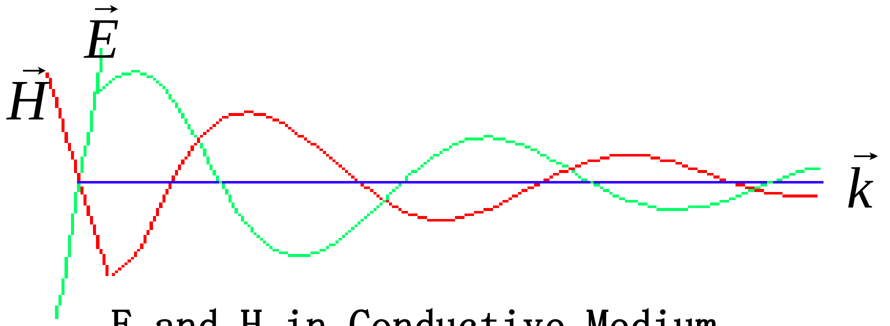
The phase of Magnetic field is different with electric field , and lag its.

intrinsic impedance $\eta_c=\sqrt{\frac{\mu}{\varepsilon_c}}=|\eta_c|e^{j\phi}$

intrinsic impedance is complex



E and H in Ideal dielectric



E and H in Conductive Medium

- **propagation parameters**

$$k_c^2 = \omega^2 \mu(\varepsilon - \mathrm{j}\frac{\sigma}{\omega})$$

$$k_c^2 = (\beta - j\alpha)^2 = \beta^2 - \alpha^2 - j2\alpha\beta \Rightarrow \begin{cases} \beta^2 - \alpha^2 = \omega^2 \mu\varepsilon \\ 2\alpha\beta = \omega\mu\sigma \end{cases}$$

$$\Rightarrow \alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2} - 1\right]}, \ \beta = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2} + 1\right]}$$

$$\lambda = \frac{2\pi}{\beta} = 1\bigg/\left\{ f\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2} + 1\right]} \right\}$$

$$v = \frac{\omega}{\beta} = 1\bigg/\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+(\frac{\sigma}{\omega\varepsilon})^2} + 1\right]}$$

Phase velocity is not only related with the medium parameters, but also related to the frequency of EM waves

2. Uniform Plane Wave in Low loss dielectrics (Special case)

Low Conductive Medium $\dfrac{\sigma}{\omega\varepsilon} << 1$

$$\boxed{(1+x)^{1/2} \approx 1 + \frac{x}{2}}$$

$$\gamma = \mathrm{j}\omega\sqrt{\mu\varepsilon}\left(1+\frac{\sigma}{\mathrm{j}\omega\varepsilon}\right)^{1/2} \approx \mathrm{j}\omega\sqrt{\mu\varepsilon} + \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}} \Rightarrow \begin{cases} \alpha \approx \dfrac{\sigma}{2}\sqrt{\dfrac{\mu}{\varepsilon}} \\ \beta \approx \omega\sqrt{\mu\varepsilon} \end{cases}$$

$$\eta_c = \sqrt{\frac{\mu}{\varepsilon_c}} = \sqrt{\frac{\mu}{\varepsilon}}\left(1+\frac{\sigma}{\mathrm{j}\omega\varepsilon}\right)^{-1/2} \approx \sqrt{\frac{\mu}{\varepsilon}}\left(1+\mathrm{j}\frac{\sigma}{2\omega\varepsilon}\right)$$

$$\boxed{(1-x)^{-1/2} \approx 1 - \frac{x}{2}}$$

- **The propagation characteristics of uniform plane wave in low conductive medium**

  - **The phase constant of low loss dielectrics is approximation to the phase constant of ideal medium**

3. Uniform Plane Wave in Good Conductor (Special case)

Good Conductor: $\dfrac{\sigma}{\omega\varepsilon} >> 1$

- **parameters of good conductive medium**

$$\gamma = \mathrm{j}\omega\sqrt{\mu\varepsilon}\left(1-\mathrm{j}\frac{\sigma}{\omega\varepsilon}\right)^{1/2} \approx \sqrt{\mathrm{j}\omega\mu\sigma} = \sqrt{\omega\mu\sigma}\,\mathrm{e}^{\mathrm{j}45^\circ} = \sqrt{\frac{\omega\mu\sigma}{2}}(1+\mathrm{j})$$

$$\alpha \approx \beta \approx \sqrt{\frac{\omega\mu\sigma}{2}} \approx \sqrt{\pi f \mu\sigma}$$

phase velocity: $v = \dfrac{\omega}{\beta} \approx \dfrac{\omega}{\sqrt{\pi f \mu\sigma}} = \sqrt{\dfrac{2\omega}{\mu\sigma}}$

$\propto \sqrt{f}$

wavelength: $\lambda = \dfrac{2\pi}{\beta} \approx \dfrac{2\pi}{\sqrt{\pi f \mu\sigma}} = 2\sqrt{\dfrac{\pi}{f\mu\sigma}}$

$\propto 1/\sqrt{f}$

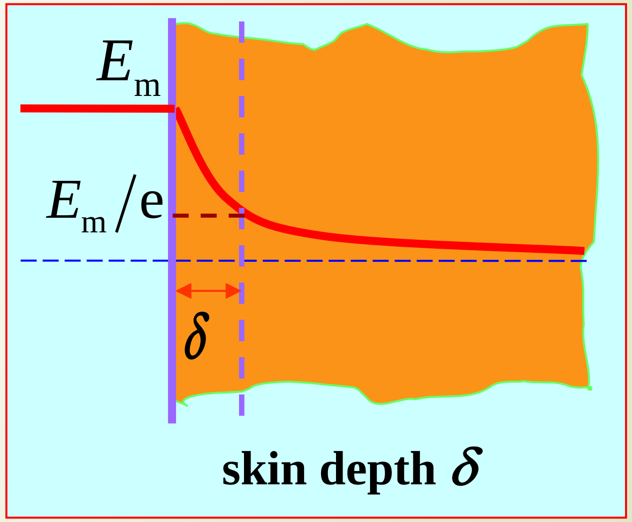
intrinsic impedance $\eta_c = \sqrt{\dfrac{\mu}{\varepsilon_c}} \approx \sqrt{\dfrac{j\omega\mu}{\sigma}} \approx \sqrt{\dfrac{2\pi f\mu}{\sigma}} e^{j45^\circ} = (1+j)\sqrt{\dfrac{\pi f\mu}{\sigma}}$

Corresponding the EM wave in good conductor, the phase of magnetic field is lagging with an angle of 45° compared with electric field

skin effect： Since the attenuation constant is larger over higher frequency, and The high frequency EM wave can only exist in the surface layer of good conductor , it's called the skin effect.

skin depth（δ):

$$E_m e^{-\alpha\delta} = \frac{E_m}{e} \xrightarrow{\alpha\delta = 1} \delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f\mu\sigma}}$$



surface impedance （square resistivity ）：

$$Z_s = R_s + jX_s = (1+j)\frac{1}{\sigma\delta} = (1+j)\sqrt{\frac{\pi f\mu}{\sigma}}$$

In the conductive medium , is the surface impedance always equal to the intrinsic resistance ?

Cu: $\mu = \mu_0 = 4\pi \times 10^{-7}\,\mathrm{H/m}$, $\sigma = 5.8 \times 10^{7}\,\mathrm{S/m}$

$$f = 50\mathrm{Hz},\quad \delta = \frac{6.6 \times 10^{-2}}{\sqrt{50}} = 9.33 \times 10^{-3}\,\mathrm{m}$$

$$f = 1\mathrm{MHz},\quad \delta = \frac{6.6 \times 10^{-2}}{\sqrt{10^{6}}} = 6.6 \times 10^{-5}\,\mathrm{m}$$

$$f = 10\mathrm{GHz},\quad \delta = \frac{6.6 \times 10^{-2}}{\sqrt{10 \times 10^{9}}} = 6.6 \times 10^{-7}\,\mathrm{m}$$

Table 1 Skin depth and surface resistance of some metal materials

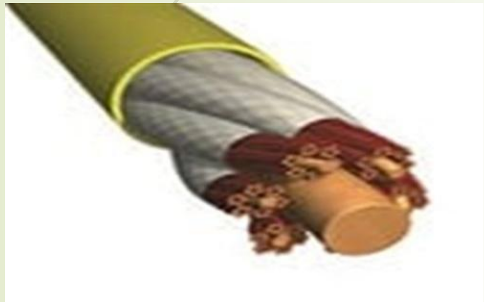
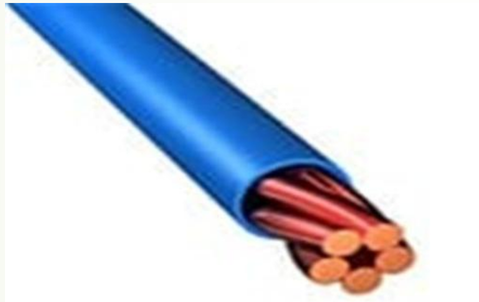
Material Name	Conductivity σ /(S/m)	Skin Depth δ /m	Surface Resistivity R_S /Ω
Silver	6.17×10^7	$0.064/\sqrt{f}$	$2.52\times10^{-7}\sqrt{f}$
Red copper	5.8×10^7	$0.066/\sqrt{f}$	$2.61\times10^{-7}\sqrt{f}$
Aluminum	3.72×10^7	$0.083/\sqrt{f}$	$3.26\times10^{-7}\sqrt{f}$
Sodium	2.1×10^7	$0.11/\sqrt{f}$	
Brass	1.6×10^7	$0.13/\sqrt{f}$	$5.01\times10^{-7}\sqrt{f}$
Stannum	0.87×10^7	$0.17/\sqrt{f}$	
Graphite	0.01×10^7	$1.6/\sqrt{f}$	

Experiment on Skin Depth

Skin Depth in Engineering

(1) Multi-insulated thin lines instead of a thick solid line

Litz: high frequency inductor、transformer、electricmoto、IT equipment、ultrsonic equipment

the total surface flowing current increased, so that R is decreased which leads to low power dissipation

(2) Surface plating with silver or gold to decrease R for current flowing along the surface of the line.

Skin Depth in Engineering

Aluminum conductor steel reinforced (ACSR) is applied in overhead electric line：

Steel: guarantee the strength

Aluminum: guarantee low resistance

The oscillating coil of High-power short-wave transmitter is made of hollow copper tube：

frequency of short-wave: tens MHz

Hollow: reduce costs according to effect of skin-depth

Skin Depth in Life

The thickness of a peice of steak in micro-oven in odor to unfreezing

Frequency of micro-oven: 2. 45GHz

Parameters of a frozen steak:

$$\varepsilon = 40\varepsilon_0 \qquad \mu = \mu_0 \qquad \sigma = 5\, S/m$$

Skin depth of a frozen steak:

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\mu}\right)^2} - 1\right]} = 40.9 \qquad d = \frac{1}{\alpha} = 20.8mm$$

E field at 8mm depth of a frozen steak vs. it on the surface:

$$\frac{E}{E_0} = e^{-\frac{8}{20.8}} = 68\%$$

E field at 8mm depth of a frozen steak vs. it on the surface:

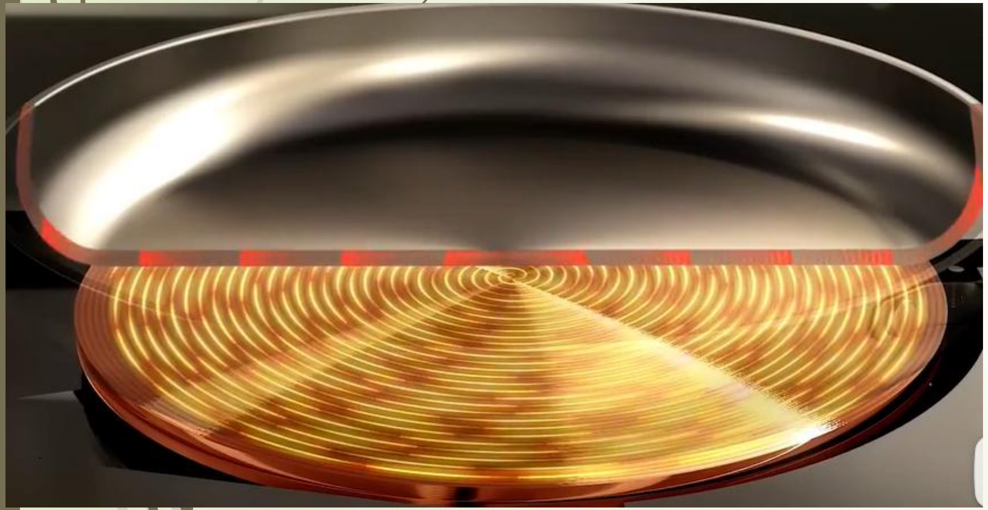
$$\frac{E}{E_0} = e^{-\frac{8}{20.8}} = 68\%$$

If the thickness is around 1.6cm, the power at the center is less than half of that on the surface.

Skin Depth in Life

Why no aluminum or copper pot on an induction cooker?

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$



EM wave in Lossy medium

Is it feasible if we communicate on and in a coal well (irrespective of safety problem)?

Parameters of coal stone:

$$\varepsilon = \frac{10}{36\pi}\times 10^{-9} F/m \quad \mu = 4\pi\times 10^{-7}\ H/\mathrm{m} \quad \sigma = 10^{-2}\ S/m$$

Frequency of 4G handphone: 2350MHz

Discussion:

Coal stone is considered as lossy medium:

$$E = E_m e^{-\alpha z}\cos(\omega t - \beta z)$$

$$\begin{cases} \alpha = \omega\sqrt{\dfrac{\mu\varepsilon}{2}[\sqrt{1+(\dfrac{\sigma}{\omega\mu})^2}-1]} \approx 0.596 \\ \beta = \omega\sqrt{\dfrac{\mu\varepsilon}{2}[\sqrt{1+(\dfrac{\sigma}{\omega\mu})^2}+1]} \approx 24.778 \end{cases}$$

When the emplitude of E field decreased to 10^{-6}, no signal will be received, so we get

$$E_m e^{-\alpha z} = 10^{-6} E_m$$

$$z = \frac{6}{\alpha} \ln 10 = 23m$$

That is the maximum transmitting distance of 4G signal is 23m, while the average depth of the coal well is around 400m. 4G signal cannot be trasnmitted.

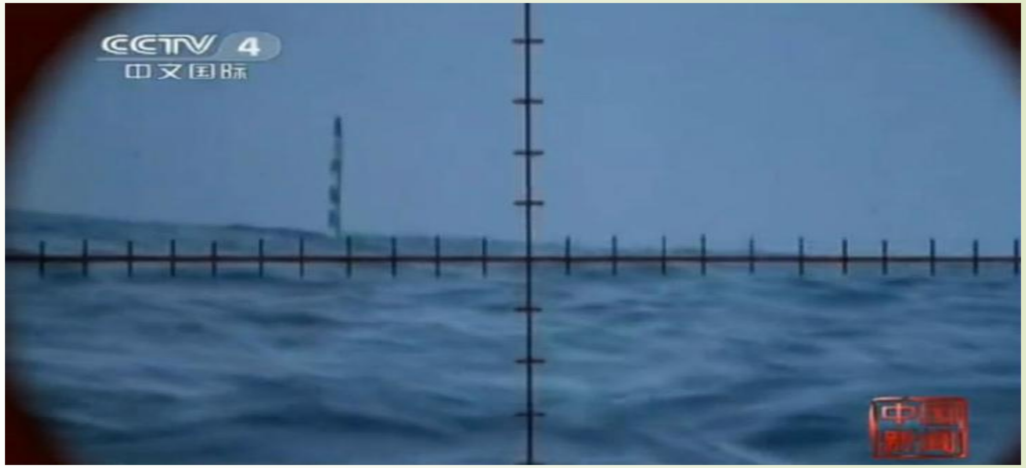
Low frequency: $f = 250Hz$

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon}{2}[\sqrt{1+(\frac{\sigma}{\omega\mu})^2} - 1]} \approx 0.0034$$

$$z = 4016m$$

Low frequency is now applied in emergency rescue.

EM wave in Lossy medium



Why there is lines on submarine?

Parameters of water of sea: $\varepsilon_r = 81$、$\mu_r = 1$、$\sigma = 4$ S/m

$$\frac{\sigma}{\omega\varepsilon} = \begin{cases} 3\times10^5 >> 1 & \text{good condutor} \\ 180 >> 1 & \text{good condutor} \\ 30 >> 1 & \text{good condutor} \end{cases}$$

$$\beta = \alpha = 0.218$$

$$\beta = \alpha = 8.89$$

$$\alpha = 21.8$$

$$e^{-\alpha z} = 10^{-6} \qquad z = -\frac{1}{\alpha}\ln 10^{-6} = \begin{cases} 63.4m & f = 3kHz \\ 1.55m & f = 5MHz \\ 0.65m & f = 30MHz \end{cases}$$

Ex. Polarized （in the X direction ）waves propagating in sea along + z . The parameter of the sea is ε_r = 81 、 μ_r =1 、 σ= 4 S/m , At z = 0, E_x = 100cos($10^7\pi t$) V/m .Please work out :

（1）The attenuation constant, phase constant, intrinsic impedance, phase velocity, wavelength and skin depth ;

（2）The distance of wave propagation, when he amplitude of the electric field intensity reduced to be 1/1000 of the value at z=0.

（3）The expression of instantaneous electric and magnetic field intensity at z = 0.8 m ;

（4）Average power per 1m^2 at z=0.8m 。

Sloution： (1) According to the problem

$$\omega = 10^7\pi \ \text{rad/s} \qquad f = \frac{\omega}{2\pi} = 5\times10^6 \quad \text{Hz}$$

So

$$\frac{\sigma}{\omega\varepsilon} = \frac{4}{10^7\pi\times(\frac{1}{36\pi}\times10^{-9})\times80} = 180 >> 1$$

In this condition, the sea can be regarded as good conductor.

So , Attenuation constant

$$\alpha = \sqrt{\pi f \mu\sigma} = \sqrt{\pi\times 5\times 10^{6}\times 4\pi\times 10^{-7}\times 4} = 8.89\ \text{Np/m}$$

Phase constant $\beta = \alpha = 8.89\ \text{rad/m}$

The characteristic impedance

$$\eta_{\mathrm{c}} = \sqrt{\frac{\omega\mu}{\sigma}}\mathrm{e}^{\mathrm{j}\frac{\pi}{4}} = \sqrt{\frac{10^{7}\pi\times 4\pi\times 10^{-7}}{4}}\mathrm{e}^{\mathrm{j}\frac{\pi}{4}} = \pi\mathrm{e}^{\mathrm{j}\frac{\pi}{4}}\ \Omega$$

Phase velocity

$$v = \frac{\omega}{\beta} = \frac{10^{7}\pi}{8.89} = 3.53\times 10^{6}\ \text{m/s}$$

Wavelebgth

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{8.89} = 0.707\ \text{m}$$

Skin depth

$$\delta = \frac{1}{\alpha} = \frac{1}{8.89} = 0.112\ \text{m}$$

（2） Let $e^{-\alpha z} = 1/1000$， i.e. $e^{\alpha z} = 1000$， can get the distance of wave propagation, when the amplitude of the electric field intensity reduced to be 1/1000 of the value at z=0

$$z = \frac{1}{\alpha}\ln 1000 = \frac{2.302}{8.89} = 0.777\ \text{m}$$

（3）According to the problem, the expression of instantaneous electric field intensity is

$$\vec{E}(z,t) = \vec{e}_x 100\mathrm{e}^{-8.89z}\cos(10^7\pi t - 8.89z)$$

So at z = 0.8 m, the expression of instantaneous electric field is

$$\vec{E}(0.8,t) = \vec{e}_x 100\mathrm{e}^{-8.89\times0.8}\cos(10^7\pi t - 8.89\times0.8)$$

$$= \vec{e}_x 0.082\cos(10^7\pi t - 7.11)\quad \text{V/m}$$

the expression of instantaneous magnetic field is

$$\vec{H}\left(0.8,t\right) = \vec{e}_y \frac{100\mathrm{e}^{-8.89\times0.8}}{\left|\eta_{\mathrm{c}}\right|}\cos(10^7\pi t - 8.89\times0.8 - \frac{\pi}{4})$$

$$= \vec{e}_y 0.026\cos(10^7\pi t - 1.61)\quad \text{A/m}$$

（4）At z = 0.8 m ,the average Poynting vector

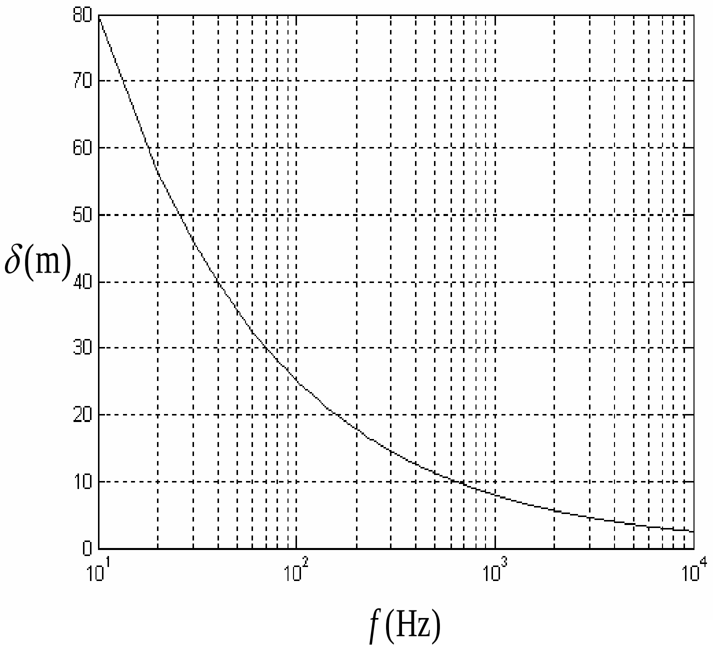
$$\vec{S}_{av} = \vec{e}_z \frac{1}{2|\eta_c|} E_{xm}^2 e^{-2\alpha z} \cos\phi$$

$$= \vec{e}_z \frac{100^2}{2\pi} e^{-2\times 8.89\times 0.8} \cos\frac{\pi}{4}$$

$$= \vec{e}_z 0.75 \quad \mathrm{mW/m^2}$$

The average power acoss The average power acoss 1m² is Pav = 0.75 mW

It can be known that the EM wave attenuation is very fast, especially at high frequency, the attenuation is more serious. This has led to a lot of difficulties for the communication between the submarines. To have lower attenuation, the working frequency must be very low, but even at the low frequency of 1 kHz, the attenuation is still obvious.



The curve of the skin depth of the sea changes with the frequency

Summary for Plane Wave in Lossy Media

- **Solution for electromagnetic field complex vector :**

$$\vec{E}(\vec{r}) = \vec{E}_{\mathrm{m}} \mathrm{e}^{-\mathrm{j}\vec{k}_c \cdot \vec{r}} \qquad \vec{H}(\vec{r}) = \vec{H}_{\mathrm{m}} \mathrm{e}^{-\mathrm{j}\vec{k}_c \cdot \vec{r}}$$

- **The phases of Electric field and magnetic field are different. In addition, electric field phase advance magnetic field phase in an angle of ϕ_0 , and its amplitude is $|\eta_c|$ times of magnetic field's .**

- **Related physical quantities :**

dispersion, skin effect, the skin depth, surface impedance, attenuation constant, phase constant, propagation constant, and result in low conductive medium and good conductors

Ex. 9.4 GHz uniform plane wave propagation in polyethylene (assuming that it's lossless medium , εr = 2.26). The amplitude of magnetic field is 7mA/m , please get the phase velocity, wavelength , wave impedance and the amplitude of electric field.

Solution： from the question $\varepsilon_r = 2.26\,,\ f = 9.4\times10^9\ \text{Hz}$

So:

$$v = \frac{v_0}{\sqrt{\varepsilon_r}} = \frac{v_0}{\sqrt{2.26}} = 1.996\times10^8 \quad \text{m/s}$$

$$\lambda = \frac{v}{f} = \frac{1.996\times10^8}{9.4\times10^9} = 2.12 \quad \text{m}$$

$$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{377}{\sqrt{2.26}} = 251 \quad \Omega$$

$$E_m = H_m\eta = 7\times10^{-3}\times251 = 1.757 \quad \text{V/m}$$

Ex. The amplitude of magnetic of uniform plane wave is 1/3π A/m , phase constant is 30 rad/m, spread in the air along the $-\vec{e}_z$ direction . If the direction of $\vec{H}$ is $-\vec{e}_y$, please write out the expression of $\vec{E}$, $\vec{H}$, and the frequency and wavelength.

Solution: Based on cosine, write directly

$$\vec{H}(z,t) = -\vec{e}_y \frac{1}{3\pi}\cos(\omega t + \beta z) \quad \text{A/m}$$

$$\vec{E}(z,t) = \eta_0 \vec{H}(z,t) \times (-\vec{e}_z) = \vec{e}_x 40\cos(\omega t + \beta z) \quad \text{V/m}$$

As β=30 rad/m, so

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{30} = 0.21 \text{ m} \ , \quad f = \frac{c}{\lambda} = \frac{3\times 10^8}{\pi/15} = \frac{45}{\pi}\times 10^8 = 1.43\times 10^9 \text{ Hz}$$

so: $$\vec{H}(z,t) = -\vec{e}_y \frac{1}{3\pi}\cos(90\times 10^8 t + 30z) \quad \text{A/m}$$

$$\vec{E}(z,t) = \vec{e}_x 40\cos(90\times 10^8 t + 30z) \quad \text{V/m}$$

Ex. 100MHz uniform EM wave propagation in lossless medium (εr = 4, μr =1) along +Z . Electric field is $\vec{E} = \vec{e}_x E_x$ .when t = 0 and z =1/8 m, the amplitude of the electric field is 10^{-4} V/m . Find the instantaneous value of electric and magnetic field.

Answer :Set the expression of the transient electric field intensity is

$$\vec{E}(z,t) = \vec{e}_x E_x = \vec{e}_x 10^{-4}\cos(\omega t - kz + \phi)$$

where $\omega = 2\pi f = 2\pi \times 10^8 \quad \text{rad/s}$

$$k = \omega\sqrt{\varepsilon\mu} = \frac{\omega}{c}\sqrt{\varepsilon_r \mu_r} = \frac{2\pi\times 10^8}{3\times 10^8}\sqrt{4} = \frac{4}{3}\pi \ \ \text{rad/m}$$

when t = 0、z =1/8 m， **the amplitude of the electric field is max ，**

$$\phi = kz = \frac{4\pi}{3}\times\frac{1}{8} = \frac{\pi}{6}$$

So $$\vec{E}(z,t) = \vec{e}_x 10^{-4}\cos(2\pi\times 10^8 t - \frac{4\pi}{3}z + \frac{\pi}{6})$$

$$= \vec{e}_x 10^{-4}\cos[2\pi\times 10^8 t - \frac{4\pi}{3}(z - \frac{1}{8})] \quad \text{V/m}$$

The instantaneous value of the transient magnetic field is:

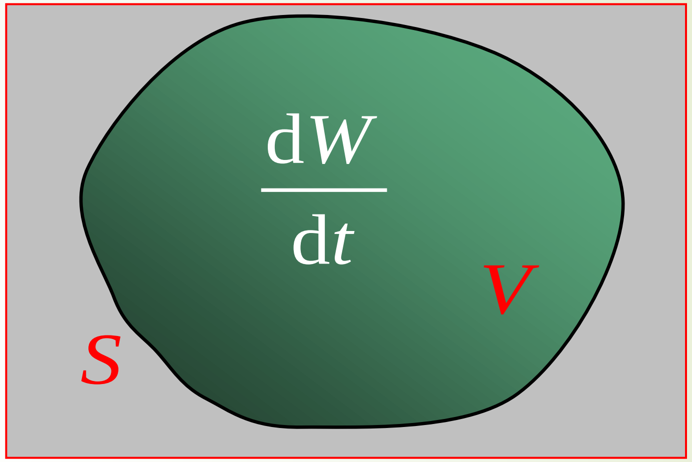
$$\vec{H} = \frac{1}{\eta}\vec{e}_z \times \vec{E} = \vec{e}_y \frac{1}{\eta}E_x$$

Where $$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \frac{\eta_0}{\sqrt{\varepsilon_r}} = 60\pi \quad \Omega$$

So $$\vec{H}(z,t) = \vec{e}_y \frac{10^{-4}}{60\pi}\cos[2\pi\times 10^8 t - \frac{4}{3}\pi(z - \frac{1}{8})] \quad \text{A/m}$$

8.3 Flow of Electromagnetic Power and the Poynting Vector

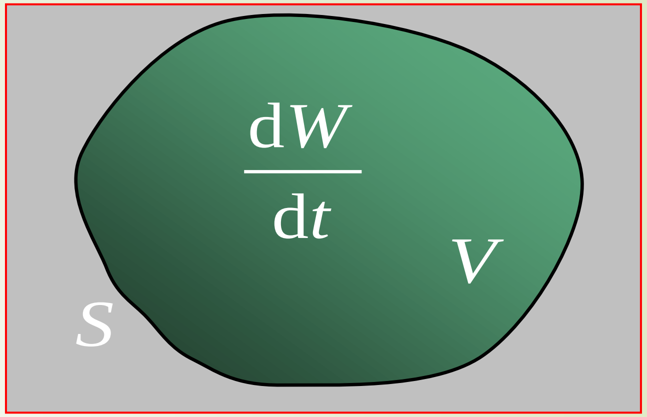
1. Conservation of energy



The energy flowing into volume V= Increased energy in the volume V + Energy loss in the volume V

Question： How to use mathematical representation ？

Storage of Electromagnetic energy in the volume V



The energy density of electric field：

$$w_e = \frac{1}{2}\vec{E}\cdot\vec{D}$$

The energy density of magnetic field：

$$w_m = \frac{1}{2}\vec{H}\cdot\vec{B}$$

The total energy density of electromagnetic field：

$$w = w_e + w_m = \frac{1}{2}\vec{E}\cdot\vec{D} + \frac{1}{2}\vec{H}\cdot\vec{B}$$

The total energy of electromagnetic field in the volume V：

$$W = \int_V w\mathrm{d}V = \int_V \left(\frac{1}{2}\vec{E}\cdot\vec{D} + \frac{1}{2}\vec{H}\cdot\vec{B}\right)\mathrm{d}V$$

Features: When the electromagnetic field changes with time, the electromagnetic field energy density at each point of the space also changes with time, which causes the flow of electromagnetic energy.

Conclusion: describe the relationship of energy according to the storage energy changed with time.

The mathematical representation for the conservation of energy ——Poynting theorem

According to

$$\begin{cases} \nabla \times \vec{H} = \vec{J} + \dfrac{\partial \vec{D}}{\partial t} \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \end{cases} \Rightarrow \begin{cases} \vec{E} \cdot \nabla \times \vec{H} = \vec{E} \cdot \vec{J} + \vec{E} \cdot \dfrac{\partial \vec{D}}{\partial t} \\ \vec{H} \cdot \nabla \times \vec{E} = -\vec{H} \cdot \dfrac{\partial \vec{B}}{\partial t} \end{cases}$$

Subtract the above two equations:

$$\vec{E} \cdot \nabla \times \vec{H} - \vec{H} \cdot \nabla \times \vec{E} = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} + \vec{H} \cdot \frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot \nabla \times \vec{A} - \vec{A} \cdot \nabla \times \vec{B}$$

$$\vec{E} \cdot \frac{\partial \vec{D}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} (\vec{E} \cdot \vec{D})$$

$$\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} (\vec{H} \cdot \vec{B})$$

$$-\nabla \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J} + \frac{\partial}{\partial t} \left(\frac{1}{2} \vec{E} \cdot \vec{D} + \frac{1}{2} \vec{H} \cdot \vec{B}\right)$$

The Poynting theorem and its physical meaning

Differential form (relationship for instantaneous power density)：

$$-\nabla\cdot(\vec{E}\times\vec{H})=\frac{\partial}{\partial t}\left(\frac{1}{2}\vec{E}\cdot\vec{D}+\frac{1}{2}\vec{H}\cdot\vec{B}\right)+\vec{E}\cdot\vec{J}$$

Integral form (relationship for instantaneous power)：

$$-\oint_S(\vec{E}\times\vec{H})\cdot\mathrm{d}\vec{S}=\frac{\mathrm{d}}{\mathrm{d}t}\int_V\left(\frac{1}{2}\vec{E}\cdot\vec{D}+\frac{1}{2}\vec{H}\cdot\vec{B}\right)\mathrm{d}V+\int_V\vec{E}\cdot\vec{J}\,\mathrm{d}V$$

the power is flowing into volume V （New physical quantity）

Electromagnetic power increased in the volume V

Power loss in the volume V

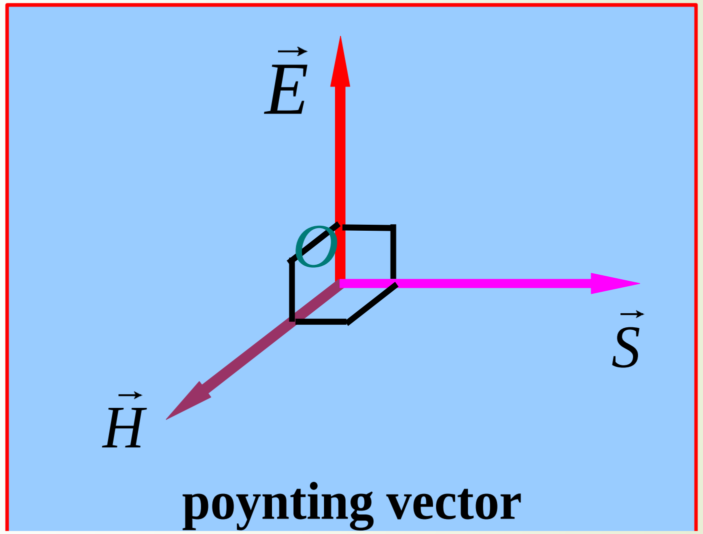
$$P_{in}=\frac{\mathrm{d}}{\mathrm{d}t}W+P_{loss}$$

The Poynting vector (energy flow density vector)

It describes the characteristics of the electromagnetic energy transmission (flow) in the time varying electromagnetic field

- **Definition:** $\vec{S} = \vec{E} \times \vec{H}$ (**W/m²**)

- **Physical meaning :** **Flow of power per unit area**



Direction of $\vec{S}$ —— The direction of electromagnetic energy transmission

Value of $\vec{S}$ —— it is the electromagnetic power per unit area at the perpendicular the transmission direction of the energy

2、Power density and Power flow density

Power density : $w = w_{\mathrm{e}} + w_{\mathrm{m}}$

The electric field energy and the magnetic field energy of uniform plane wave in ideal medium is equal.

where, $w_{\mathrm{e}} = \frac{1}{2}\varepsilon\left|\vec{E}(\vec{r},t)\right|^2 \quad w_{\mathrm{m}} = \frac{1}{2}\mu\left|\vec{H}(\vec{r},t)\right|^2 \xrightarrow{|\vec{\mathbf{E}}| = \eta|\vec{\mathbf{H}}|} w_{\mathrm{e}} = w_{\mathrm{m}}$

$$w = 2w_{\mathrm{e}} = 2w_{\mathrm{m}} = \varepsilon\left|\vec{E}(\vec{r},t)\right|^2 = \mu\left|\vec{H}(\vec{r},t)\right|^2$$

$$w_{\mathrm{av}} = \frac{1}{2}\varepsilon E_{\mathrm{m}}^2 = \frac{1}{2}\mu H_{\mathrm{m}}^2$$

Power flow density : $\vec{S} = \vec{E}(\vec{r},t)\times\vec{H}(\vec{r},t) \xrightarrow{\vec{\mathbf{H}} = \frac{1}{\eta}\vec{e}_n\times\vec{\mathbf{E}}} \vec{S} = \vec{e}_n\frac{1}{\eta}(\vec{E}\bullet\vec{E}) - \vec{E}\frac{1}{\eta}(\vec{e}_n\bullet\vec{E})$

$$\vec{S} = \vec{e}_n\frac{1}{\eta}\left|\vec{E}(\vec{r},t)\right|^2 = \vec{e}_n\eta\left|\vec{H}(\vec{r},t)\right|^2$$

$$\vec{S}_{\mathrm{av}} = \vec{e}_n\frac{1}{2\eta}E_{\mathrm{m}}^2 = \vec{e}_n\frac{1}{2}\eta H_{\mathrm{m}}^2$$

Relationship : $\vec{v}_e = \vec{S}/w$

$$\longrightarrow \vec{v}_e = \vec{v}_p = \vec{e}_n\frac{1}{\sqrt{\mu\varepsilon}}$$

The energy velocity and the phase velocity of uniform plane wave in ideal medium is equal

Instantaneous energy density and energy flow density vector

Instantaneous power density relation:

$$-\nabla \cdot \vec{S} = \frac{\partial}{\partial t}\left(\frac{1}{2}\vec{\mathbf{E}} \cdot \vec{\mathbf{D}} + \frac{1}{2}\vec{\mathbf{H}} \cdot \vec{\mathbf{B}}\right) + \vec{\mathbf{E}} \cdot \vec{\mathbf{J}}$$

Instantaneous power relation :

$$-\oint_S \vec{\mathbf{S}} \cdot \mathrm{d}\vec{\mathbf{S}} = \frac{\partial}{\partial t}\int_V \left(\frac{1}{2}\vec{\mathbf{E}} \cdot \vec{\mathbf{D}} + \frac{1}{2}\vec{\mathbf{H}} \cdot \vec{\mathbf{B}}\right)\mathrm{d}V + \int_V \vec{\mathbf{E}} \cdot \vec{\mathbf{J}}\,\mathrm{d}V$$

Notice :

- **all the variables in the formula must be represented by real form.**
- **the real form of each variable need to be used when doing calculation.**

$$\vec{S} = \overrightarrow{E}(\vec{r},t) \times \overrightarrow{H}(\vec{r},t)$$

Ex： The calculation of the instantaneous Poynting vector

$$\vec{S}(\vec{r},t)=\vec{E}(\vec{r},t)\times\vec{H}(\vec{r},t)$$
$$=\mathrm{Re}[\vec{E}(\vec{r})e^{j\omega t}]\times\mathrm{Re}[\vec{H}(\vec{r})e^{j\omega t}]$$

Ex： Calculation of the electric field energy density of instantaneous storage

$$w_e=\frac{1}{2}\vec{\mathbf{E}}(\vec{r},t)\cdot\vec{\mathbf{D}}(\vec{r},t)$$
$$=\frac{1}{2}\mathrm{Re}[\vec{\mathbf{E}}(\vec{r})e^{j\omega t}]\cdot\mathrm{Re}[\vec{\mathbf{D}}(\vec{r})e^{j\omega t}]$$

When solving the problems ,the instantaneous expression of the field must be written ！

Ex： Calculation of instantaneous power density consumption

$$p_{loss}=\vec{\mathbf{E}}(\vec{r},t)\cdot\vec{\mathbf{J}}(\vec{r},t)$$
$$=\mathrm{Re}[\vec{\mathbf{E}}(\vec{r})e^{j\omega t}]\cdot\mathrm{Re}[\vec{\mathbf{J}}(\vec{r})e^{j\omega t}]$$

Average energy / power density and energy flow density vector

Time average definition of time functions:

$$A_{av}(\vec{r},t) = \lim_{T\to\infty}\frac{1}{T}\int_0^T A(\vec{r},t)dt$$

$$= \frac{1}{T}\int_0^T A(\vec{r},t)dt$$

it is a periodic function with time T

✓ **The average Poynting vector （energy flow density vector）**

$$\vec{S}_{av}(\vec{r}) = \frac{1}{T}\int_0^T \vec{S}(\vec{r},t)dt$$

$$= \frac{1}{T}\int_0^T \vec{E}(\vec{r},t)\times\vec{H}(\vec{r},t)dt$$

$$= \frac{1}{T}\int_0^T \mathrm{Re}[\vec{E}(\vec{r})e^{j\omega t}]\times\mathrm{Re}[\vec{H}(\vec{r})e^{j\omega t}]dt$$

$$= \frac{1}{2}\mathrm{Re}[\vec{E}(\vec{r})\times\vec{H}^*(\vec{r})]$$

When solving the problems ,the instantaneous expression of the source is not required to be written ！

✓ Average electric field energy density

$$w_{Eav}(\vec{r}) = \frac{1}{T}\int_0^T w_E(\vec{r},t)dt$$

$$= \frac{1}{T}\int_0^T \frac{1}{2}\vec{D}(\vec{r},t)\bullet\vec{E}(\vec{r},t)dt$$

$$= \frac{1}{T}\int_0^T \frac{1}{2}\mathrm{Re}[\vec{D}(\vec{r})e^{j\omega t}]\bullet\mathrm{Re}[\vec{E}(\vec{r})e^{j\omega t}]dt$$

$$= \frac{1}{4}\mathrm{Re}[\vec{D}(\vec{r})\bullet\vec{E}^*(\vec{r})]$$

$$= \frac{1}{4}\varepsilon'\vec{E}(\vec{r})\bullet\vec{E}^*(\vec{r})$$

When solving the problems ,the instantaneous expression of the source is not required to be written ！

✓ Average magnetic field energy density

$$w_{Mav}(\vec{r}) = \frac{1}{T}\int_0^T w_M(\vec{r},t)dt$$

$$= \frac{1}{T}\int_0^T \frac{1}{2}\vec{B}(\vec{r},t)\bullet\vec{H}(\vec{r},t)dt$$

$$= \frac{1}{T}\int_0^T \frac{1}{2}\mathrm{Re}[\vec{B}(\vec{r})e^{j\omega t}]\bullet\mathrm{Re}[\vec{H}(\vec{r})e^{j\omega t}]dt$$

$$= \frac{1}{4}\mathrm{Re}[\vec{B}(\vec{r})\bullet\vec{H}^*(\vec{r})]$$

$$= \frac{1}{4}\mu'\vec{H}(\vec{r})\bullet\vec{H}^*(\vec{r})$$

When solving the problems ,the instantaneous expression of the source is not required to be written ！

✓ Average Joule loss power density

$$p_{Jav}(\vec{r}) = \frac{1}{T}\int_0^T p_J(\vec{r},t)dt$$

$$= \frac{1}{T}\int_0^T \vec{J}(\vec{r},t)\bullet\vec{E}(\vec{r},t)dt$$

$$= \frac{1}{T}\int_0^T \mathrm{Re}[\sigma\vec{E}(\vec{r})e^{j\omega t}]\bullet\mathrm{Re}[\vec{E}(\vec{r})e^{j\omega t}]dt$$

$$= \frac{1}{2}\mathrm{Re}[\sigma\vec{E}(\vec{r})\bullet\vec{E}^*(\vec{r})]$$

$$= \frac{1}{2}\sigma\vec{E}(\vec{r})\bullet\vec{E}^*(\vec{r})$$

When solving the problems ,the instantaneous expression of the source is not required to be written ！

Ex. The electric field intensityof plane wave in free space

$$\vec{E} = \vec{e}_x 50\cos(\omega t - kz)\ \text{V/m}$$

find the average power of vertical cross a round plane (radius=2.5m) at z = z0.

Sloution : The complex expression of electric field is $\vec{E} = \vec{e}_x 50\mathrm{e}^{-\mathrm{j}kz}$

The intrinsic impedance of free space is $\eta_0 = 120\pi \quad \Omega$

Get the expression of magnetic field $\vec{H} = \vec{e}_y \dfrac{E}{\eta_0} = \vec{e}_y \dfrac{5}{12\pi}\mathrm{e}^{-\mathrm{j}kz}\ \text{A/m}$

So , the average Poynting vector is

$$\vec{S}_{\text{av}} = \frac{1}{2}\mathrm{Re}(\vec{E} \times \vec{H}^*) = \vec{e}_z \frac{1}{2} \times 50 \times \frac{5}{12\pi} = \vec{e}_z \frac{125}{12\pi} \quad \text{W/m}^2$$

The average power of vertical cross a round plane is

$$P_{\text{av}} = \int_S \vec{S}_{\text{av}} \cdot \mathrm{d}\vec{S} = \frac{125}{12\pi} \times \pi R^2 = \frac{125}{12\pi} \times \pi \times 2.5^2 = 65.1 \quad \text{W}$$

Ex. In the electromagnetic measurement, in order to prevent the indoor electronic equipment from the outside world the influence of electromagnetic field，Copper plate can be used to construct shielding room, usually require copper thickness greater than 5δ . If the require frequency range is from 10KHz to 100MHZ , Calculate how thick copper plate at least。the parameter of Cu :$\mu=\mu_0$、$\varepsilon=\varepsilon_0$、$\sigma = 5.8 \times 10^7$ S/m。

Answer : For the frequency range of low-end f_L =10kHz ,

$$\frac{\sigma}{\omega_L \varepsilon} = \frac{5.8\times10^{7}}{2\pi\times10^{4}\times\dfrac{1}{36\pi}\times10^{-9}} = 1.04\times10^{14} >> 1$$

For the frequency range of high-end f_H =100MHz ,

$$\frac{\sigma}{\omega_H \varepsilon} = \frac{5.8\times10^{7}}{2\pi\times10^{8}\times\dfrac{1}{36\pi}\times10^{-9}} = 1.04\times10^{10} >> 1$$

this shows , copper can be treat as good conductor in the required frequency range of ,

$$\delta_L = \frac{1}{\sqrt{\pi f_L \mu\sigma}} = \frac{1}{\sqrt{\pi\times10^4\times4\pi\times10^{-7}\times5.8\times10^7}} = 0.66\ \text{mm}$$

$$\delta_H = \frac{1}{\sqrt{\pi f_H \mu\sigma}} = \frac{1}{\sqrt{\pi\times10^8\times4\pi\times10^{-7}\times5.8\times10^7}} = 6.6\ \mu\text{m}$$

In order to satisfy a given frequency range of the shielding requirements, The thickness of the copper plate at least should be

$$d = 5\delta_L = 3.3 \quad \text{mm}$$

Extended reading1： polarization of waves transmitting in arbitrary direction

1. 旋向的判断 $\vec{E}=\left(\vec{e}_x-\vec{e}_y-\vec{e}_y\mathrm{j}+\vec{e}_z\mathrm{j}\right)\mathrm{e}^{-\mathrm{j}k(x+y+z)}$

(a) 根据电场的表达式，分别求出$\vec{k}$，$\vec{\mathbf{E}}_{\mathbf{mr}}$及$\vec{\mathbf{E}}_{\mathbf{mi}}$；

(b) 根据下式结果判断旋向：

$$\vec{k}\cdot\left(\vec{\mathbf{E}}_{\mathbf{mi}}\times\vec{\mathbf{E}}_{\mathbf{mr}}\right)=\begin{cases}>0 & \text{右旋极化}\\ =\mathbf{0} & \text{不旋转，即线极化}\\ <0 & \text{左旋极化}\end{cases}$$

2. 对于非线极化情况，需要进一步确定是否为圆极化

如果下列两式满足，则为圆极化，否则为椭圆极化：

$$\begin{cases}\left|\vec{\mathbf{E}}_{\mathbf{mr}}\right|=\left|\vec{\mathbf{E}}_{\mathbf{mi}}\right|\\ \vec{\mathbf{E}}_{\mathbf{mr}}\cdot\vec{\mathbf{E}}_{\mathbf{mi}}=0\end{cases}$$

1. 已知一平面波的电场强度 $\vec{E}=(4\hat{a}_x+j5\hat{a}_y+3\hat{a}_z)e^{-jk_o(3x-4z)}$，该平面波的传播方向上的单位矢量________，极化方式为 ________。

1. 旋向的判断

(a) 根据电场的表达式，分别求出 $\vec{k}$，$\vec{\mathbf{E}}_{\mathbf{mr}}$ 及 $\vec{\mathbf{E}}_{\mathbf{mi}}$；

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2. 已知一平面波的电场强度 $\vec{E}=(4\hat{a}_x+j5\hat{a}_y+3\hat{a}_z)e^{-jk_o(3x-4z)}$，该平面波的传播方向上的单位矢量 $\hat{a}_n=(0.6,0,-0.8)$，极化方式为 右旋圆极化 。

$$\begin{cases}|\mathbf{E}_{\mathbf{mr}}|=|\mathbf{E}_{\mathbf{mi}}|\\ \vec{\mathbf{E}}_{\mathbf{mr}}\cdot\vec{\mathbf{E}}_{\mathbf{mi}}=0\end{cases}$$

3. 某电磁波从真空中进入介质后，发生变化的物理量有__________。

A. 波长和频率； B. 波长和波速；C. 频率和波速； D. 频率和能量

5. 某电磁波从真空中进入介质后，发生变化的物理量有____B______。

A. 波长和频率； B. 波长和波速；C. 频率和波速； D. 频率和能量

1、设在理想电介质（$\mu_r=1$）中传播的均匀平面波的电场强度为

$$\vec{E}=(\hat{a}_x+A\hat{a}_y+j\hat{a}_z)e^{-jk_0(x+y+z)}$$ （V/m）其中 k_0 为自由空间的波数。

（1）求该平面波传播方向上的单位矢量，确定常数 A 和介质的相对介电常数 ε_r；

（2）求该平面波的磁场强度的瞬时矢量表达式、平均能流密度矢量和平均电磁场能量密度；

（3）判断该平面波的极化方式。

解：

（1） $\overrightarrow{a_n}=(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}})$

$A=-(1+j)$，

$k=k_0\sqrt{3}=k_0\sqrt{\varepsilon_r}$，　　$\varepsilon_r=3$

(2)

$$\vec{H}=\frac{1}{\eta}\vec{e}_n\times\vec{E}$$

$$\vec{H}=\frac{1}{120\pi}\left[\hat{a}_x(2j+1)+\hat{a}_y(1-j)-\hat{a}_z(2+j)\right]e^{-jk_0(x+y+z)}$$

$$\overrightarrow{H(t)}=\frac{1}{120\pi}\left[(1+2j)\hat{a}_x+(1-j)\hat{a}_y-(2+j)\hat{a}_z\right]\cos\left(\omega t-k_0(x+y+z)\right)$$

$$\overrightarrow{S_{av}}=\mathrm{Re}\left[\vec{S}\right]=\frac{1}{2}\mathrm{Re}(\vec{E}\times\vec{H}^*)$$

$$=\frac{1}{60\pi}(\overrightarrow{a_x}+\overrightarrow{a_y}+\overrightarrow{a_z})$$

$$w_e=w_m=\frac{1}{2}\mathrm{Re}(\varepsilon\vec{E}\bullet\vec{E}^*)=\varepsilon=3\varepsilon_0$$

总的平均电磁场能量密度为 $w=w_e+w_m=6\varepsilon_0$

$$（3）\quad \vec{E}=\left[(\hat{a}_x-\hat{a}_y)+j(\hat{a}_z-\hat{a}_y)\right]e^{-jk_0(x+y+z)}$$

右旋椭圆极化

1. 旋向的判断

(a) 根据电场的表达式，分别求出$\vec{k}$，$\vec{E}_{mr}$及$\vec{E}_{mi}$；

(b) 根据下式结果判断旋向：

$$\vec{k}\cdot(\vec{E}_{mi}\times\vec{E}_{mr})=\begin{cases}>0 & \text{右旋极化}\\ =0 & \text{不旋转，即线极化}\\ <0 & \text{左旋极化}\end{cases}$$

2. 对于非线极化情况，需要进一步确定是否为圆极化

如果下列两式满足，则为圆极化，否则为椭圆极化：

$$\begin{cases}|\vec{E}_{mr}|=|\vec{E}_{mi}|\\ \vec{E}_{mr}\cdot\vec{E}_{mi}=0\end{cases}$$

Extended reading2： Dispersion and Group velocity

- **Dispersion： the pahse velocity changes with freqneucy**

In idea dielectric medium

$$v_p = \frac{\omega}{k} \qquad k = \omega\sqrt{\mu\varepsilon}$$

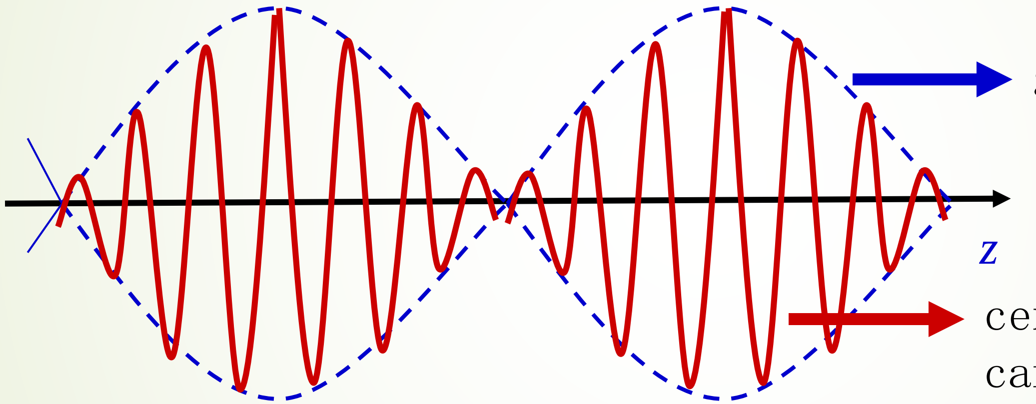
In conductive medium

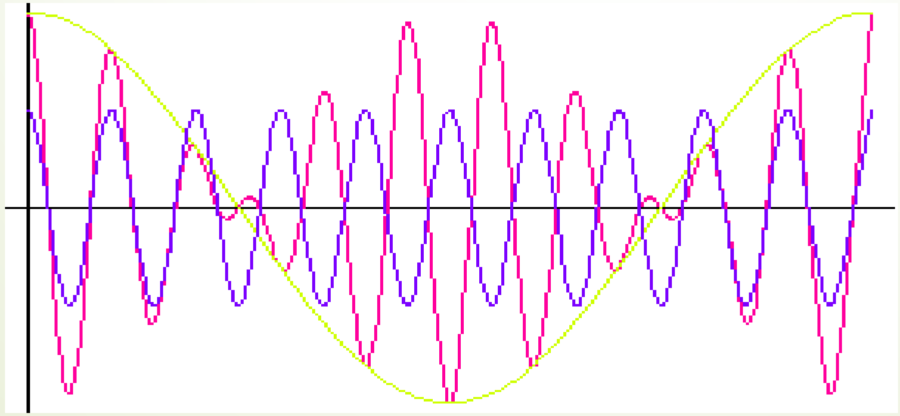
$$v_p = \frac{\omega}{k} \qquad k = \beta - j\alpha$$

β isfrequency dependent, that is, conductive medium is dispersion medium

Extended reading2： Dispersion and Group velocity

- **group velocity： a modulated singnal always applies a certain of frequencyband at some center frequency. The velocity of signal including the whole frequency band is group velocity.**





Consider the composite wave of two frequency $\omega_0+\Delta\omega$ *and* $\omega_0-\Delta\omega$, *assuming their emplitudes are both* E_m , theie phase constant are:

$$\beta_1 = \beta + \Delta\beta , \beta_2 = \beta - \Delta\beta$$

$$\vec{E}_1(z,t) = \vec{e}_x E_\mathrm{m} \cos[(\omega_0 + \Delta\omega)t - (\beta_0 + \Delta\beta)z]$$

$$\vec{E}_2(z,t) = \vec{e}_x E_\mathrm{m} \cos[(\omega_0 - \Delta\omega)t - (\beta_0 - \Delta\beta)z]$$

The composite wave is

In real: $\vec{E}(z,t) = \vec{E}_1(z,t) + \vec{E}_2(z,t)$

$$= \vec{e}_x 2E_\mathrm{m} \cos(\Delta\omega t - \Delta\beta z)\cos(\omega_0 t - \beta_0 z)$$

In complex: $\vec{E}(z) = \boxed{A_m \cos(\Delta\omega t - \Delta kz)}\,e^{j(\omega t - kz)}$

Amplitude of composite, low frequency travelling wave at $\Delta\omega$

travelling wave at ω

Group velocity:

$$v_g = \frac{dz}{dt} = \frac{\Delta\omega}{\Delta k} \approx \frac{d\omega}{dk} = \frac{d(v_p k)}{dk} = v_p + k\frac{dv_p}{dk} = v_p + \frac{\omega}{v_p}\frac{dv_p}{d\omega}v_g$$

$$\Rightarrow v_g = \frac{v_p}{1 - \frac{\omega}{v_p}\frac{dv_p}{d\omega}}$$

Discussion:

(1): **When** $\frac{dv_p}{d\omega} = 0,\ v_g = v_p$

In idea dielectric, group velocity=phase velocity, no dispersion.

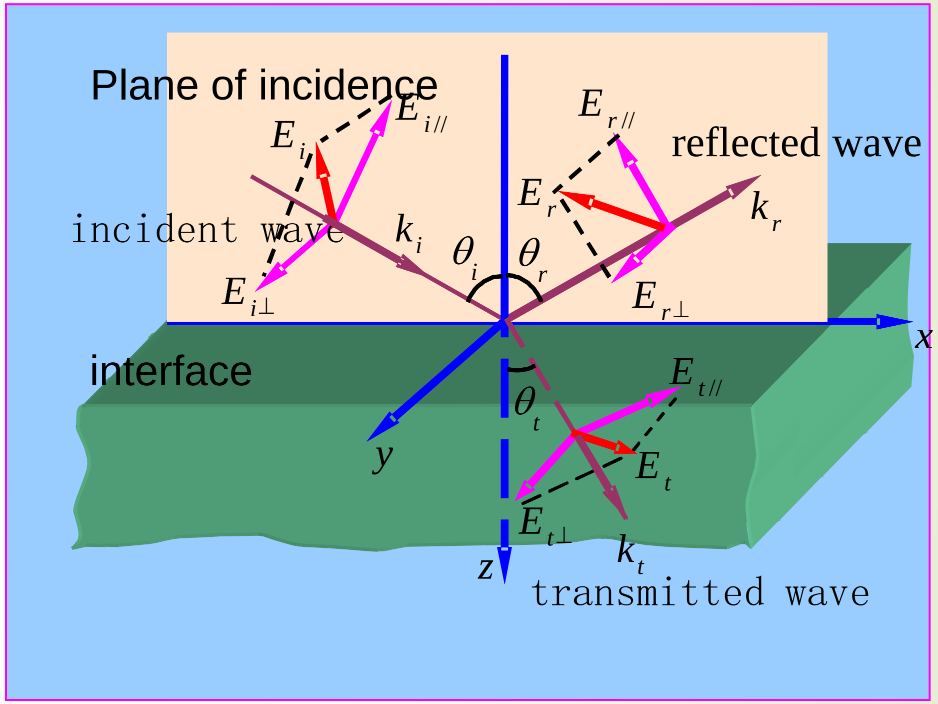
(2): **When** $\frac{dv_p}{d\omega} > 0,\ v_g > v_p$ Anomalous dispersion.

(3): **When** $\frac{dv_p}{d\omega} < 0,\ v_g < v_p$ Dispersion.

8.4 Normal Incidence at a Plane Conducting Boundary

- ## Phenomenon：

In the space , at the side of the incident wave of electromagnetic wave added a reflected wave ; the transmitted wave can exist on the other side .



- ## The incident way：

vertical incidence 、

oblique incidence

- ## Medium：

Conductive medium, ideal conductor, the ideal dielectric

The basic problem：

To solve the incident wave and transmission wave electromagnetic fields respectively

Incident wave space： $\vec{E}_1(\vec{r}) = \vec{E}_i(\vec{r}) + \vec{E}_r(\vec{r}) = \vec{E}_{im} e^{-j\vec{k}_i \cdot \vec{r}} + \vec{E}_{rm} e^{-j\vec{k}_r \cdot \vec{r}}$

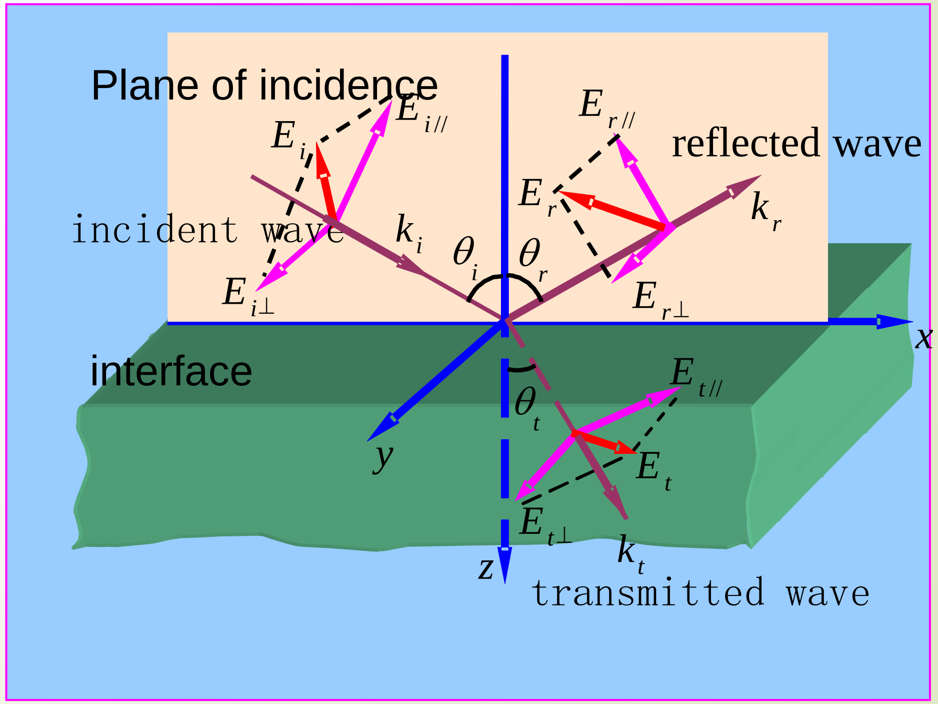
Transmission wave space： $\vec{E}_2(\vec{r}) = \vec{E}_t(\vec{r}) = \vec{E}_{tm} e^{-j\vec{k}_t \cdot \vec{r}}$

- **problem：**

Known: $\vec{E}_{im},\ \vec{k}_i$

Solve: $\vec{E}_{rm}, \vec{E}_{tm};\ \ \vec{k}_r, \vec{k}_t$

Find the direction and the value of the corresponding physical quantities.



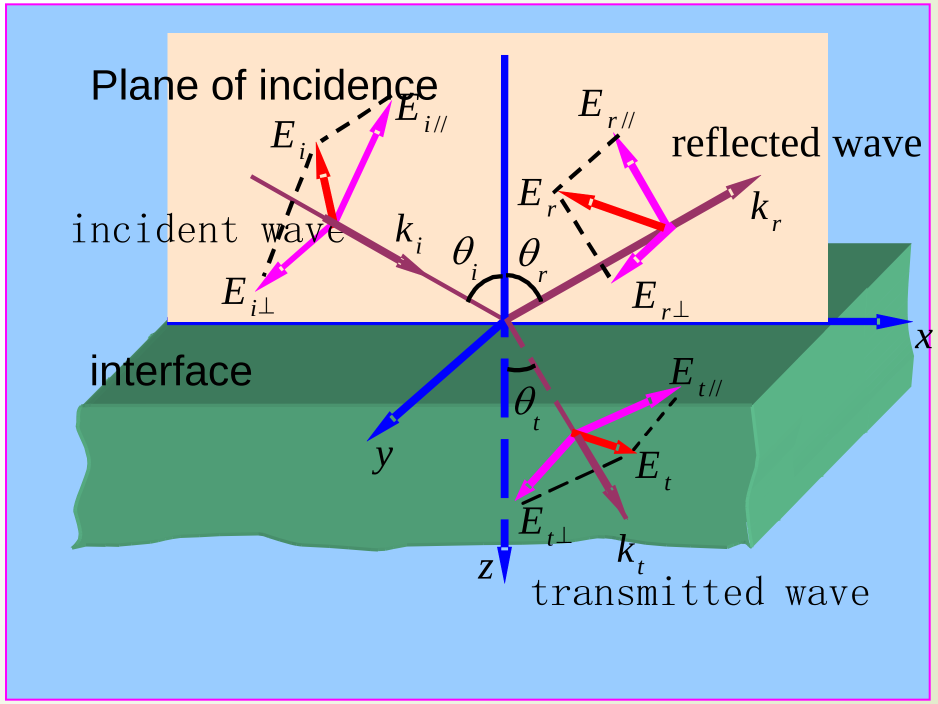
Analysis Method：

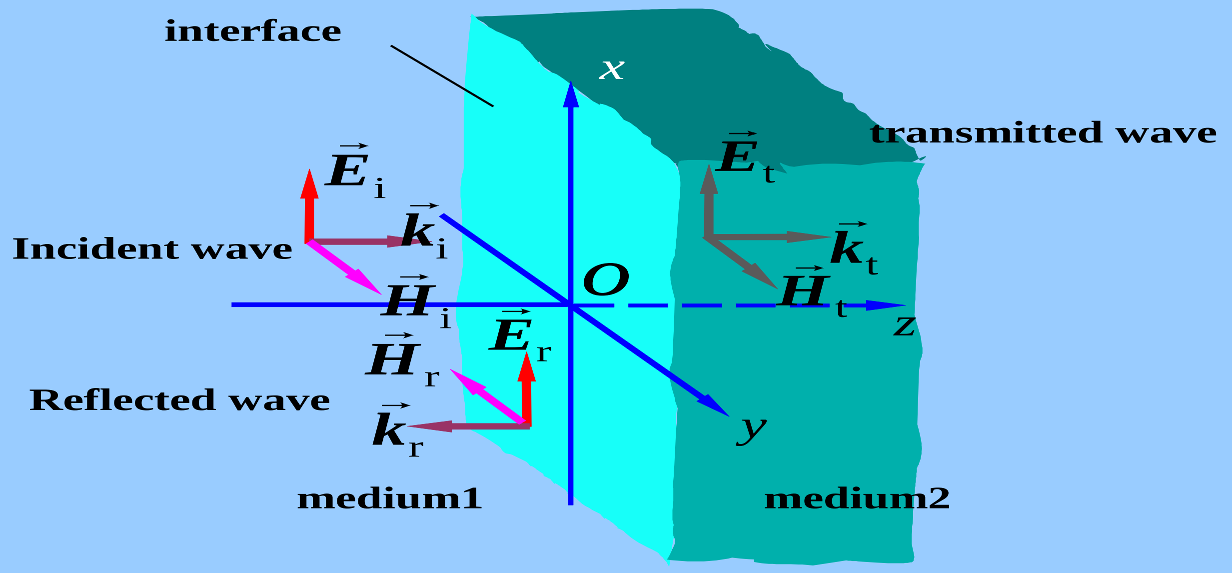
Establish the relationship of various physical quantities on the boundary

incident wave（known）＋reflected wave（unknown） $\xleftrightarrow{\text{boundary condition}}$ **transmitted wave (unknown)**

$$\vec{e}_n \times (\vec{E}_1 - \vec{E}_2)\Big|_S = 0$$

$$\vec{e}_n \times (\vec{H}_1 - \vec{H}_2)\Big|_S = 0$$





Uniform plane wave vertical incidence to the interface between two different media plane

feature: **(Determine the direction of physical quantities of reflected and transmission wave !)**

- The reflection wave propagation along the -z direction. the transmitted wave propagation along the +z direction
- the electric field only has x component

why?

problem: solve the amplitude of reflected wave and transmission wave

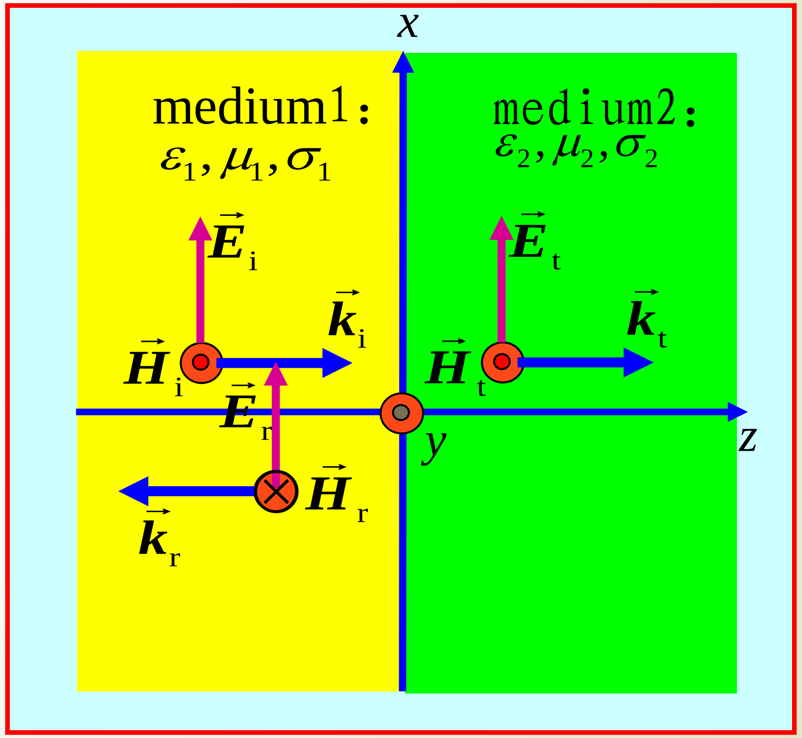
method: write out the expression and use the boundary conditions

- **Establish coordinate system as shown**

- **at $z < 0$ region, the parameters of conductive medium1** μ_1、ε_1、σ_1

- **at $z > 0$ region, the parameters of conductive medium 2** μ_2、ε_2、σ_2

- **Assuming that the incident wave is linear polarized wave at x direction**



without loss of generality! why?

How is for circular polarized wave?

incident wave in medium 1:

$$\vec{E}_{\mathrm{i}}(z)=\vec{e}_x E_{\mathrm{im}} \mathrm{e}^{-\gamma_1 z}$$

$$\vec{H}_{\mathrm{i}}(z)=\vec{e}_y \frac{E_{\mathrm{im}}}{\eta_{1\mathrm{c}}} \mathrm{e}^{-\gamma_1 z}$$

reflected wave in medium 1 :

$$\vec{E}_{\mathrm{r}}(z)=\vec{e}_x E_{\mathrm{rm}} \mathrm{e}^{\gamma_1 z}$$

$$\vec{H}_{\mathrm{r}}(z)=-\vec{e}_y \frac{E_{\mathrm{rm}}}{\eta_{1\mathrm{c}}} \mathrm{e}^{\gamma_1 z}$$

transmitted wave in medium 2:

$$\vec{E}_{\mathrm{t}}(z)=\vec{e}_x E_{\mathrm{tm}} \mathrm{e}^{-\gamma_2 z}$$

$$\vec{H}_{\mathrm{t}}(z)=\vec{e}_y \frac{E_{\mathrm{tm}}}{\eta_{2\mathrm{c}}} \mathrm{e}^{-\gamma_2 z}$$

$$\gamma_1=\mathrm{j}k_{1\mathrm{c}}=\mathrm{j}\omega\sqrt{\mu_1\varepsilon_{1\mathrm{c}}}$$

$$=\mathrm{j}\omega\sqrt{\mu_1\varepsilon_1}\left(1-\mathrm{j}\frac{\sigma_1}{\omega\varepsilon_1}\right)^{1/2}$$

$$\eta_{1\mathrm{c}}=\sqrt{\frac{\mu_1}{\varepsilon_{1\mathrm{c}}}}=\sqrt{\frac{\mu_1}{\varepsilon_1}}\left(1-\mathrm{j}\frac{\sigma_1}{\omega\varepsilon_1}\right)^{-1/2}$$

$$=\eta_1\left(1-\mathrm{j}\frac{\sigma_1}{\omega\varepsilon_1}\right)^{-1/2}$$

$$\gamma_2=\mathrm{j}\omega\sqrt{\mu_2\varepsilon_2}\left(1-\mathrm{j}\frac{\sigma_2}{\omega\varepsilon_2}\right)^{1/2}$$

$$\eta_{2\mathrm{c}}=\eta_2\left(1-\mathrm{j}\frac{\sigma_2}{\omega\varepsilon_2}\right)^{-1/2}$$

$$\vec{e}_n \times (\vec{E}_1 - \vec{E}_2)\big|_S = 0$$

$$\vec{e}_n \times (\vec{H}_1 - \vec{H}_2)\big|_S = 0$$

$$\Rightarrow \begin{cases} E_1(0) = E_2(0) \\ H_1(0) = H_2(0) \end{cases}$$

$$\vec{E}_1(0) = \vec{E}_\mathrm{i}(0) + \vec{E}_\mathrm{r}(0) = \vec{e}_x(E_\mathrm{im} + E_\mathrm{rm}) \qquad \vec{E}_2(0) = \vec{E}_\mathrm{t}(0) = \vec{e}_x E_\mathrm{tm}$$

$$\vec{H}_1(0) = \vec{H}_\mathrm{i}(0) + \vec{H}_\mathrm{r}(0) = \vec{e}_y\left(\frac{E_\mathrm{im}}{\eta_{1\mathrm{c}}} - \frac{E_\mathrm{rm}}{\eta_{1\mathrm{c}}}\right) \qquad \vec{H}_2(0) = \vec{H}_\mathrm{t}(0) = \vec{e}_y \frac{E_\mathrm{tm}}{\eta_{2\mathrm{c}}}$$

$$\Rightarrow \begin{cases} E_\mathrm{im} + E_\mathrm{rm} = E_\mathrm{tm} \\ \dfrac{1}{\eta_{1\mathrm{c}}}(E_\mathrm{im} - E_\mathrm{rm}) = \dfrac{1}{\eta_{2\mathrm{c}}} E_\mathrm{tm} \end{cases} \Rightarrow \begin{cases} E_\mathrm{rm} = \Gamma E_\mathrm{im} \\ E_\mathrm{tm} = \tau E_\mathrm{im} \end{cases}$$

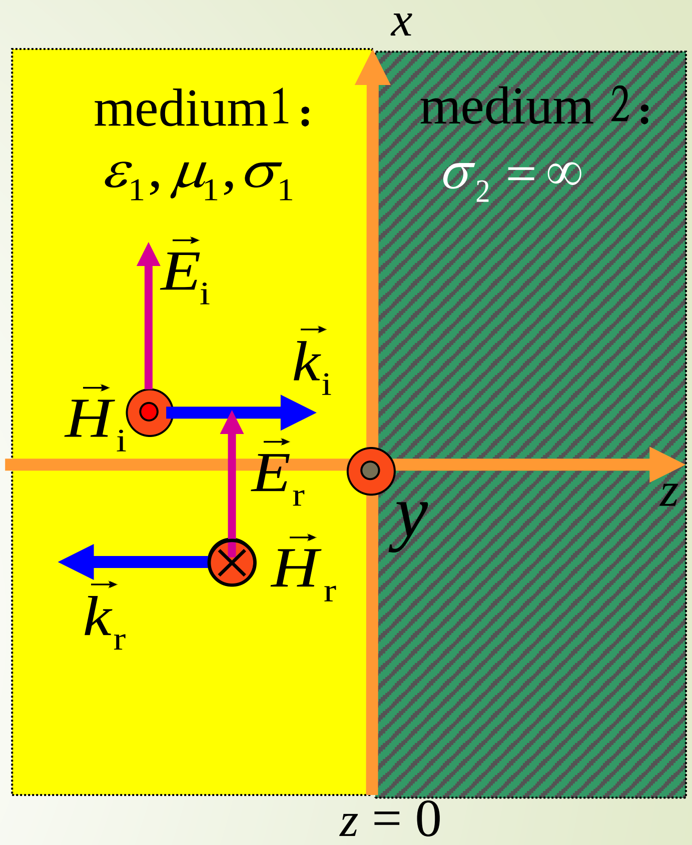
where：

$$\Gamma = \frac{\eta_{2\mathrm{c}} - \eta_{1\mathrm{c}}}{\eta_{2\mathrm{c}} + \eta_{1\mathrm{c}}} \qquad \tau = \frac{2\eta_{2\mathrm{c}}}{\eta_{2\mathrm{c}} + \eta_{1\mathrm{c}}}$$

Reflection coefficient on the interface   Transmission coefficient on the interface

medium 1 is ideal dielectric，$\sigma_1 = 0$

medium 2 is ideal conductor，$\sigma_2 = \infty$



then $\beta_1 = \omega\sqrt{\mu_1\varepsilon_1}, \quad \eta_1 = \sqrt{\dfrac{\mu_1}{\varepsilon_1}}, \quad \eta_{2c} = 0$

so $\quad \Gamma = -1、 \tau = 0$

$$E_{\mathrm{rm}} = -E_{\mathrm{im}}$$

on the interface， the Phase difference between the incident wave and reflected wave is π

incident wave in medium 1：

$$\vec{E}_{\mathrm{i}}(z) = \vec{e}_x E_{\mathrm{im}} \mathrm{e}^{-\mathrm{j}\beta_1 z}, \quad \vec{H}_{\mathrm{i}}(z) = \vec{e}_y \frac{E_{\mathrm{im}}}{\eta_1} \mathrm{e}^{-\mathrm{j}\beta_1 z}$$

reflected wave in medium 1：

$$\vec{E}_{\mathrm{r}}(z) = -\vec{e}_x E_{\mathrm{im}} \mathrm{e}^{\mathrm{j}\beta_1 z}, \quad \vec{H}_{\mathrm{r}}(z) = \vec{e}_y \frac{E_{\mathrm{im}}}{\eta_1} \mathrm{e}^{\mathrm{j}\beta_1 z}$$

the electromagnetic field of total field in miduim 1

$$\vec{E}_1(z)=\vec{e}_x E_{\mathrm{im}}(\mathrm{e}^{-\mathrm{j}\beta_1 z}-\mathrm{e}^{\mathrm{j}\beta_1 z})=-\vec{e}_x \mathrm{j}2E_{\mathrm{im}}\sin(\beta_1 z)$$

$$\vec{H}_1(z)=\vec{e}_y\frac{E_{\mathrm{im}}}{\eta_1}(\mathrm{e}^{-\mathrm{j}\beta_1 z}+\mathrm{e}^{\mathrm{j}\beta_1 z})=\vec{e}_y\frac{2E_{\mathrm{im}}\cos(\beta_1 z)}{\eta_1}$$

Is it true that H_1 calculated by

$$\vec{H}_1=\frac{1}{\eta_1}\vec{e}_z\times E_1$$

Instantaneous value form

$$\vec{E}_1(z,t)=\mathrm{Re}[\vec{E}_1(z)\mathrm{e}^{\mathrm{j}\omega t}]=\vec{e}_x 2E_{\mathrm{im}}\sin(\beta_1 z)\sin(\omega t)$$

$$\vec{H}_1(z,t)=\mathrm{Re}[\vec{H}_1(z)\mathrm{e}^{\mathrm{j}\omega t}]=\vec{e}_y\frac{2E_{\mathrm{im}}}{\eta_1}\cos(\beta_1 z)\cos(\omega t)$$

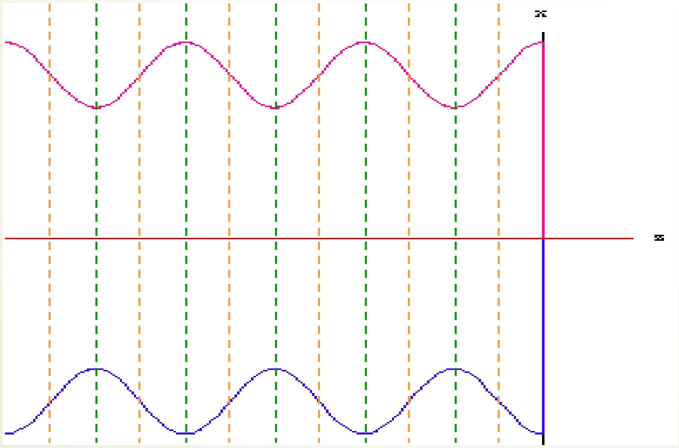
the relationship of the total fields:

- ✓ time phase difference is $\pi/2$
- ✓ space length difference is $\lambda/4$

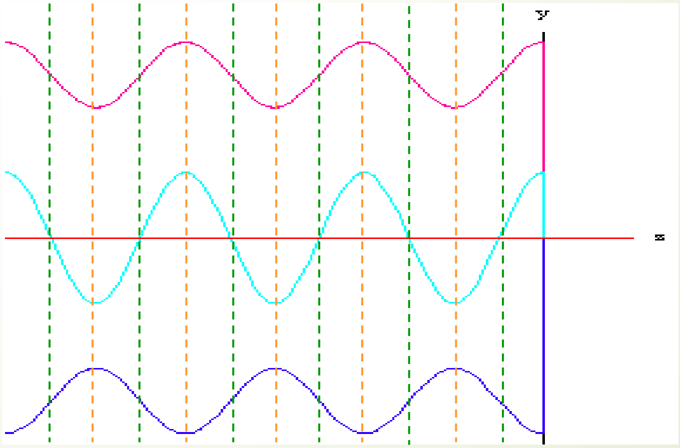
incident wave

total wave

reflected wave

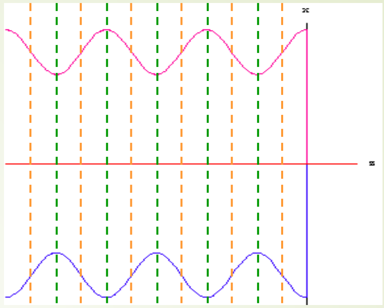


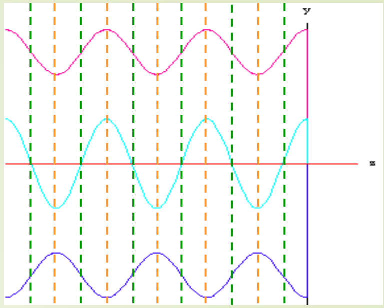
electric field intensity



magnetic field intensity

• The relevant physical quantities and concepts





✓ the average poynting vector of the total field is :

$$\vec{S}_{\mathrm{av}} = \frac{1}{2}\mathrm{Re}[\vec{E}_1 \times \vec{H}_1^*] = \frac{1}{2}\mathrm{Re}\left[-\vec{e}_x \mathrm{j}2E_{\mathrm{im}} \sin(\beta_1 z) \times \vec{e}_y \left(\frac{2E_{\mathrm{im}} \cos(\beta_1 z)}{\eta_1} \right)^* \right] = 0$$

✓ the induced current density on the surface of the ideal conductor

$$\vec{J}_S = \vec{e}_{\mathrm{n}} \times \vec{H}_1(z)\big|_{z=0} = -\vec{e}_z \times \vec{e}_y \frac{2E_{\mathrm{im}} \cos(\beta_1 z)}{\eta_1}\Big|_{z=0} = \vec{e}_x \frac{2E_{\mathrm{im}}}{\eta_1}$$

✓ **travelling wave**： Electromagnetic wave propagated along a certain direction in the space (moving)

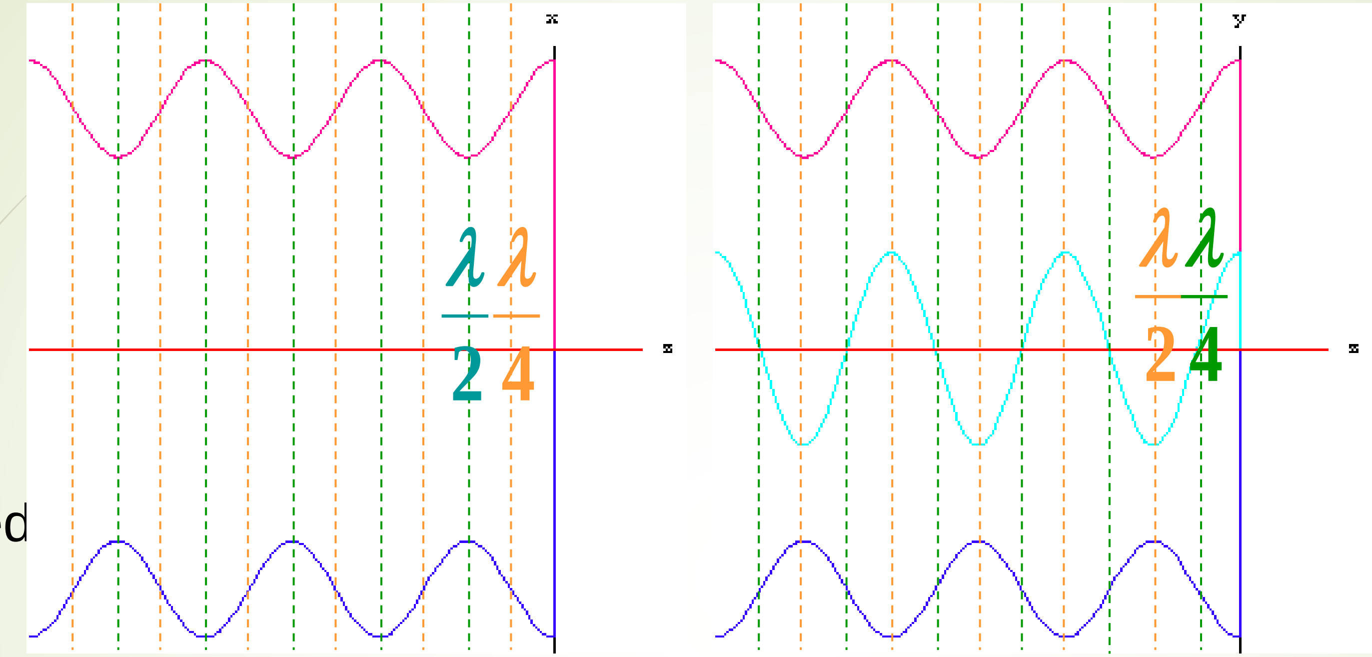
✓ **Standing wave**： Electromagnetic wave is not propagated in the space， there are antinodes and nodes at fixed point.

✓ **travelling-standing wave** ： part of the electromagnetic wave in space spread, and part of don't spread.

incident wave

total wave

reflected wave



travelling wave along z

standing wave

travelling wave along -z

electric field intensity   magnetic field intensity

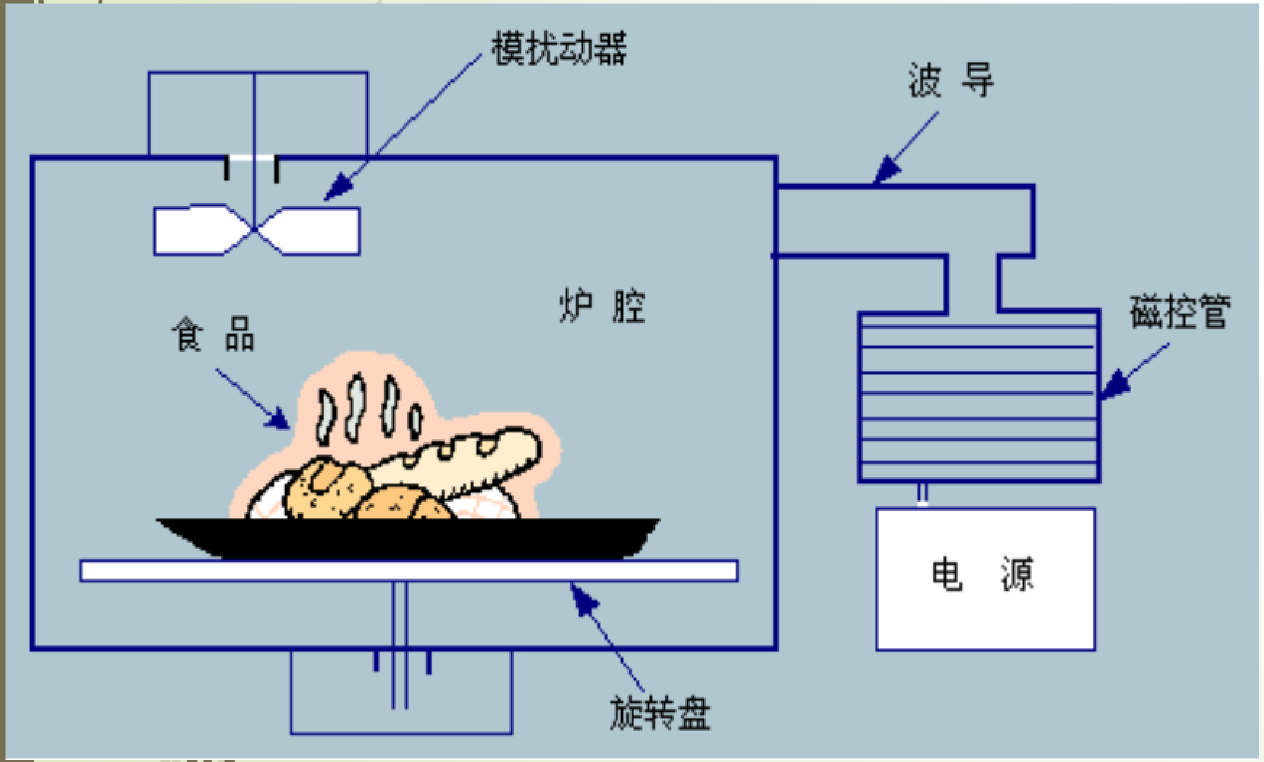
the point of antinodes (the point of the maximum value of electric field)：

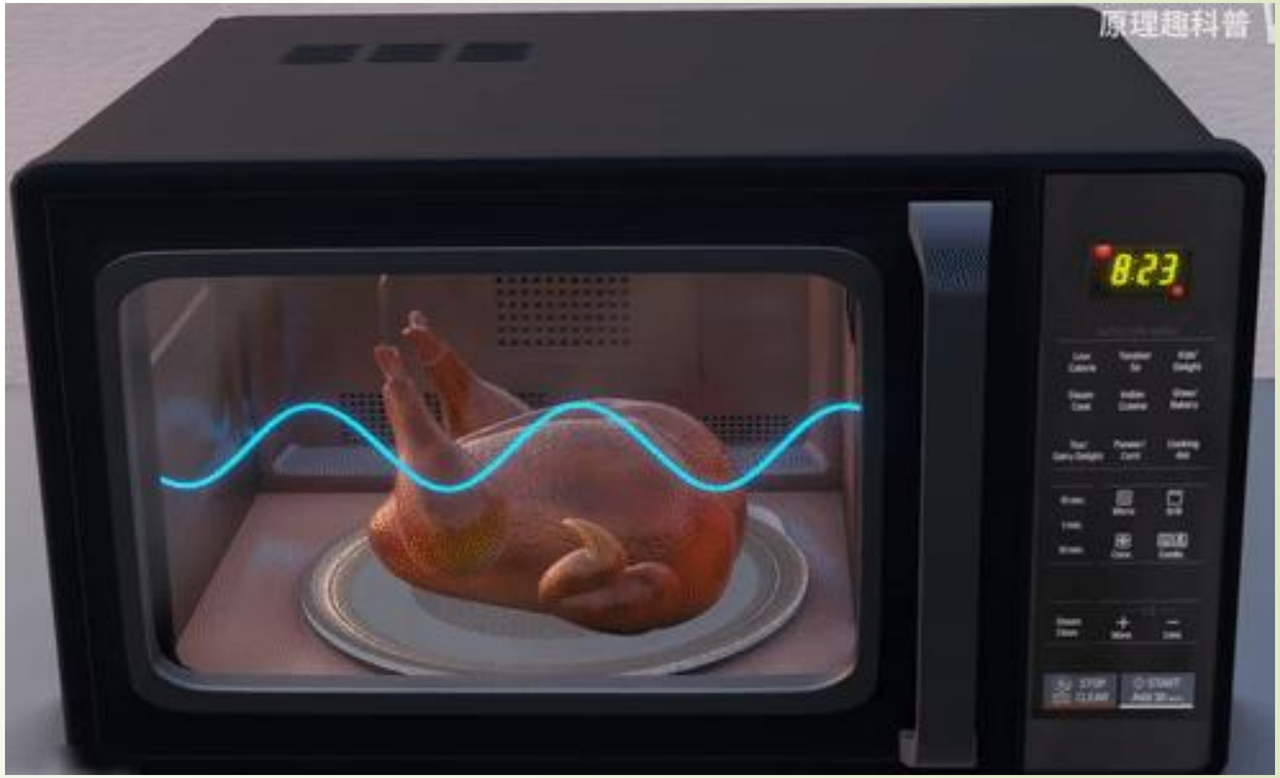
The distance from conducting plate is $z_{\max} = -\dfrac{(2n+1)\lambda_1}{4} \quad (n = 0,1,2,3,\cdots)$

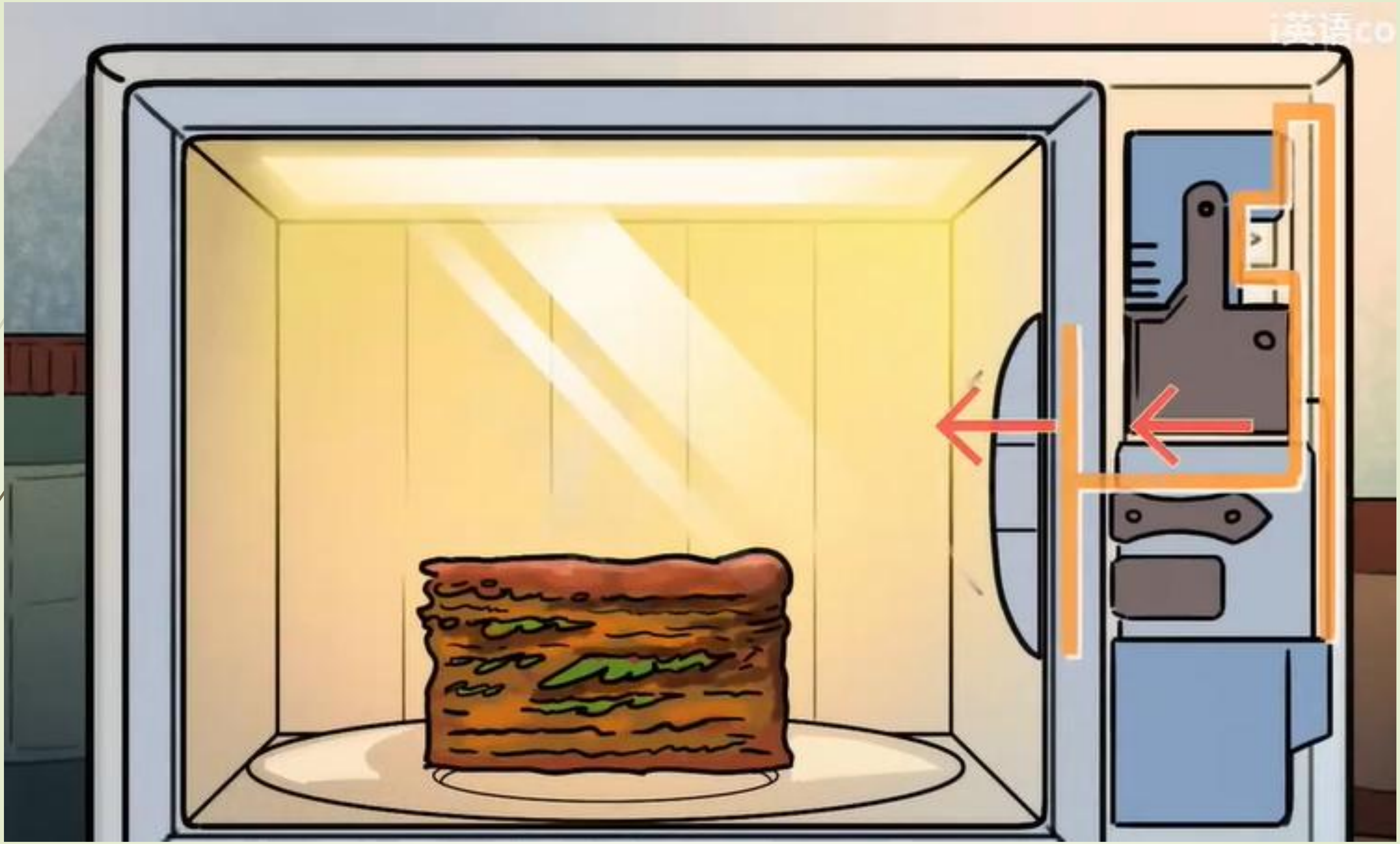
the point of nodes（the point of the minimum value of electric field）：

The distance from conducting plate is $z_{\min} = -\dfrac{n\lambda_1}{2} \quad (n = 0,1,2,3,\cdots)$

Microwave Oven: a multi-mode oscillator







Ex. uniform plane wave propagate along +Z, the electric field intensity vector is

$$\vec{E}_{\mathrm{i}} = \vec{e}_x 100\sin(\omega t - \beta z) + \vec{e}_y 200\cos(\omega t - \beta z) \quad \mathrm{V/m}$$

Find:

（1）the corresponding magnetic field intensity ;

（2）if on the direction of propagation at the point of *z* = 0，place an infinite plate of ideal conductor, write out the electric and magnetic field intensity in the *z* < 0 region ;

（3）the electric current density on the surface of ideal conductor.

Solution: (1) The expressions of the complex of the electric field intensity

$$\vec{E}_{\mathrm{i}} = \vec{e}_x 100 \mathrm{e}^{-\mathrm{j}\beta z} \mathrm{e}^{-\mathrm{j}\pi/2} + \vec{e}_y 200 \mathrm{e}^{-\mathrm{j}\beta z}$$

so
$$\vec{H}_{\mathrm{i}}(z) = \frac{1}{\eta_0} \vec{e}_z \times \vec{E}_{\mathrm{i}} = \frac{1}{\eta_0}(-\vec{e}_x 200 \mathrm{e}^{-\mathrm{j}\beta z} + \vec{\boldsymbol{e}}_y 100 \mathrm{e}^{-\mathrm{j}\beta z} \mathrm{e}^{-\mathrm{j}\pi/2})$$

the instantaneous expression:

$$\vec{H}_{\mathrm{i}}(z,t) = \mathrm{Re}[\vec{H}_{\mathrm{i}}(z)\mathrm{e}^{\mathrm{j}\omega t}]$$

$$= \frac{1}{\eta_0}[-\vec{e}_x 200\cos(\omega t - \beta z) + \vec{e}_y 100\cos(\omega t - \beta z - \frac{1}{2}\pi)]$$

(2) the electric field of the reflected wave is

$$\vec{E}_{\mathrm{r}}(z) = -\vec{e}_x 100\mathrm{e}^{\mathrm{j}\beta z}\mathrm{e}^{-\mathrm{j}\pi/2} - \vec{e}_y 200\mathrm{e}^{\mathrm{j}\beta z}$$

the magnetic field of the reflected wave is

$$\vec{H}_{\mathrm{r}}(z) = \frac{1}{\eta_0}(-\vec{e}_z \times \vec{E}_{\mathrm{r}}) = \frac{1}{\eta_0}(-\vec{e}_x 200\mathrm{e}^{\mathrm{j}\beta z} + \vec{e}_y 100\mathrm{e}^{\mathrm{j}\beta z}\mathrm{e}^{-\mathrm{j}\pi/2})$$

the expressions of electric and magnetic field of the total field in the $z < 0$ region, respectively

$$\vec{E}_1 = \vec{E}_\mathrm{i} + \vec{E}_\mathrm{r} = -\vec{e}_x \mathrm{j}200\mathrm{e}^{-\mathrm{j}\pi/2}\sin(\beta z) - \vec{e}_y \mathrm{j}400\sin(\beta z)$$

$$\vec{H}_1 = \vec{H}_\mathrm{i} + \vec{H}_\mathrm{r} = \frac{1}{\eta_0}[-\vec{e}_x 400\cos(\beta z) + \vec{e}_y 200\mathrm{e}^{-\mathrm{j}\pi/2}\cos(\beta z)]$$

(3) the electric current density on the surface of ideal conductor is

$$\vec{J}_S = -\vec{e}_z \times \vec{H}_1\Big|_{z=0}$$

$$= \vec{e}_x \frac{200}{\eta_0}\mathrm{e}^{-\mathrm{j}\pi/2} + \vec{e}_y \frac{400}{\eta_0} = -\vec{e}_x \mathrm{j}0.53 + \vec{e}_y 1.06$$

(4) State the polarization method of incident wave and the reflected wave. (Give the answer in Mook)

易拉罐能不能增强WiFi信号?

今天才知道，路由器上放两个易拉罐，wife信号可以增强三倍

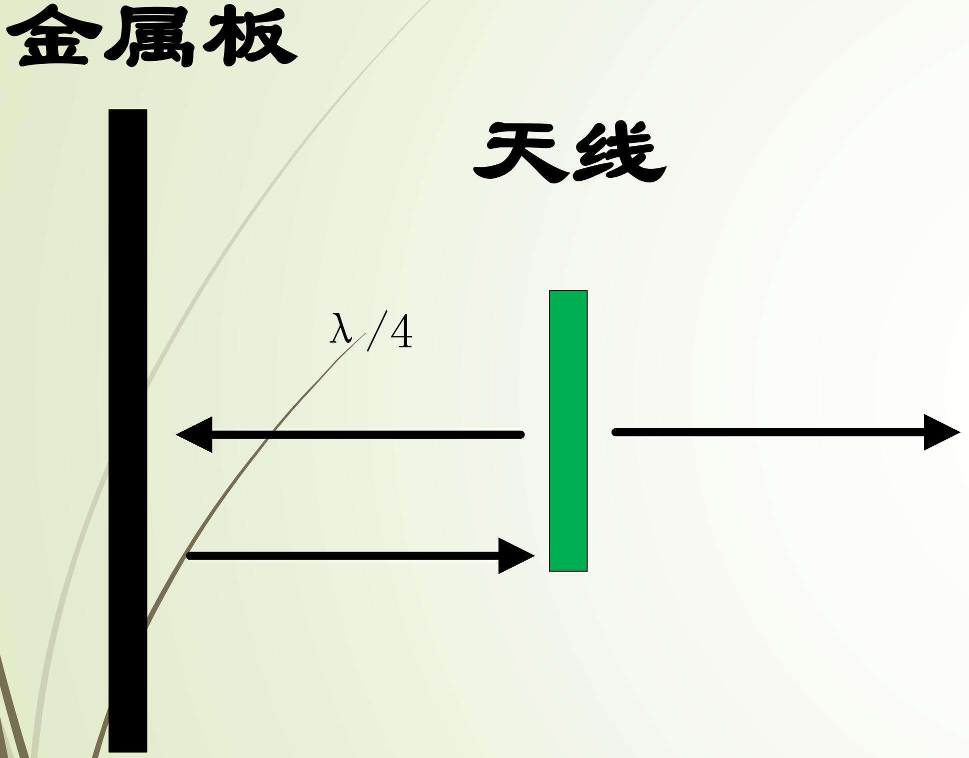
妙招小贴士

2019-11-13 15:51 | 传媒高管,手工领域爱好者

关注

是真的吗？

易拉罐能不能增强WiFi信号?



- 分析:

(1) WiFi天线是全向天线，现在只考虑左右两个方向，波会向左和向右传播

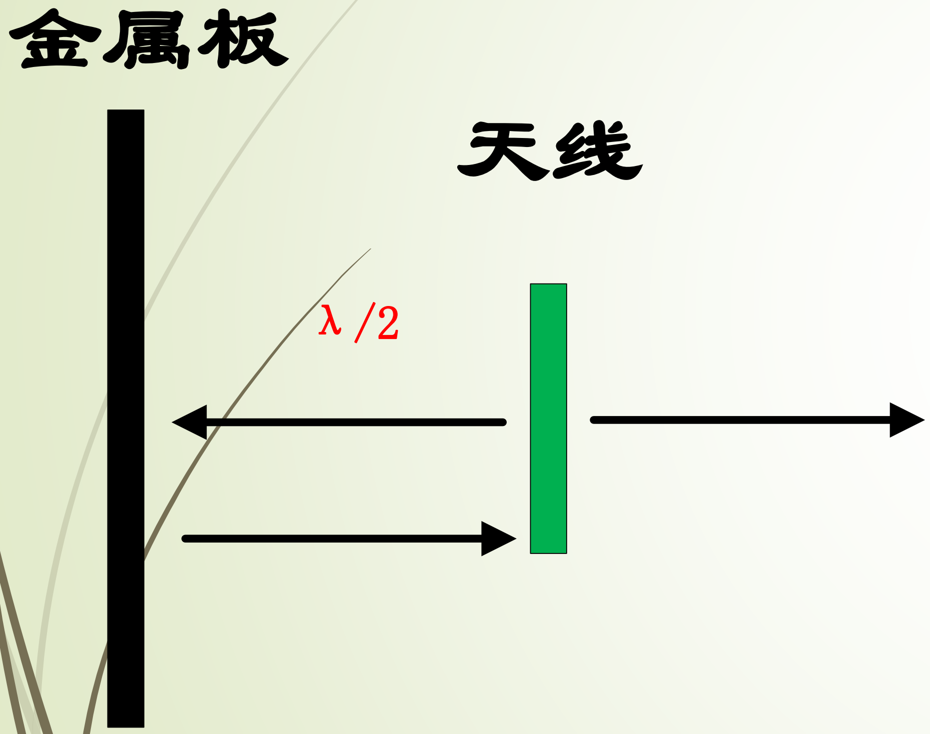
(2) 假设天线距离左边金属板的距离是 $\lambda/4$，那么波传播到左边金属板时相位相对于天线处滞后90°

(3) 金属板反射，滞后180°

(4) 波向右传播到天线处又滞后90°

(5) 总滞后360°，相当于和向右的波同相叠加，幅度增加为原来的两倍，信号变强。

易拉罐能不能增强WiFi信号？



- 分析：

（1） WiFi天线是全向天线，现在只考虑左右两个方向，波会向左和向右传播

（2）假设天线距离左边金属板的距离是 λ/2，那么波传播到左边金属板时相位相对于天线处滞后180°

（3）金属板反射，滞后180°

（4）波向右传播到天线处又滞后180°

（5）总滞后540° =180°，相当于和向右的波反相叠加，幅度变为零，信号减弱。

易拉罐能不能增强WiFi信号？

今天才知道，路由器上放两个易拉罐，wife信号可以增强三倍

妙招小贴士

2019-11-13 15:51 | 传媒高管,手工领域爱好者

关注

（1）易拉罐能不能增强Wifi信号？

能，但是有条件。有可能增强，也可能减弱

（2）WiFi信号能不能增强三倍？

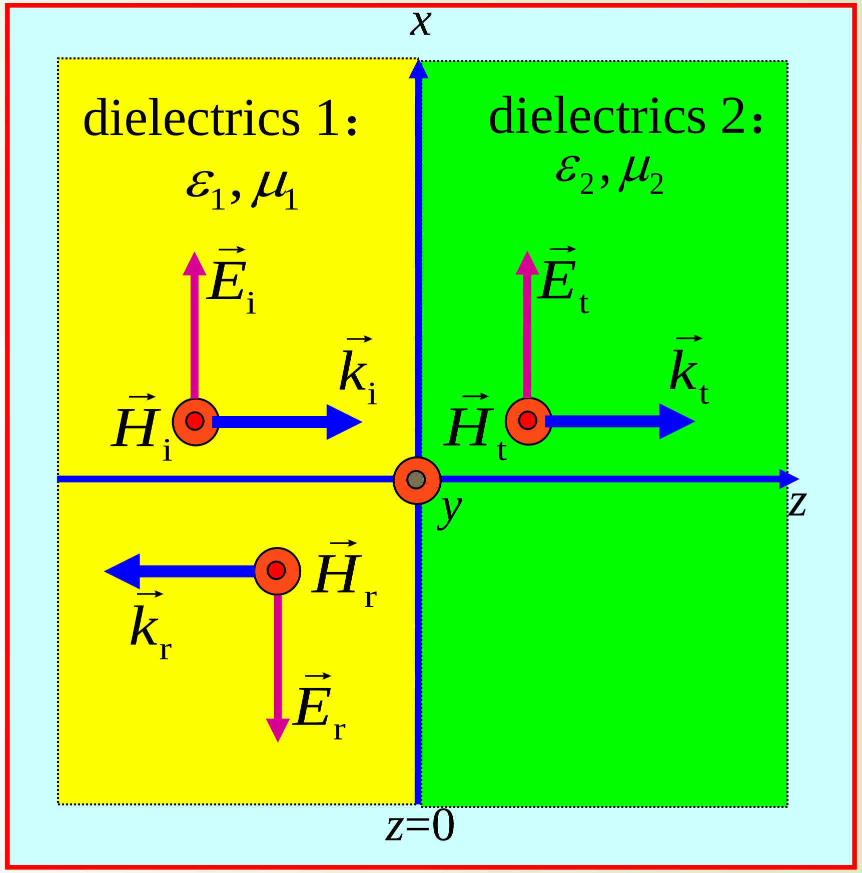
不能，最大增强两倍。

（3）增强的本质？

类似于在灯泡后面加一面镜子。

全向天线变成定向天线，在天线设计中经常用到反射板。

8.5 Normal Incidence at a Plane Dielectric Boundary



Assume that two kinds of medium are the ideal dielectrics，$\sigma_1 = \sigma_2 = 0$

so

$$\gamma_1 = \mathrm{j}\beta_1 = \mathrm{j}\omega\sqrt{\mu_1\varepsilon_1}$$

$$\gamma_2 = \mathrm{j}\beta_2 = \mathrm{j}\omega\sqrt{\mu_2\varepsilon_2}$$

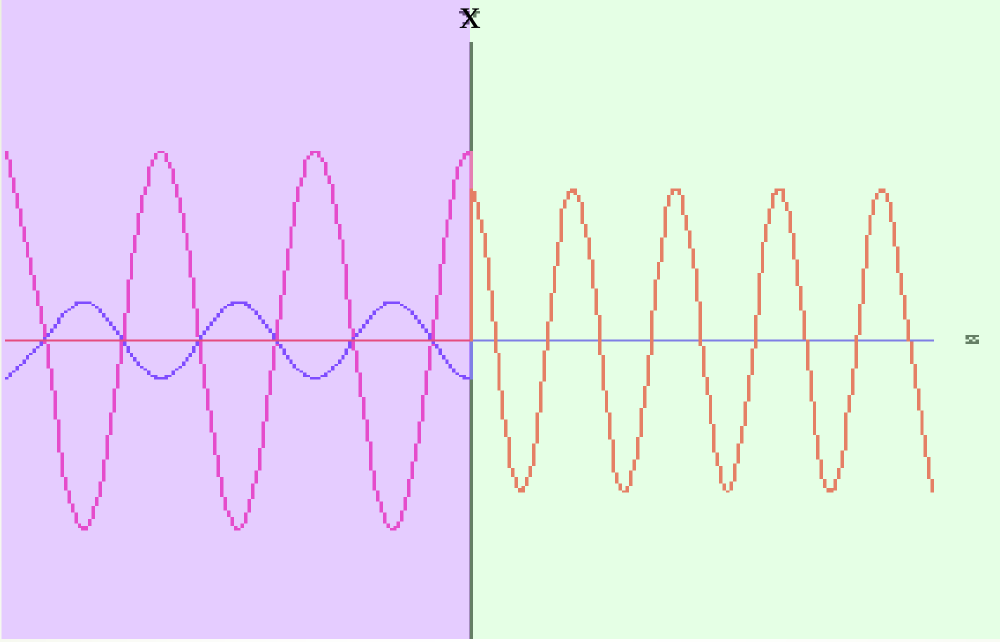
$$\eta_{1c} = \eta_1 = \sqrt{\frac{\mu_1}{\varepsilon_1}}, \quad \eta_{2c} = \eta_2 = \sqrt{\frac{\mu_2}{\varepsilon_2}}$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}, \quad \tau = \frac{2\eta_2}{\eta_2 + \eta_1}$$

- Discuss
  - **when $\eta_2 > \eta_1$，$\Gamma > 0$，the electric field of reflected wave and incidence wave is the same in phase.**
  - **when $\eta_2 < \eta_1$，$\Gamma < 0$，the electric field of reflected wave and incidence wave is reverse in phase.**

incidence wave

reflected wave



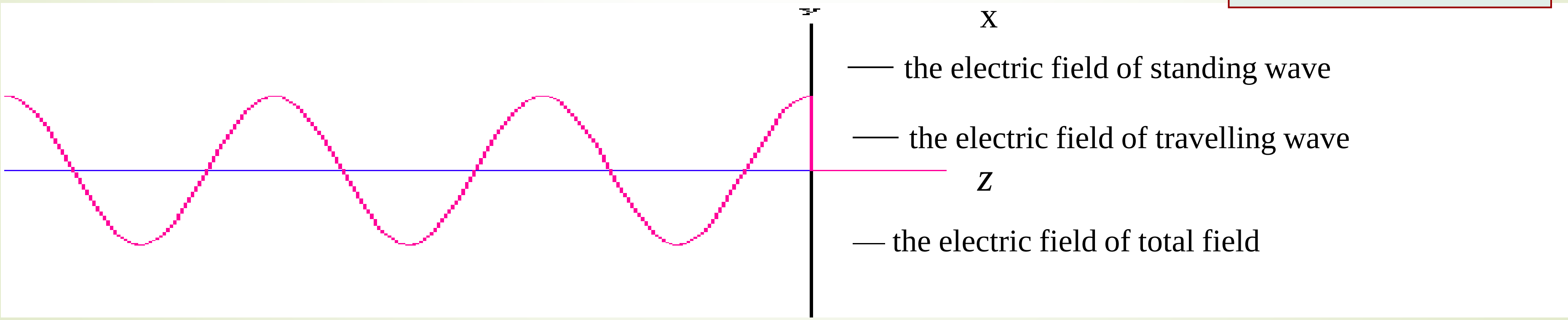
transmitted wave

total：

$$\vec{E}_1(z) = \vec{e}_x E_{\mathrm{im}} (\mathrm{e}^{-\mathrm{j}\beta_1 z} + \Gamma \mathrm{e}^{\mathrm{j}\beta_1 z}) = \vec{e}_x E_{\mathrm{im}} \left[(1+\Gamma)\mathrm{e}^{-\mathrm{j}\beta_1 z} + \Gamma (\mathrm{e}^{\mathrm{j}\beta_1 z} - \mathrm{e}^{-\mathrm{j}\beta_1 z}) \right]$$

$$= \vec{e}_x E_{\mathrm{im}} \left[(1+\Gamma)\mathrm{e}^{-\mathrm{j}\beta_1 z} + \mathrm{j}2\Gamma \sin(\beta_1 z) \right]$$

travelling-standing wave



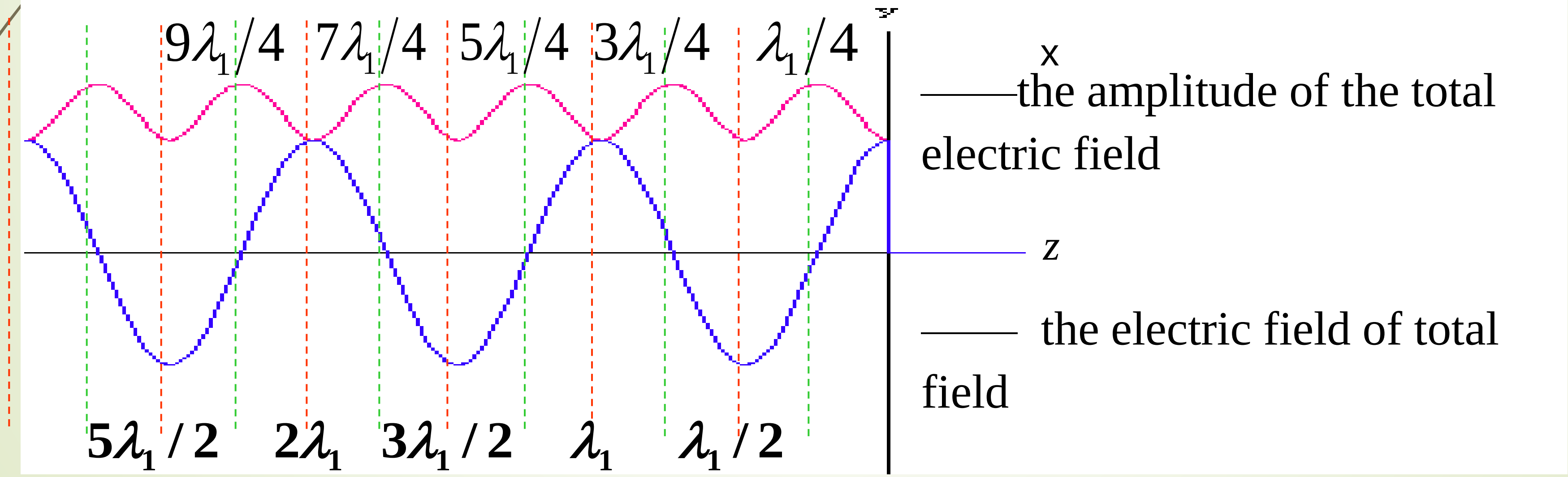
The amplitude change of the total electric field in the ideal dielectric

$$\left|\vec{E}_1(z)\right| = E_{\mathrm{im}}\left|1+\Gamma \mathrm{e}^{\mathrm{j}2\beta_1 z}\right| = E_{\mathrm{im}}\sqrt{1+\Gamma^2+2\Gamma\cos(2\beta_1 z)}$$

- **optically denser medium（$\Gamma < 0$）**

$$\left|\vec{E}_1(z)\right|_{\max} = E_{\mathrm{im}}\left|1-\Gamma\right| \qquad z = -(n/2+1/4)\lambda_1 \qquad (n = 0,1,2,\cdots)$$

$$\left|\vec{E}_1(z)\right|_{\min} = E_{\mathrm{im}}\left|1+\Gamma\right| \qquad z = -n\lambda_1/2 \qquad (n = 0,1,2,\cdots)$$



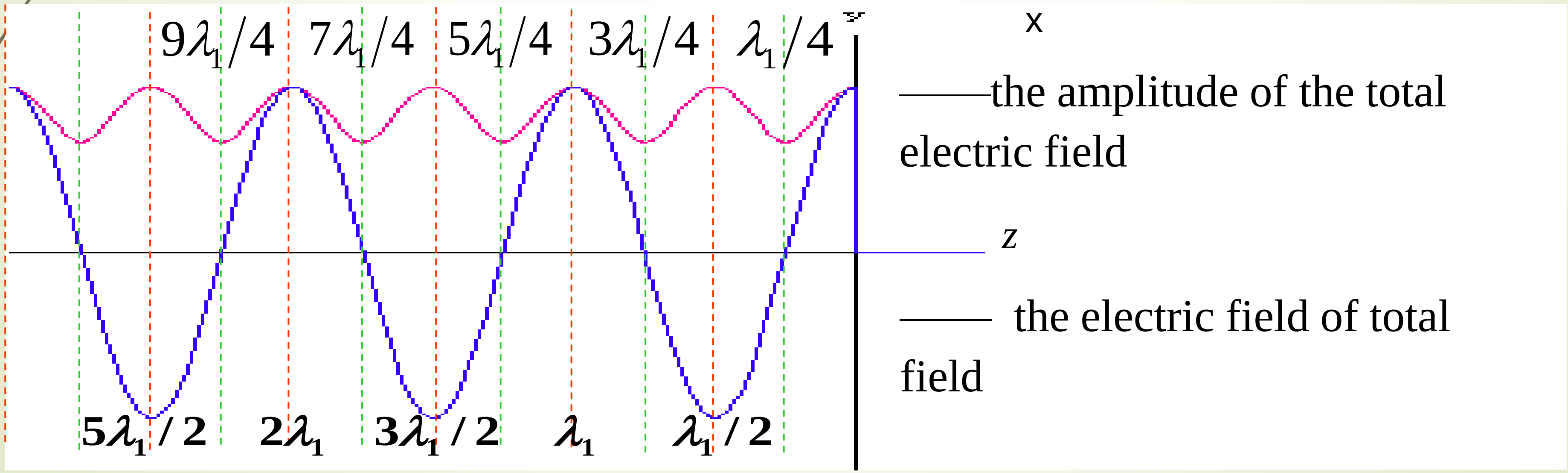
The change rule in the space for the amplitude of the total electric field in the ideal dielectric

$$\left|\vec{E}_1(z)\right| = E_{\text{im}}\left|1+\Gamma e^{j2\beta_1 z}\right| = E_{\text{im}}\sqrt{1+\Gamma^2+2\Gamma\cos(2\beta_1 z)}$$

- **Light thin medium（$\Gamma > 0$）**

$$\left|\vec{E}_1(z)\right|_{\max} = E_{\text{im}}\left|1+\Gamma\right| \qquad z = -n\lambda_1/2 \qquad (n = 0,1,2,\cdots)$$

$$\left|\vec{E}_1(z)\right|_{\min} = E_{\text{im}}\left|1-\Gamma\right| \qquad z = -(n/2+1/4)\lambda_1 \qquad (n = 0,1,2,\cdots)$$



Question and Thinking：

1） Whether the amplitude of travelling wave in the ideal dielectric will change with different space position?

2）What is the standing wave?

3）Reflected wave is travelling wave or standing wave?

4）what wave is the total field wave？

5）Why the amplitude of total field will by changed with different space position?

6）What is the reason for standing wave？

reflection ⟷ standing wave

Introducing a new quantity to express standing wave or the degree of reflection

- **standing-wave ratio (SWR) *S***

$$S=\frac{\left|\vec{E}\right|_{\max}}{\left|\vec{E}\right|_{\min}}=\frac{1+|\Gamma|}{1-|\Gamma|} \Longleftrightarrow |\Gamma|=\frac{S-1}{S+1}$$

- Discuss
  - **when $\Gamma=0$, $S=1$, travelling wave.**
  - **when $\Gamma=\pm 1$, $S=\infty$, Pure standing-wave.**
  - **when $0<|\Gamma|<1$, $1<S<\infty$, Mixed wave. With the value of S increase, the component of standing wave increase ,and the component of the travellling wave reduce.**

Electromagnetic energy flow density

The average power density propagate along Z in the dielectric 1

$$\vec{S}_{\mathrm{iav}} = \frac{1}{2}\mathrm{Re}\left[\vec{E}_{\mathrm{i}} \times \vec{H}_{\mathrm{i}}^{*}\right] = \vec{e}_z \boxed{\frac{1}{2\eta_1} E_{\mathrm{im}}^2}$$

$$\vec{S}_{\mathrm{rav}} = \frac{1}{2}\mathrm{Re}\left[\vec{E}_{\mathrm{r}} \times \vec{H}_{\mathrm{r}}^{*}\right] = -\vec{e}_z \frac{1}{2\eta_1} \Gamma^2 E_{\mathrm{im}}^2$$

$$\vec{S}_{\mathrm{1av}} = \frac{1}{2}\mathrm{Re}\left[\vec{E}_1 \times \vec{H}_1^{*}\right] = \vec{e}_z \frac{E_{\mathrm{im}}^2}{2\eta_1}(1-\Gamma^2)$$

$$\boxed{\vec{S}_{\mathrm{1av}} = \vec{S}_{\mathrm{2av}}}$$

The average power density in the dielectric 2

$$\vec{S}_{\mathrm{2av}} = \frac{1}{2}\mathrm{Re}\left[\vec{E}_2 \times \vec{H}_2^{*}\right] = \vec{e}_z \boxed{\frac{E_{\mathrm{im}}^2}{2\eta_2}\tau^2}$$

due $1-\Gamma^2 = (1+\Gamma)(1-\Gamma) = \dfrac{\eta_1}{\eta_2}\tau^2$

Ex. In the free space, uniform plane wave vertical incidence to the semi infinite plane of lossless dielectric. Given the SWR=3 in the free space, the wavelength in the dielectric is 1/6 of that in the space. The point of the minimum value of standing wave is on the interface. Work out the relative permeability and relative permittivity of the dielectric.

Answer： Due to SWR is $S=\dfrac{1+|\Gamma|}{1-|\Gamma|}=3 \Rightarrow |\Gamma|=\dfrac{1}{2}$

as the point of the minimum value of standing wave is on the interface $\Gamma=-\dfrac{1}{2}$

as reflection coefficient $\Gamma=\dfrac{\eta_2-\eta_1}{\eta_2+\eta_1}$

where $\eta_1=\eta_0,\ \eta_2=\sqrt{\dfrac{\mu_2}{\varepsilon_2}}=\sqrt{\dfrac{\mu_r}{\varepsilon_r}}\eta_0$

$$\Rightarrow \eta_2=\frac{1}{3}\eta_0 \Rightarrow \frac{\mu_r}{\varepsilon_r}=\frac{1}{9}$$

because the wavelength in 2 region $\lambda_2=\dfrac{\lambda_0}{\sqrt{\mu_r\varepsilon_r}}=\dfrac{\lambda_0}{6} \Rightarrow \varepsilon_r\mu_r=36 \Rightarrow \begin{cases}\mu_r=2\\ \varepsilon_r=18\end{cases}$

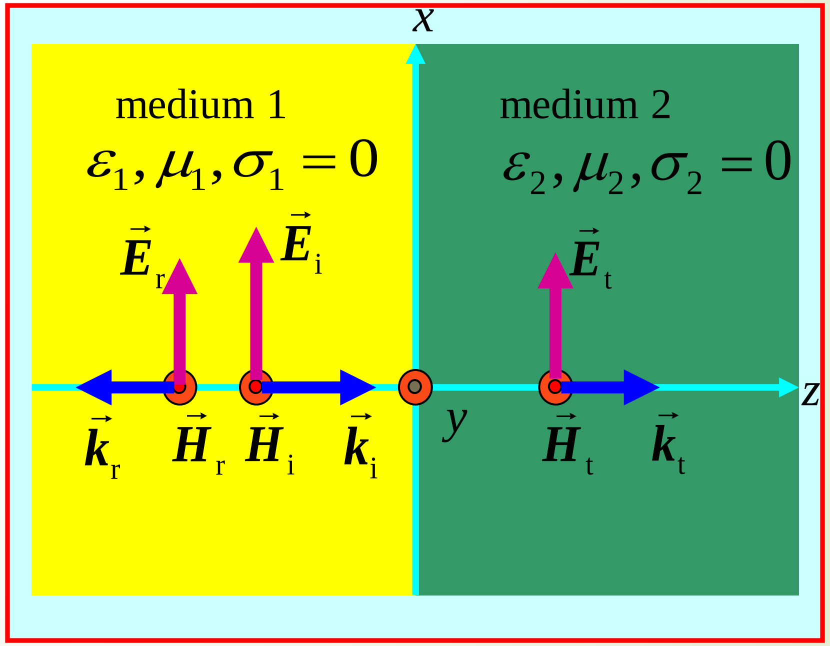
Ex. **The electric field of incidence wave is** $\vec{E}_{\mathrm{i}} = \vec{e}_x 100\cos(3\pi 10^9 t - 10\pi z)\ \ \mathrm{V/m}$ **.**

it vertical incident to the interface at $z = 0$ from the region ($z < 0$) of air (μ_r=1 ε_r = 4) .

Find the electric and magnetic fields in the region of $z > 0$.

Solution: the intrinsic impedance in the $z > 0$ region

$$\eta_2 = \sqrt{\frac{\mu_2}{\varepsilon_2}} = \eta_0 \sqrt{\frac{\mu_{r2}}{\varepsilon_{r2}}} = \frac{120\pi}{2} = 60\pi\ \ \Omega$$



Transmission coefficient

$$\tau = \frac{2\eta_2}{\eta_1 + \eta_2} = \frac{2 \times 60\pi}{120\pi + 60\pi} = 0.667$$

phase constant

$$\beta_2 = \omega\sqrt{\mu_2\varepsilon_2} = \omega\sqrt{\mu_0\varepsilon_0}\sqrt{\varepsilon_{r2}} = \frac{3\pi\times10^9}{3\times10^8}\times2 = 20\pi \quad \text{rad/m}$$

so

$$\vec{E}_2 = \vec{e}_x E_{2m}\cos(\omega t - \beta_2 z) = \vec{e}_x \tau E_{im}\cos(\omega t - \beta_2 z)$$

$$= \vec{e}_x 0.667\times10\cos(3\pi\times10^9 t - 20\pi z)$$

$$= \vec{e}_x 6.67\cos(3\pi\times10^9 t - 20\pi z) \quad \text{V/m}$$

$$\vec{H}_2 = \frac{1}{\eta_2}\vec{e}_z\times\vec{E}_2$$

$$= \vec{e}_y \frac{6.67}{60\pi}\cos(3\pi\times10^9 t - 20\pi z)$$

$$= \vec{e}_y 0.036\cos(3\pi\times10^9 t - 20\pi z) \quad \text{A/m}$$

Ex. Given the parameters of medium 1 are ε_{r1}=4、μ_{r1}=1、σ_1=0； medium 2 : ε_{r2}=10、μ_{r2} = 4、σ_2= 0 . A uniform plane wave which its angular frequency is $\omega=5\times10^8$ rad /s vertical incidence to the interface from medium 1 . The incidence wave is linear polarized wave in the direction of x axis , when t=0、z=0， the amplitude of incidence wave is 2.4 V/m . Find:

(1) β_1 and β_2 ;

(2) reflection coefficient Γ_1 and Γ_2 ;

(3) the electric field in the region 1 $\vec{E}_1(z,t)$;

(4) the electric field in the region 2 $\vec{E}_2(z,t)$.

Solution:（1）

$$\beta_1=\omega\sqrt{\mu_1\varepsilon_1}=\omega\sqrt{\mu_0\varepsilon_0}\sqrt{\mu_{r1}\varepsilon_{r1}}=\frac{5\times10^8}{3\times10^8}\times2=3.33\ \ \text{rad/m}$$

$$\beta_2=\omega\sqrt{\mu_0\varepsilon_0}\sqrt{\mu_{r2}\varepsilon_{r2}}=\frac{5\times10^8}{3\times10^8}\sqrt{10\times4}=10.54\ \ \text{rad/m}$$

（2）

$$\eta_1=\sqrt{\frac{\mu_1}{\varepsilon_1}}=\eta_0\sqrt{\frac{\mu_{r1}}{\varepsilon_{r1}}}=\eta_0\frac{1}{2}=60\pi\ \ \Omega$$

$$\eta_2=\sqrt{\frac{\mu_2}{\varepsilon_2}}=\eta_0\sqrt{\frac{\mu_{r2}}{\varepsilon_{r2}}}=\eta_0\sqrt{\frac{4}{10}}\approx 75.9\pi\ \ \Omega$$

$$\Gamma=\frac{\eta_2-\eta_1}{\eta_2+\eta_1}=\frac{75.9-60}{60+75.9}=0.117$$

（3） the electric field in the region 1

$$\vec{E}_1(z)=\vec{E}_i(z)+\vec{E}_r(z)=\vec{e}_x E_{im}(e^{-j\beta_1 z}+\Gamma e^{j\beta_1 z})$$

$$=\vec{e}_x E_{im}[(1+\Gamma)e^{-j\beta_1 z}+j2\Gamma\sin(\beta_1 z)]$$

$$=\vec{e}_x 2.4[1.117e^{-j3.33z}+j0.234\sin(3.33z)]$$

or $$\vec{E}_1(z)=\vec{E}_{\mathrm{i}}(z)+\vec{E}_{\mathrm{r}}(z)=\vec{e}_x 2.4\mathrm{e}^{-\mathrm{j}3.33z}+\vec{e}_x 0.281\mathrm{e}^{\mathrm{j}3.33z}$$

$$\vec{E}_1(z,t)=\mathrm{Re}\left[\vec{E}_1(z)\mathrm{e}^{\mathrm{j}\omega t}\right]$$

$$=\vec{e}_x 2.4\cos(5\times10^8 t-3.33z)+\vec{e}_x 0.281\cos(5\times10^8 t+3.33z)$$

（4） $$\tau=\frac{2\eta_2}{\eta_1+\eta_2}\approx 1.12$$

so $$\vec{E}_2(z)=\vec{e}_x E_{\mathrm{tm}}\mathrm{e}^{-\mathrm{j}\beta_2 z}=\vec{e}_x\tau E_{\mathrm{im}}\mathrm{e}^{-\mathrm{j}\beta_2 z}$$

$$=\vec{e}_x 1.12\times2.4\mathrm{e}^{-\mathrm{j}10.54z}=\vec{e}_x 2.68\mathrm{e}^{-\mathrm{j}10.54z}$$

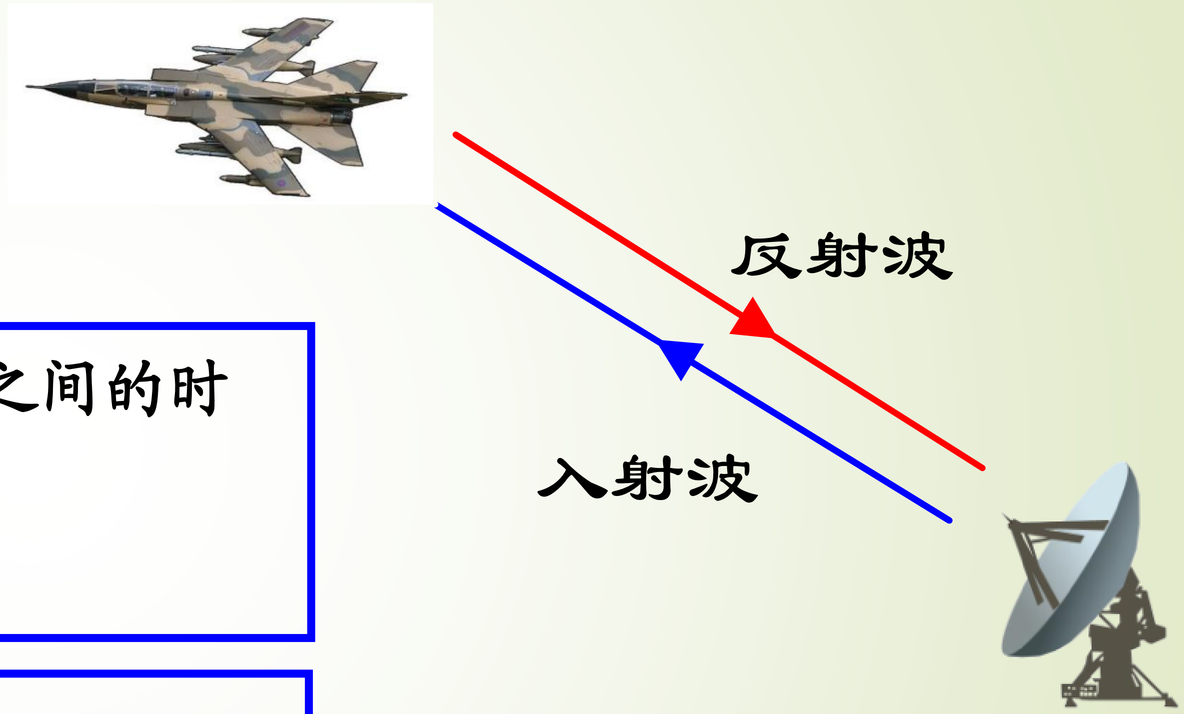
$$\vec{E}_2(z,t)=\vec{e}_x 2.68\cos(5\times10^8 t-10.54z)$$

雷达测距和雷达低空盲区

雷达工作原理:

（1）发射电磁波，接收电磁波。

（2）根据反射波特性判断物体的特性。



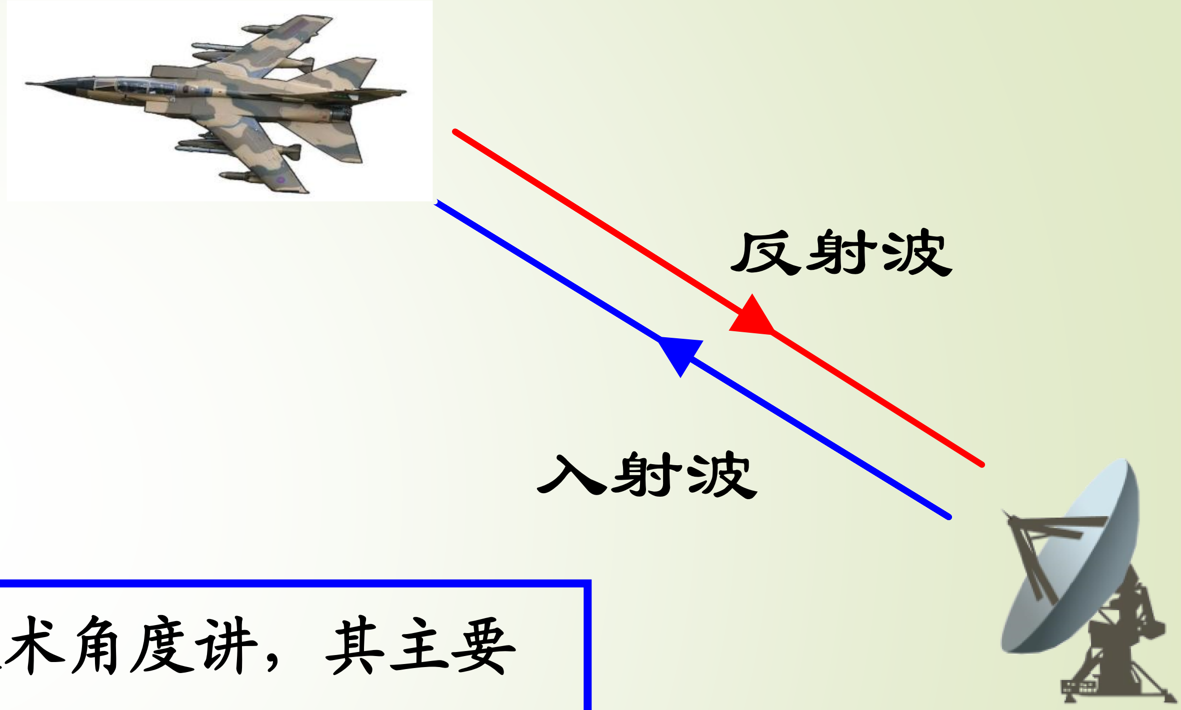
雷达与目标之间距离为s，发射波与回波之间的时间间隔为t，电磁波以光速c传播

目标距离雷达之间的距离为：$S=ct/2$

测距满足条件:

1. 电磁波直线传播
2. 有反射波存在
3. 入射波和反射波等速

雷达测距和雷达低空盲区



如何不被雷达探测到?

（1）隐身（不是我们讨论的重点）

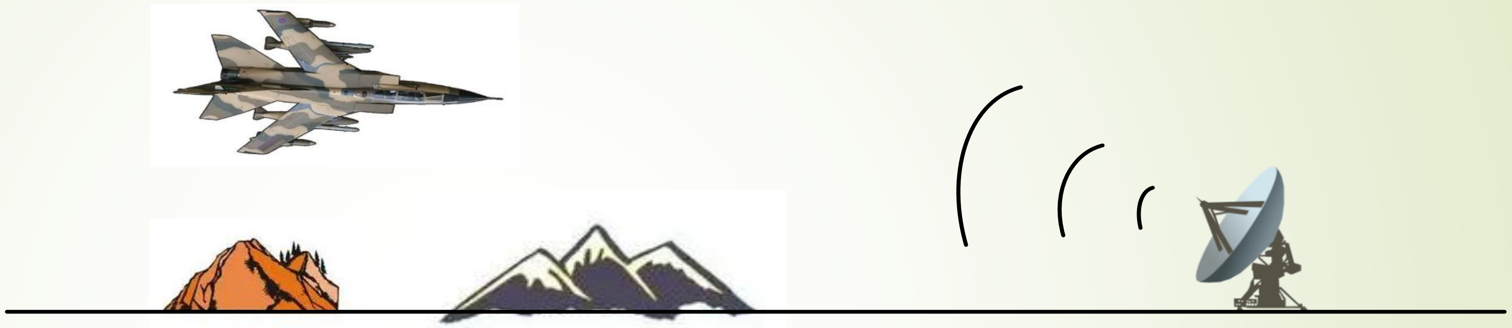
（2）低空突防

1. 隐身：一是减小飞机的雷达反射面，从技术角度讲，其主要措施有设计合理的飞机外型、使用吸波材料、主动对消、被动对消等;二是降低红外辐射，主要是对飞机上容易产生红外辐射的部位采取隔热、降温等措施;三是运用隐蔽色降低肉眼可视度

雷达测距和雷达低空盲区

如何不被雷达探测到？

（2）低空突防

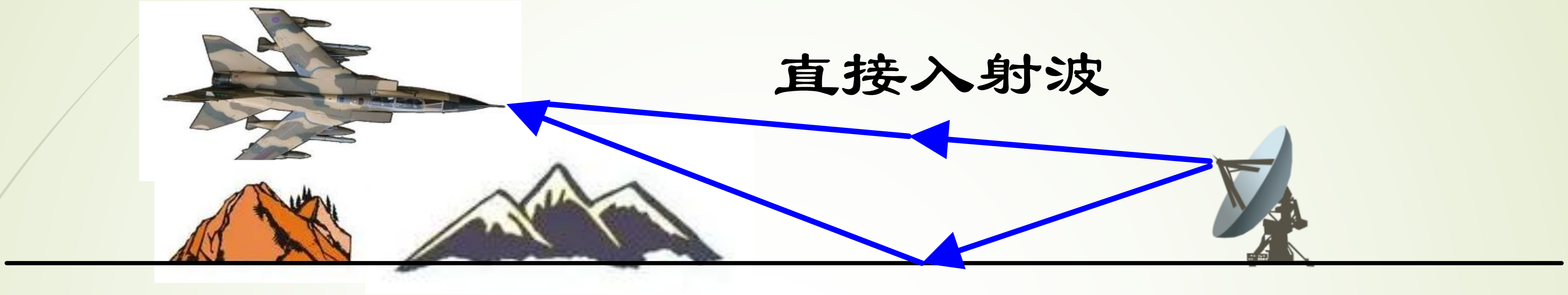


2．低空突防：雷达指向低空目标时存在盲区，低空指1000米以下。

实战案例：（1）在1982年马岛战争中，阿根廷使用法国研制的“飞鱼”机载巡航导弹，击沉英国的“谢菲尔德”号导弹驱逐舰。该巡航导弹就是低空巡航导弹。

（2）1991年美国向伊拉克发射了数百枚“战斧”式巡航导弹，大都精确击中目标。战斧导弹也是超低空飞行的。导弹在海面飞行高度7～15米，平坦陆地为50米以下，山区和丘陵地带为100米以下.

雷达测距和雷达低空盲区



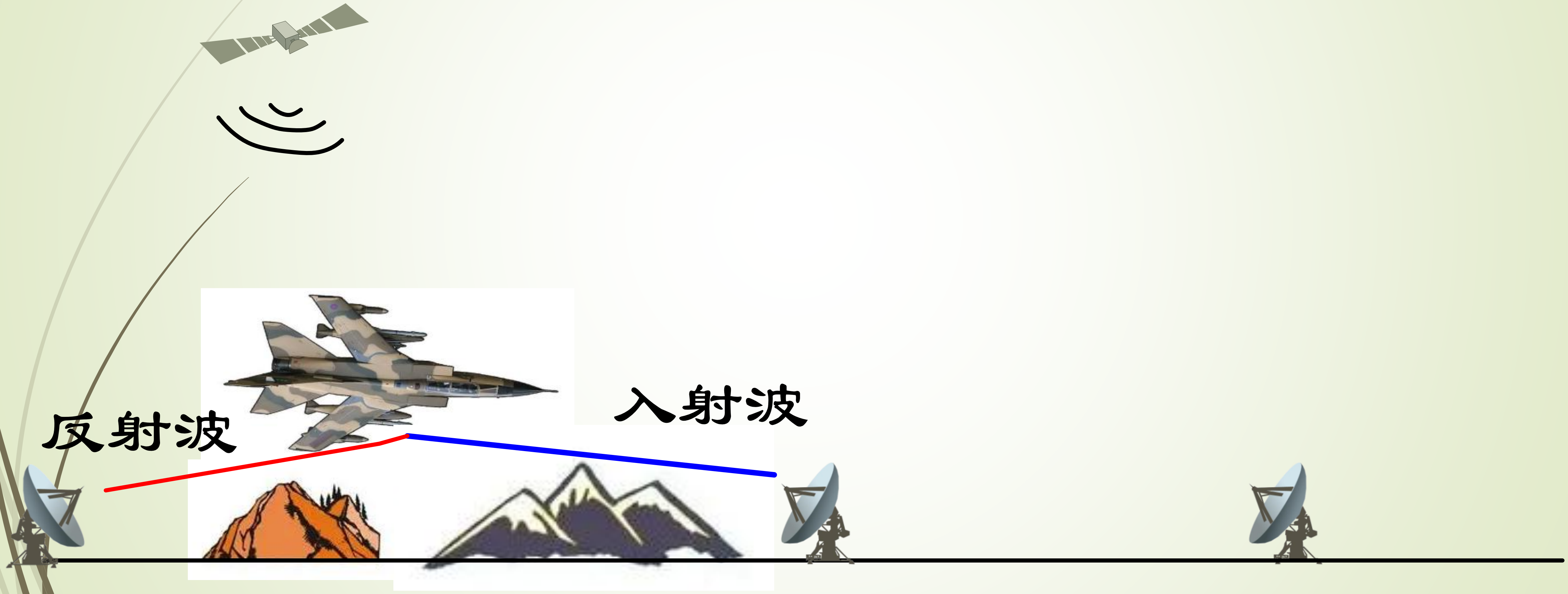
由于雷达波几乎于90度入射，反射波处于斜滑入射方向，反射系数为-1. 地面反射波与直接波近似等值反相，合成波大大消弱。故而成了雷达的盲区，无法发现目标。

$$\begin{cases} \theta_i \to \dfrac{\pi}{2} \\ \Gamma_\perp = \dfrac{\eta_2\cos\theta_i - \eta_1\cos\theta_t}{\eta_2\cos\theta_i + \eta_1\cos\theta_t} \approx -1 \end{cases}$$

$$\begin{cases} \theta_i \to \dfrac{\pi}{2} \\ \Gamma_\parallel = \dfrac{\eta_1\cos\theta_i - \eta_2\cos\theta_t}{\eta_1\cos\theta_i + \eta_2\cos\theta_t} \approx -1 \end{cases}$$

雷达测距和雷达低空盲区

如何解决低空盲区？



Summary

- In the space of incidence wave there are incidence wave and reflected wave , only transmitted wave in the transmitted space
- The direction of incidence、reflected and transmitted wave both in the normal direction of the interface.
- The direction of electromagnetic field of incidence、reflected and transmitted wave both in the same coordinate axis direction.
- **Incidence, reflection and transmission wave respectively satisfy the propagation rules in a single dielectric space.**
- Related concepts and physical quantities：

Travelling wave 、standing wave 、travelling-standing wave, reflection coefficient 、transmission coefficient, standing wave coefficient (SWR), antinodes, node, light dense medium, light thinner medium

Key points for the problem solving method:

✓ Basis on the incidence wave, written expressions of reflected wave and transmitted wave respectively. Then analysis and discuss the related total wave.

✓ According to the definition and the law to work out the physical quantities.

Homework

- 8-5, 8-6, 8-12, 8-17, 8-22, 8-27